Architecting Physical Intelligence:

Cross-Stack Co-Design from Systems to Silicon

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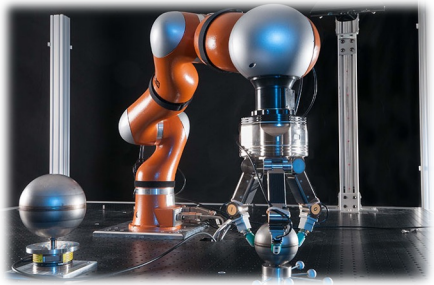
CS6998 Week 7: Accelerating Neuro-Symbolic and Compositional AI

What Do We Mean By Physical Intelligence?

Embodied Robots

Autonomous Systems

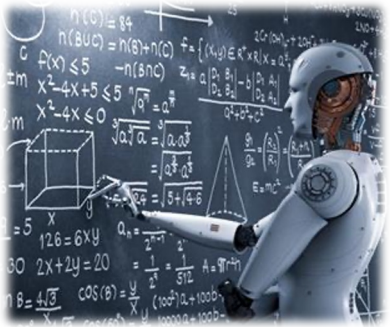


Manipulation

AR/VR

Assistive Systems

Smart Homes



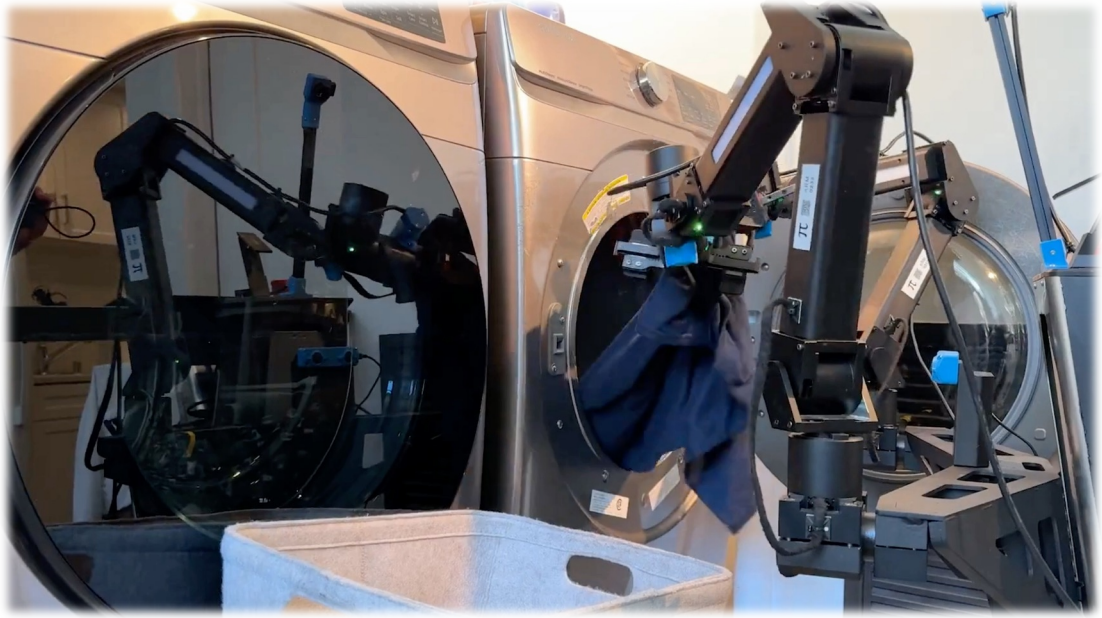
Education

Agriculture

Manufacturing

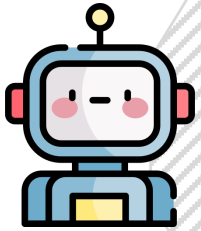
Healthcare

What Do We Mean By Physical Intelligence?



Physical Intelligence refers to embodied agents that **perceive, reason, and act** in the **physical world** to accomplish complex **long-horizon tasks**

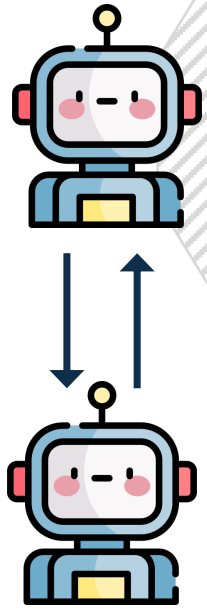
What Does Physical Intelligence Compute?



Perception & Localization → Reasoning & Planning → Control & Action

Environment interaction / feedback

What Does Physical Intelligence Compute?



Perception & Localization → Reasoning & Planning → Control & Action

Environment interaction / feedback

What Does Physical Intelligence Compute?

Physical AI

Cognition

Autonomy

Tight coupling across timescales

Perception & Localization → Reasoning & Planning → Control & Action

Environment interaction / feedback

Why Computing for Physical Intelligence Is Hard?

Physical AI

Cognition

Autonomy

Tight coupling across timescales

Real-Time Performance

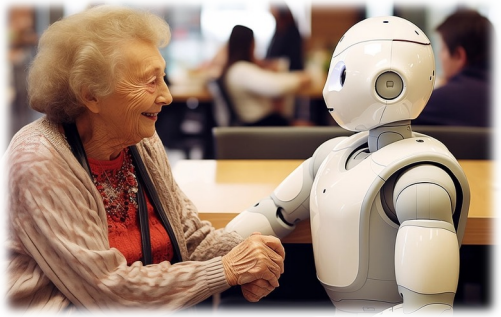
	Goal	Current
Cognition Loop:	>1 Hz	<0.1 Hz
Autonomy Loop:	>100 Hz	<1 Hz

✗

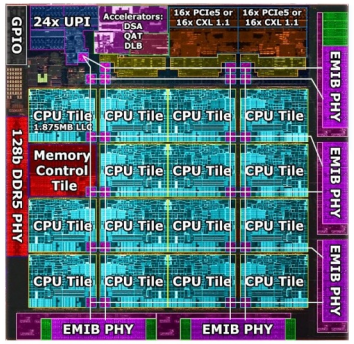
Strict Energy Budgets

Battery:
Battery Capacity = 64 Wh
Average Runtime = 20 mins
Standby Time = 40 mins
Recharge Time = 13 mins

Critical Safety & Reliability



Complex Design Space



Design Space:
Algorithm: $O(10^{11})$
Hardware: $O(10^{8})$
Dynamics: $O(10^{3})$
Sensors: $O(10^{2})$

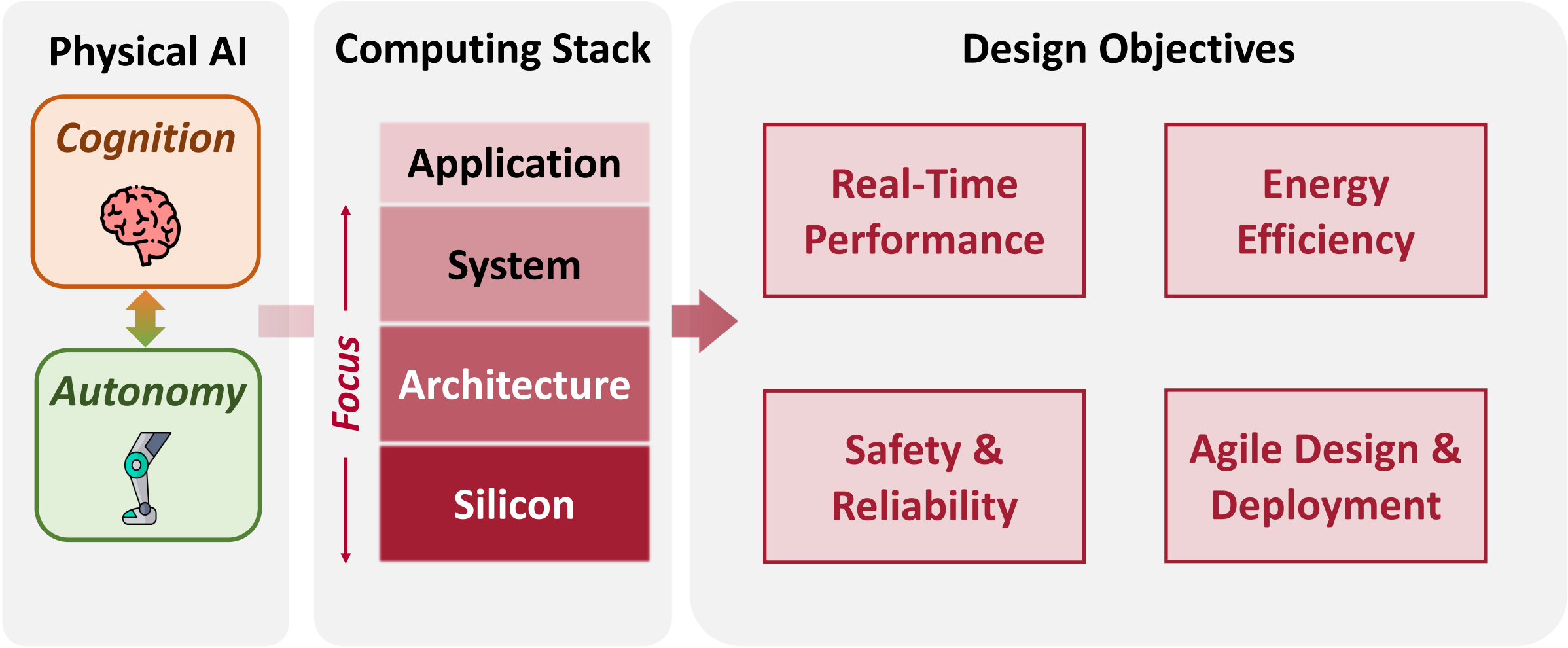
Fundamental mismatches exist between today's systems and physical intelligence requirements

How should we design computing hardware that enable intelligence to reason and act in physical world efficiently, safely, and in real time?

Cross-Stack System-Architecture-Silicon Co-Design

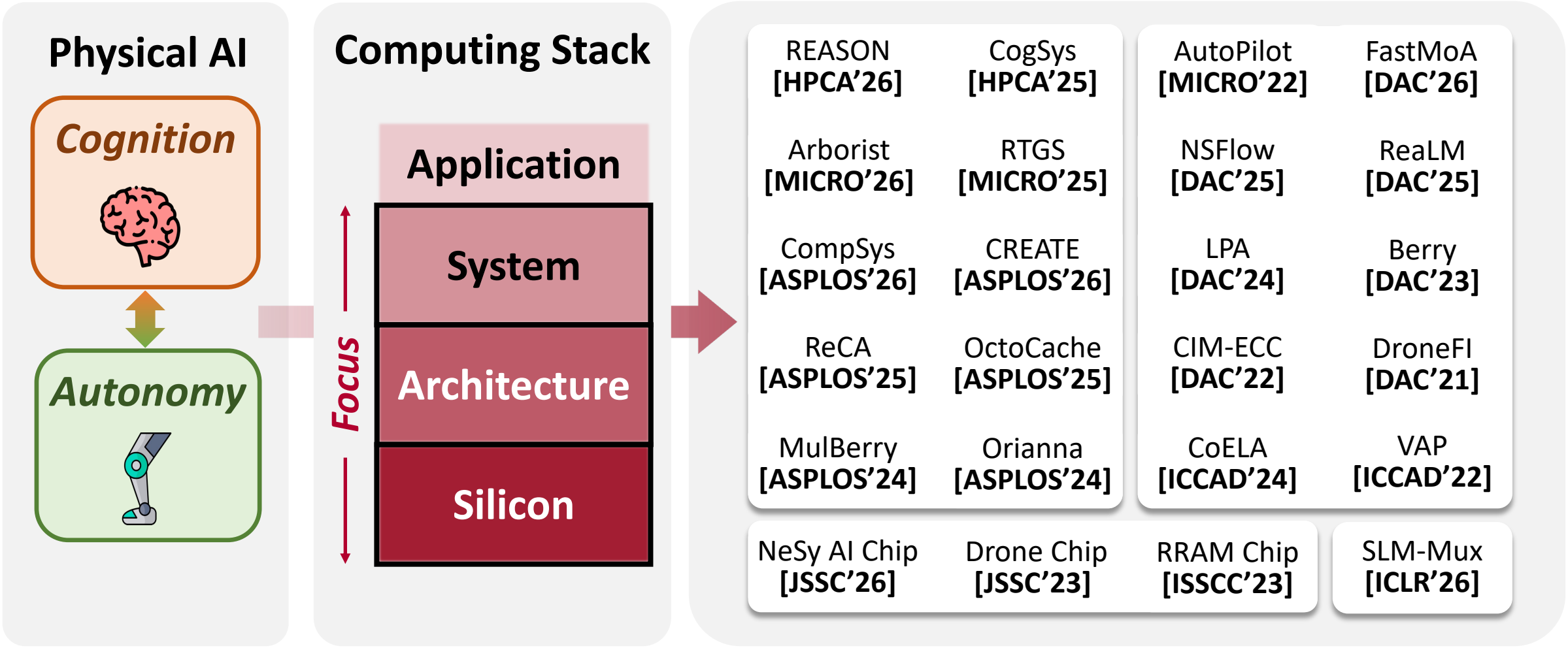
Architecting Computing System for Physical AI

Cross-Stack Co-Design from Systems to Silicon



Architecting Computing System for Physical AI

Cross-Stack Co-Design from Systems to Silicon



Architecting Computing System for Physical AI

Cross-Stack Co-Design from Systems to Silicon

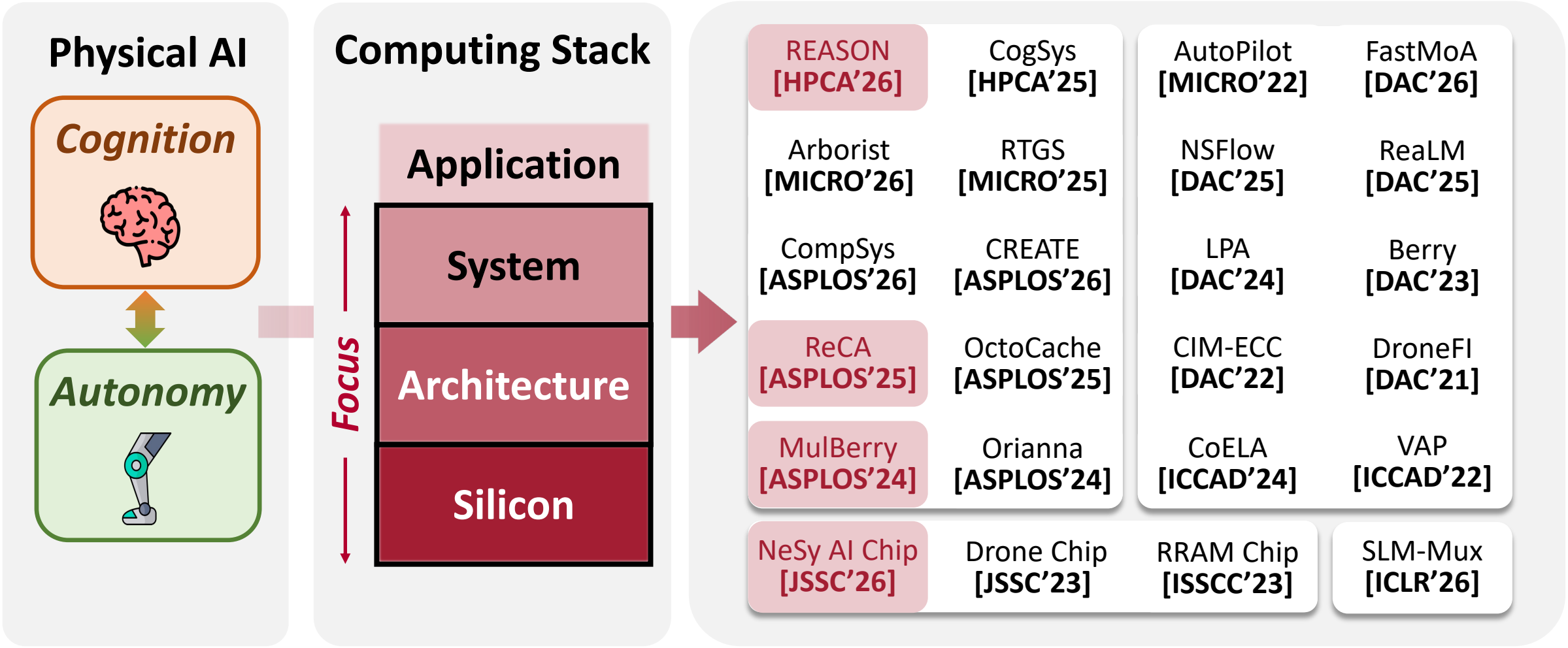
Statement & Vision

This research builds cross-stack system, architecture, and silicon foundations for physical intelligence —

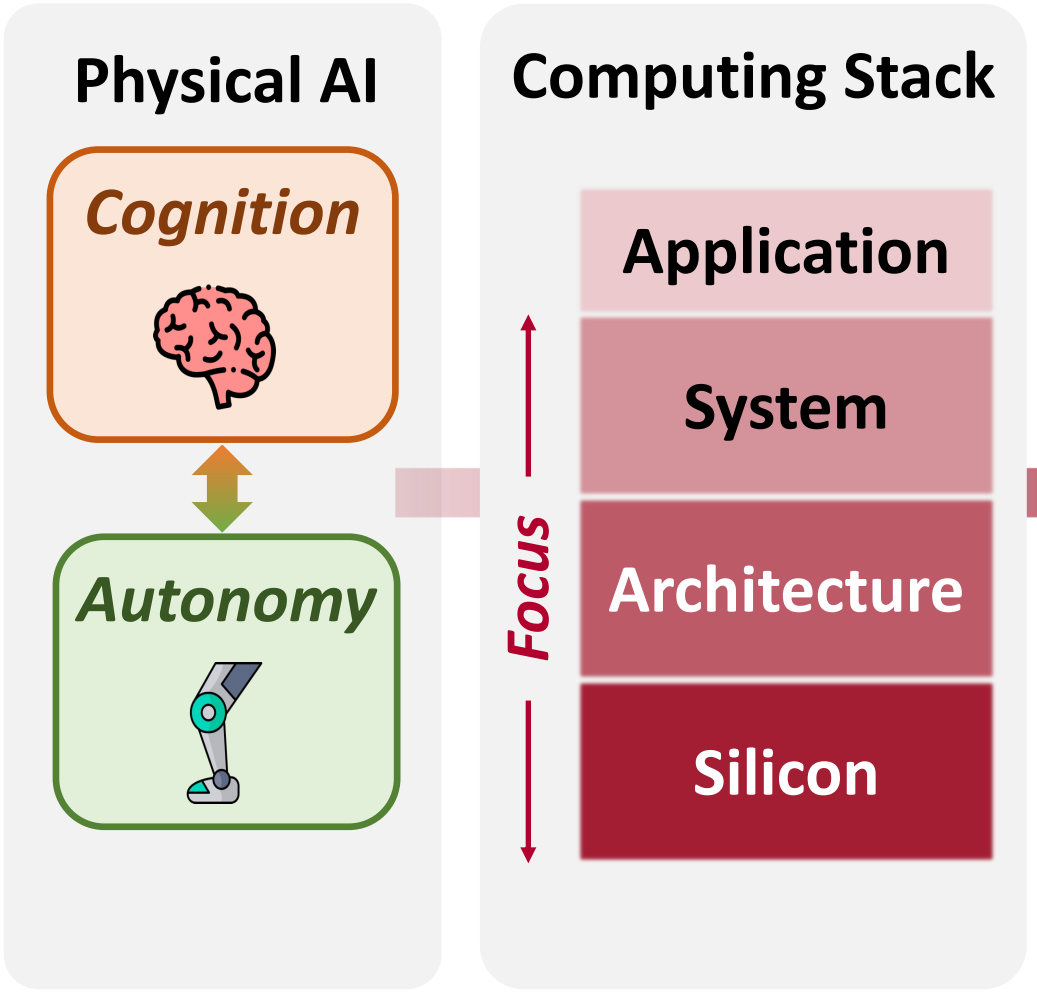
enabling embodied agents to reason, plan, and act efficiently, safely, and in real time

Architecting Computing System for Physical AI

Cross-Stack Co-Design from Systems to Silicon



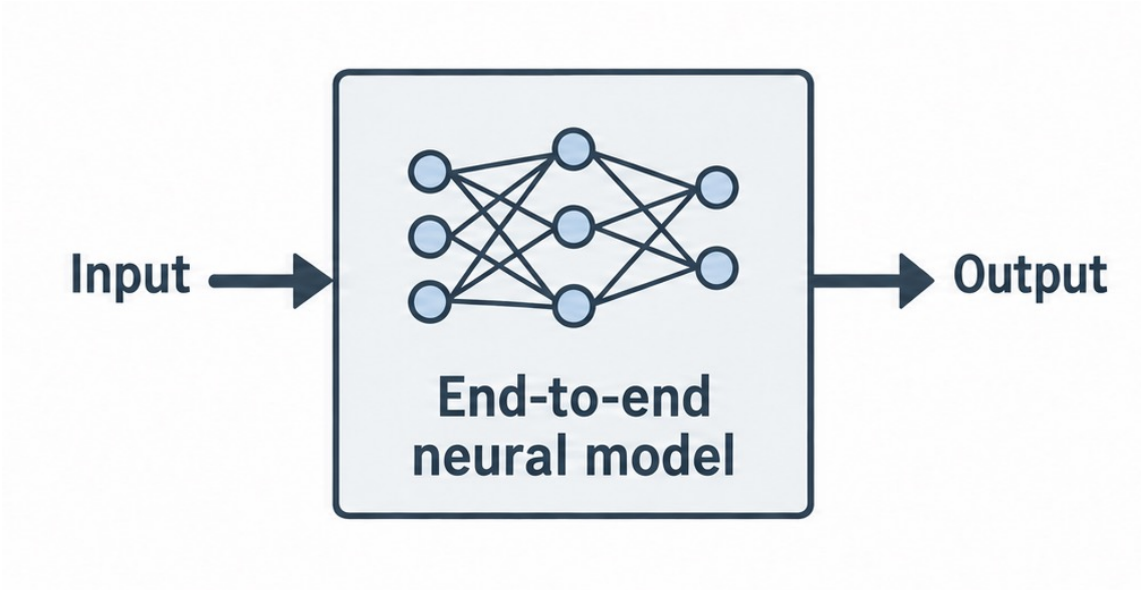
Outline



- Background & Motivation
- **Thrust 1: Hardware for *Cognition***
  - REASON: Flexible architecture for neuro-symbolic cognition
- Thrust 2: Hardware for *Autonomy*
- Future Research

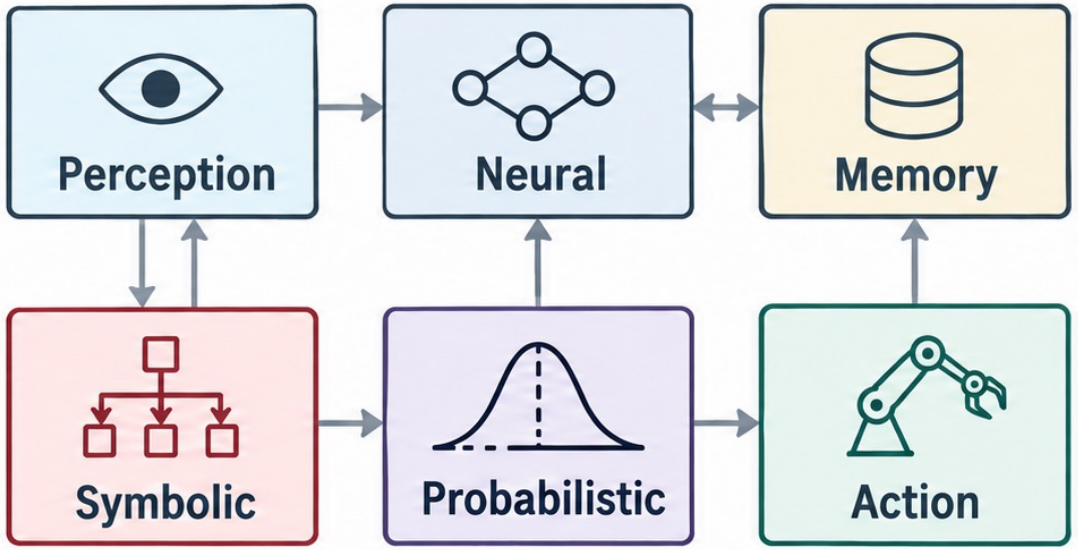
Cognitive AI Systems Are Becoming Compositional

Neuro-only



- Opaque end-to-end execution
- Limited verifiability and robustness
- No explicit physical grounding
- Poor generalization under shift

Compositional

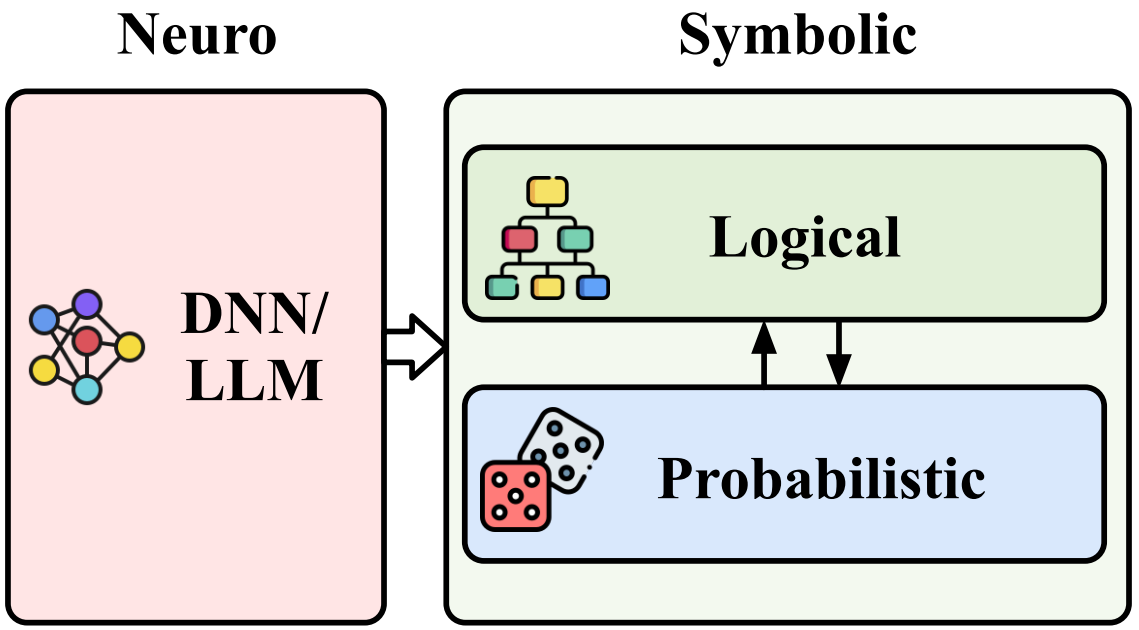


- Interpretable and modular structure
- Verifiable and trustworthy by design
- Explicit grounding in physical state
- Data efficient with stronger generalization

Compositional AI pairs learned models with structured reasoning

Neuro-Symbolic Computing as a Compositional System

Perception, representation, abstraction learning

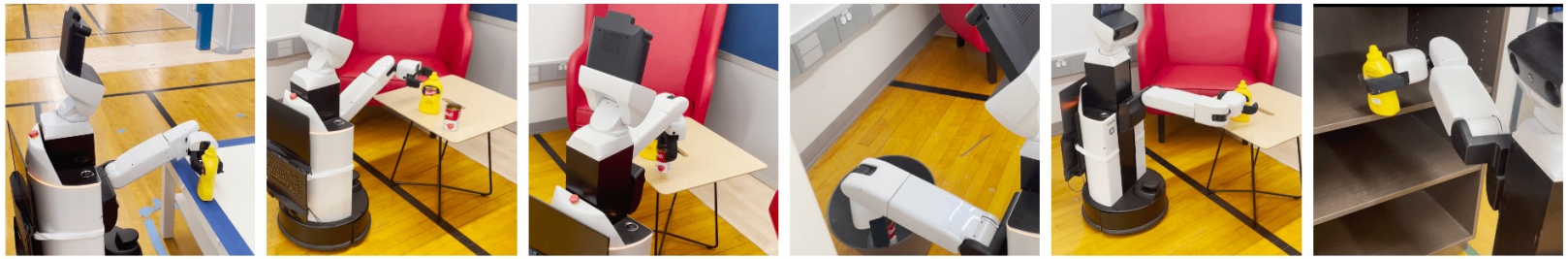


Explicit logic, rules, and constraints, interpretable and verifiable reasoning

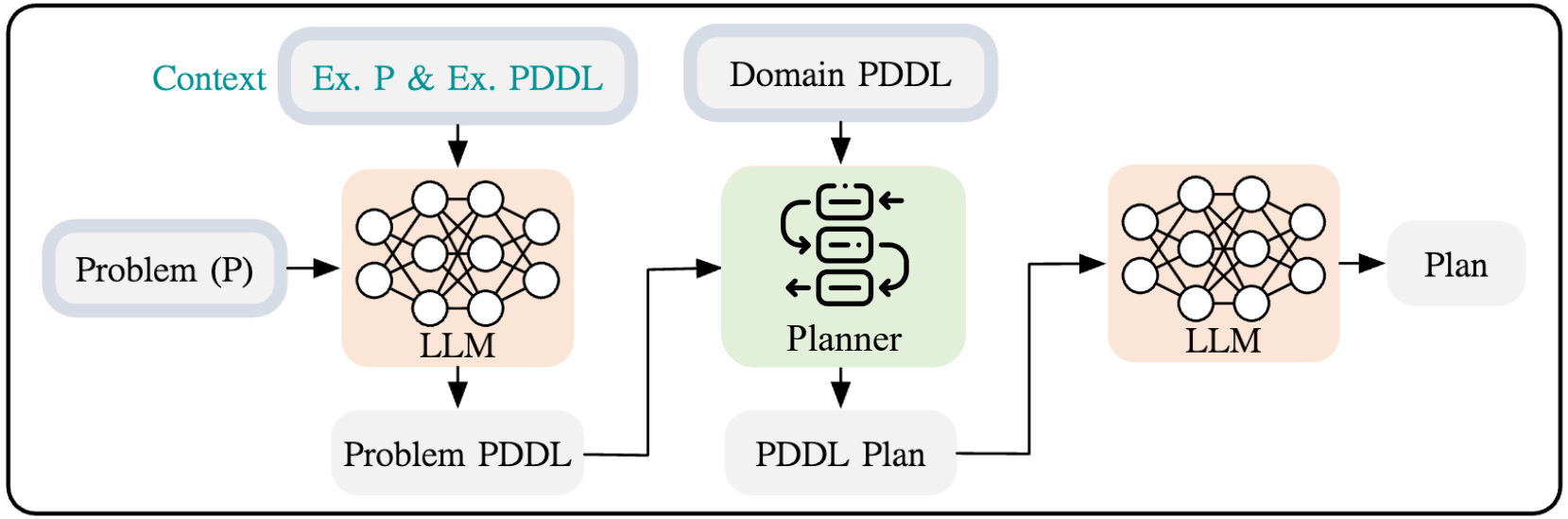
Model uncertainty, belief updates, and reason under ambiguity

Neuro-symbolic cognition integrates complementary reasoning mechanisms

Motivating Example: Neuro-Symbolic Planning



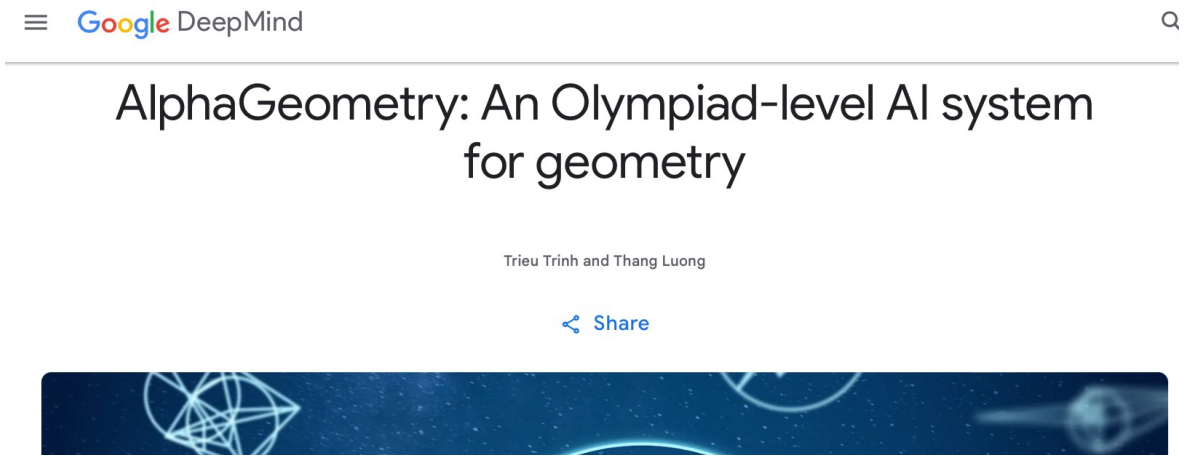
Grasp bottle | Free gripper | Grasp can | Place can | Re-grasp bottle | Place bottle

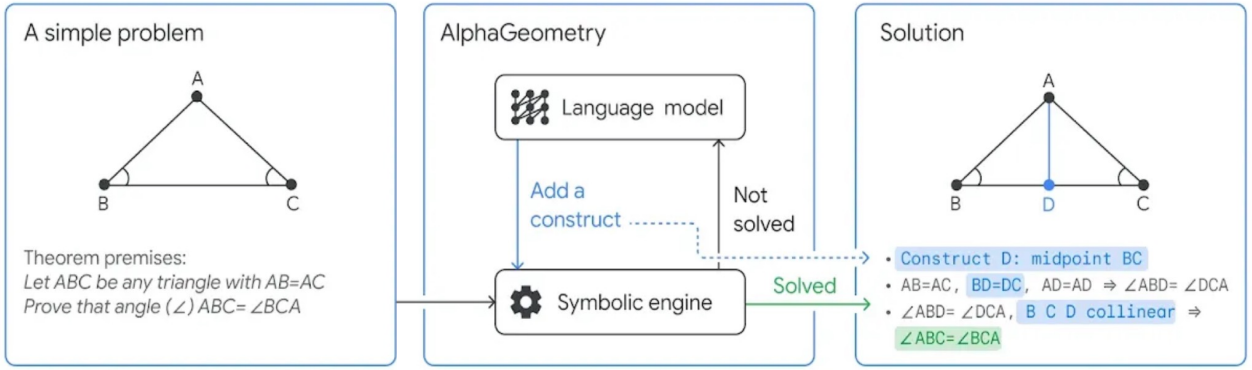


- **Task**: long-horizon multi-step planning in unseen environments
- **Modules**:
- Neuro: natural language interpretation, task decomposition, plan structure generation
- Symbolic: constraint-based planner to ensure feasibility & correctness

Trustworthy planning emerges from combining neural flexibility with symbolic guarantees

Motivating Example: Neuro-Symbolic Reasoning





- **Task**: Olympiad-level geometry solving
- **Modules**:
  - Neuro: generate auxiliary constructions
  - Symbolic: deductive reasoning & verification

Trinh et al, "Solving Olympiad Geometry without Human Demonstrations", Nature 2024

The signals investors must watch as AI moves from scale to reasoning

Feb 5, 2026 • Min Zhou

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2023, AI

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For regulated industries, AWS's neurosymbolic AI promises safe, explainable agent automation

Emilia David

August 6, 2025

How neuro-symbolic AI bridges the trust gap for physicians

Author(s) Karim Galil, MD

Listen

EY-Parthenon launches neurosymbolic A capabilities

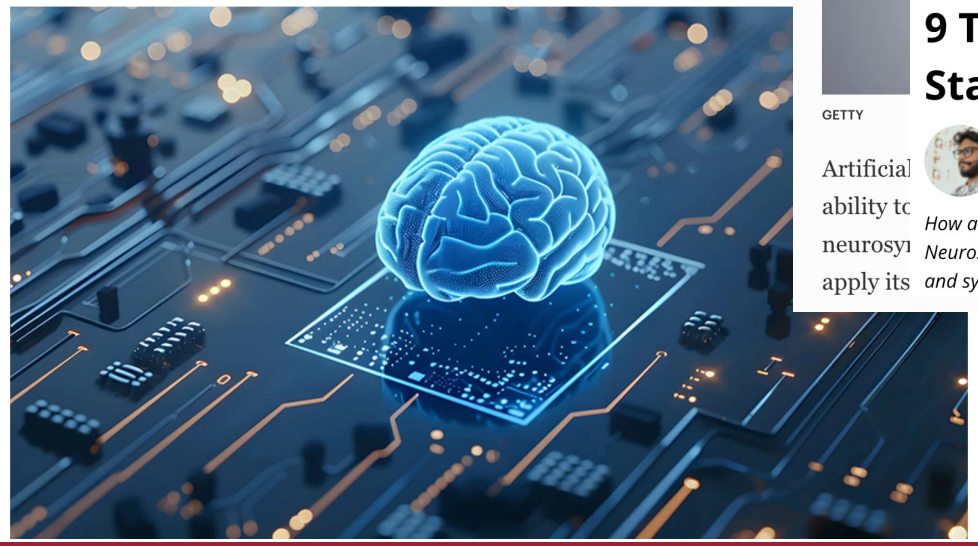
01 October 2025 | Consulting.us

Key Takea

- Neuro-s healthca
- This tec data anc

SHOW MORE

A more human intelligence.



Trinh et al, "Sol ration

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Intelligent Partnership in War: DARPA's Neuro-Symbolic AI Program

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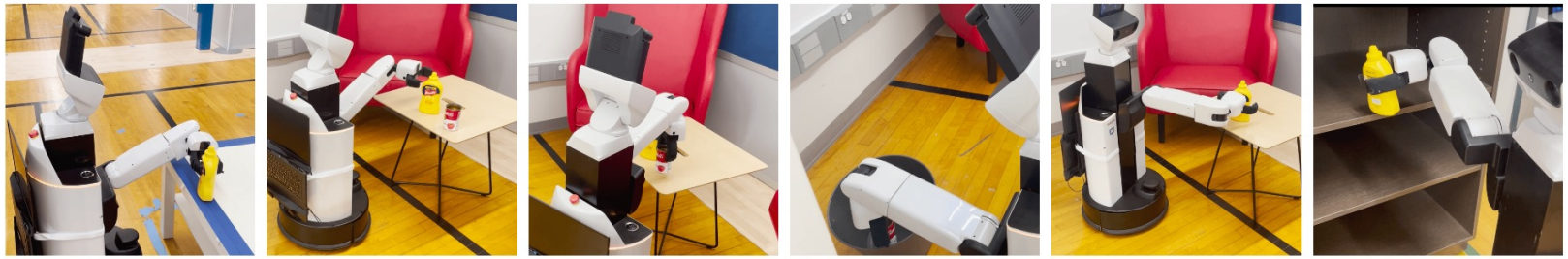
But Systems Hit a Wall



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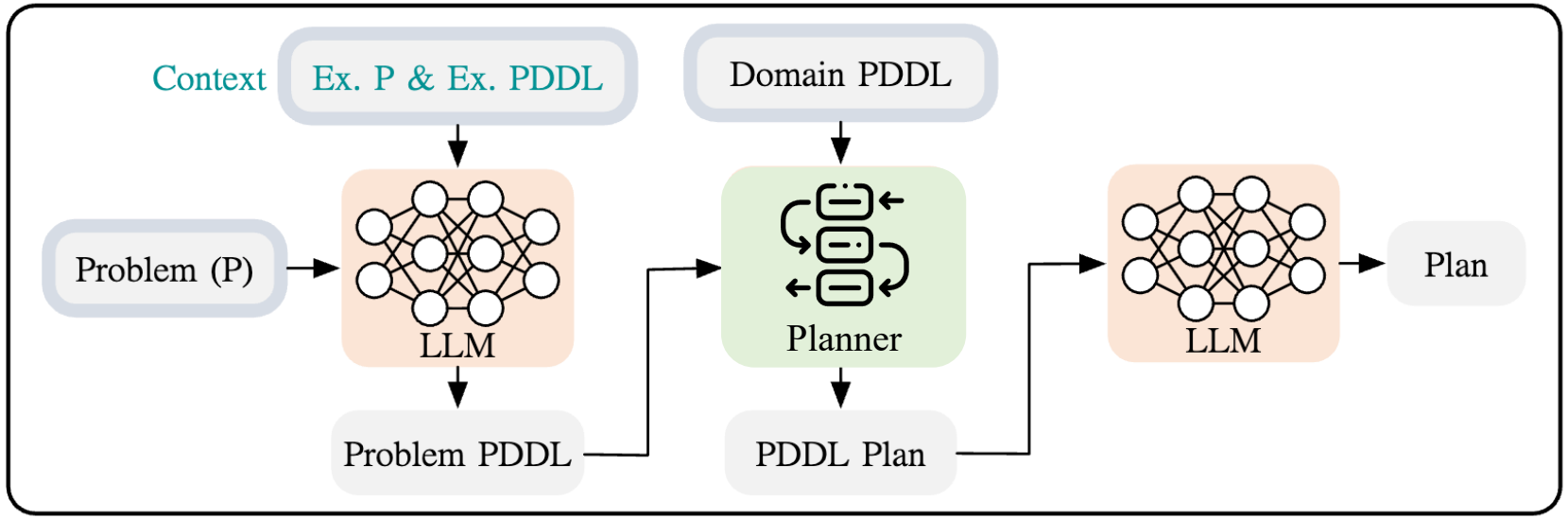
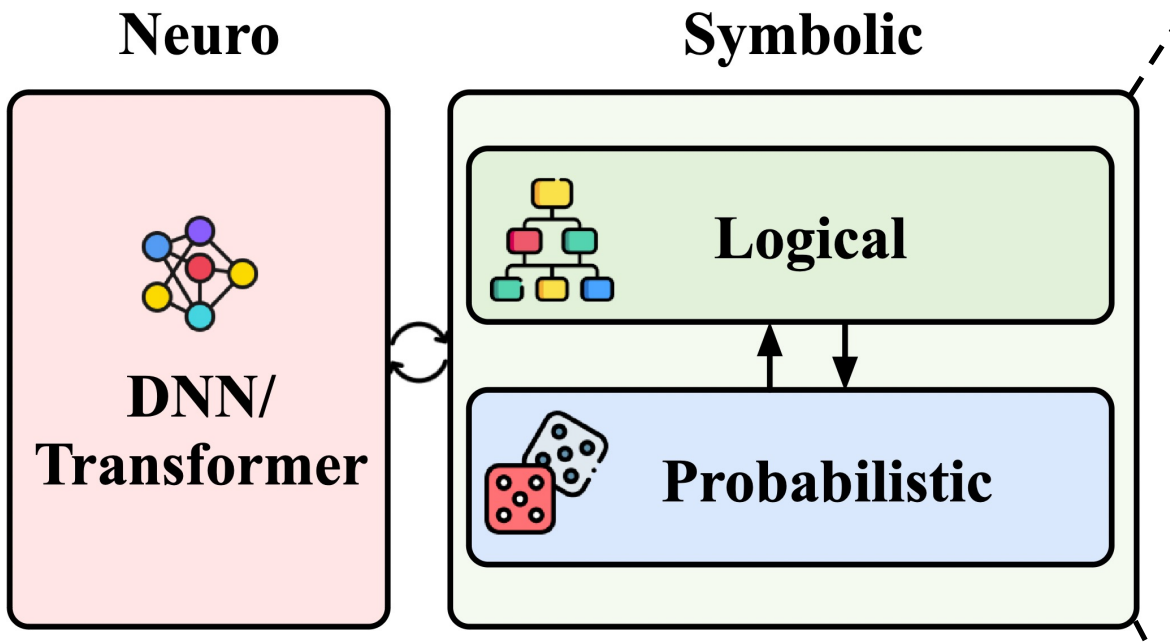


- **Task**: long-horizon multi-step planning in unseen environments
- **Modules**:
- Neuro: natural language interpretation, task decomposition, plan structure generation

Symbolic components incurs **>20 seconds** on CPU/GPU platforms

Symbolic reasoning is slow, latency explodes with task complexity

Symbolic = Logical + Probabilistic



Wan, Yang, Qian, Raj, Park, Wang, Raychowdhury, Krishna, *"Compositional AI Beyond LLMs: System Implications of Neuro-Symbolic Architectures"*, **ASPLOS 2026**
(Best Paper, SRC JUMP2.0 CoCoSys 2025)

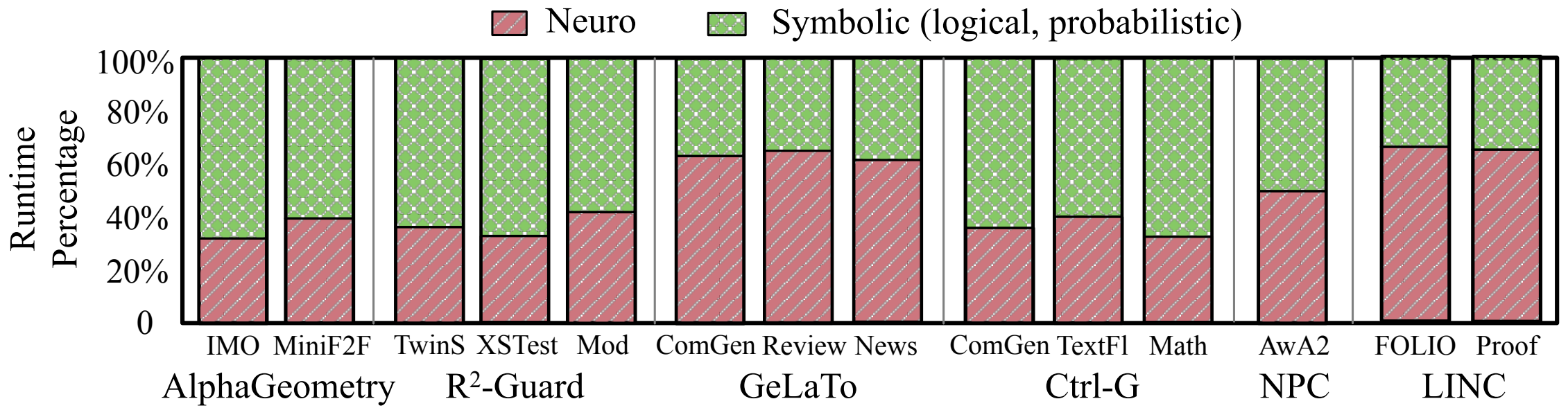
Support Symbolic Operations:

Logical:
- First-Order Logic (FOL)
- Boolean Satisfiability (SAT)
- Symbolic Search
- Vector-Symbolic Architecture (VSA)
- Symbolic Circular Convolution

Probabilistic:
- Probabilistic Circuit (PC)
- Hidden Markov Model (HMM)
- Probabilistic Graphical Model (PGM)

Core computational primitives: logical reasoning, probabilistic reasoning

Where Does End-to-End Latency Come From?



Neuro-Symbolic Models	Applications	Neural Operations	Symbolic operations
AlphaGeometry	Math Reasoning	LLM (8B)	First-order logic, SAT
R^2-Guard	Unsafety detection	LLM (3B)	First-order logic, Probabilistic Circuit
GeLaTo	Constrained generation	LLM (7B)	First-order logic, SAT, Markov Model
Ctrl-G	Interactive editing	LLM (3B)	Markov Model, Probabilistic Circuit
LINC	Logical reasoning	LLM (7B)	First-order logic, SAT

Latency bottlenecks arise from symbolic (logical + probabilistic) kernels

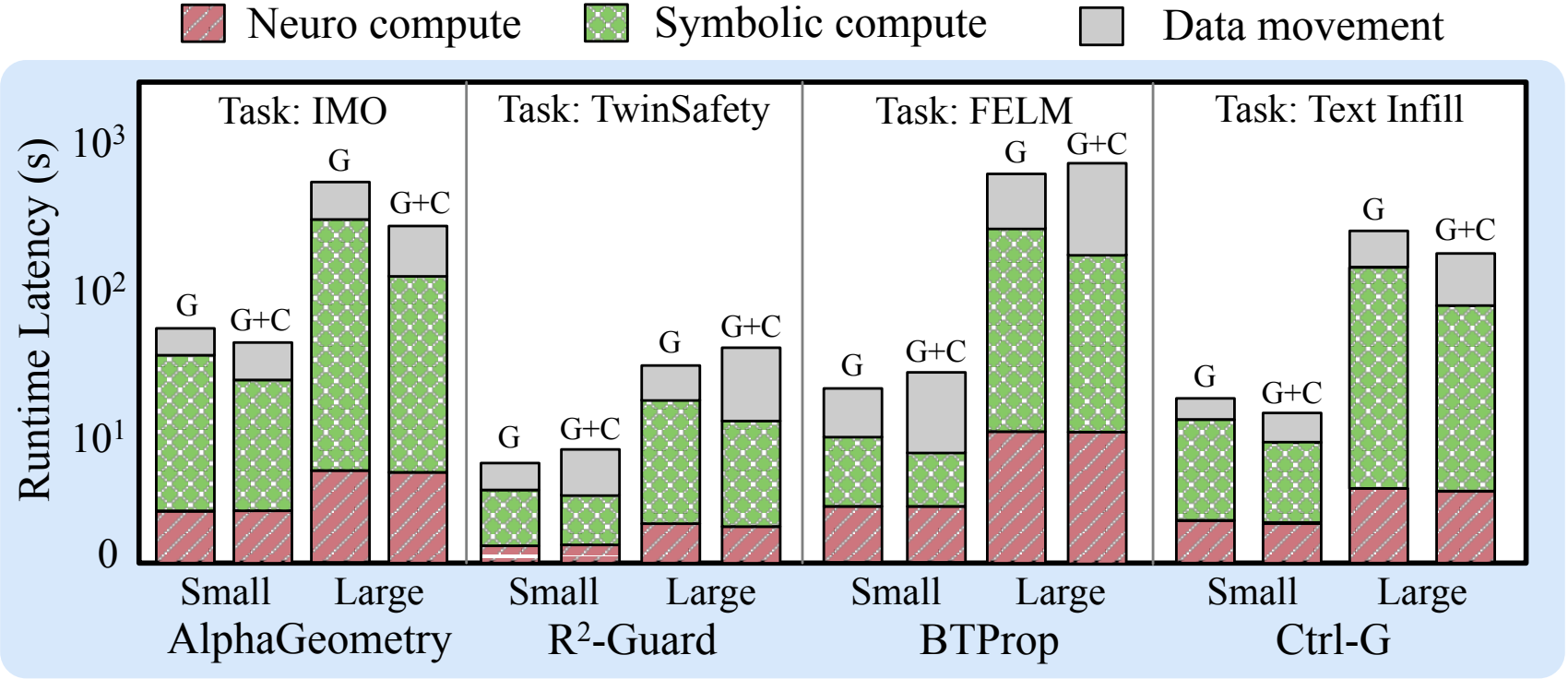
Why GPUs Inefficient for Symbolic Kernels?

	Neural Kernel		Logical Kernel		Probabilistic Kernel	
	MatMul	Softmax	Sparse MatVec	FOL	Marginal	Bayesian
Compute Efficiency						
Compute Throughput (%)	96.8	62.2	32.5	14.7	35.0	31.1
ALU Utilization (%)	98.4	72.0	43.9	29.3	48.5	52.8
Memory Behavior						
L1 Cache Throughput (%)	82.4	58.0	27.1	20.6	32.4	37.1
L2 Cache Throughput (%)	41.7	27.6	18.3	12.4	24.2	27.5
L1 Cache Hit Rate (%)	88.5	85.0	53.6	37.0	42.4	40.7
L2 Cache Hit Rate (%)	73.4	66.7	43.9	32.7	50.2	47.6
DRAM BW Utilization (%)	39.8	28.6	57.4	70.3	60.8	68.0
Control Divergence and Scheduling						
Warp Execution Efficiency (%)	96.3	94.1	48.8	54.0	59.3	50.6
Branch Efficiency (%)	98.0	98.7	60.0	58.1	63.4	66.9
Eligible Warps/Cycle (%)	7.2	7.0	2.4	2.1	2.8	2.5

Symbolic kernels suffer from low ALU utilization, low cache hit rate, high data movement, complex control flow, and low warp and branch efficiency

Even optimized GPU kernels suffer from low utilization due to irregular symbolic operations

How About CPU for Symbolic, GPU for Neuro?



(G: GPU | G+C: GPU+CPU)

- Symbolic remain slow on CPUs: low arithmetic intensity, control flow overhead, poor locality
- Tight neuro-symbolic coupling introduces CPU-GPU communication overhead

CPU+GPU suffer from inefficient symbolic execution and frequent neuro-symbolic interaction

Computation Patterns of Symbolic Reasoning

	First-Order Logic (FOL) / Boolean SAT	Probabilistic Circuit (PC)	Hidden Markov Model (HMM)	Neural Network
Core operators	Boolean logic, Constraint propagation	Sum-product, Log-sum-exp	Dynamic programming (sum/max)	MatMul, Conv, Attention
Structure	Irregular graphs	DAG (heterogeneous nodes)	Chain	Layered DAG
Control Flow	Highly dynamic, branch-heavy	Graph traversal	Sequential recurrence	Static
Parallelism	Limited, Search-dependent	Structure-limited	Time-step limited	Massive data parallelism
Memory Access	Pointer-heavy, random	Irregular	State-table streaming	Dense, contiguous

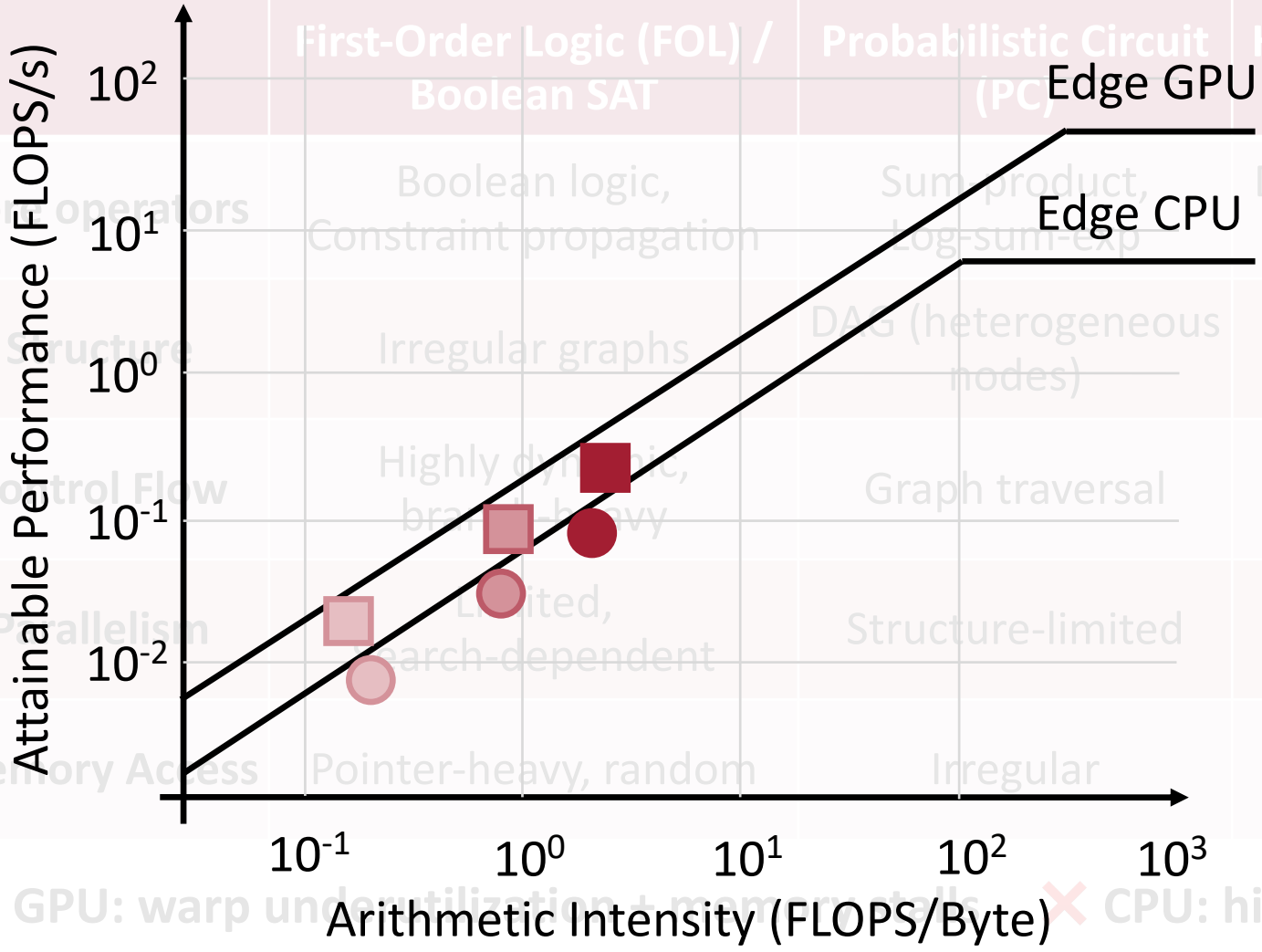
❌ **GPU: warp underutilization + memory stalls**

❌ **CPU: high control overhead + low throughput**

❌ **ML Accelerator: structural mismatch**

❌ **Graph Accelerator: semantic mismatch**

Roofline Analysis



SAT @ Edge CPU
PC @ Edge CPU
HMM @ Edge CPU

SAT @ Edge GPU
PC @ Edge GPU
HMM @ Edge GPU

Design Principles for Symbolic Reasoning Architecture

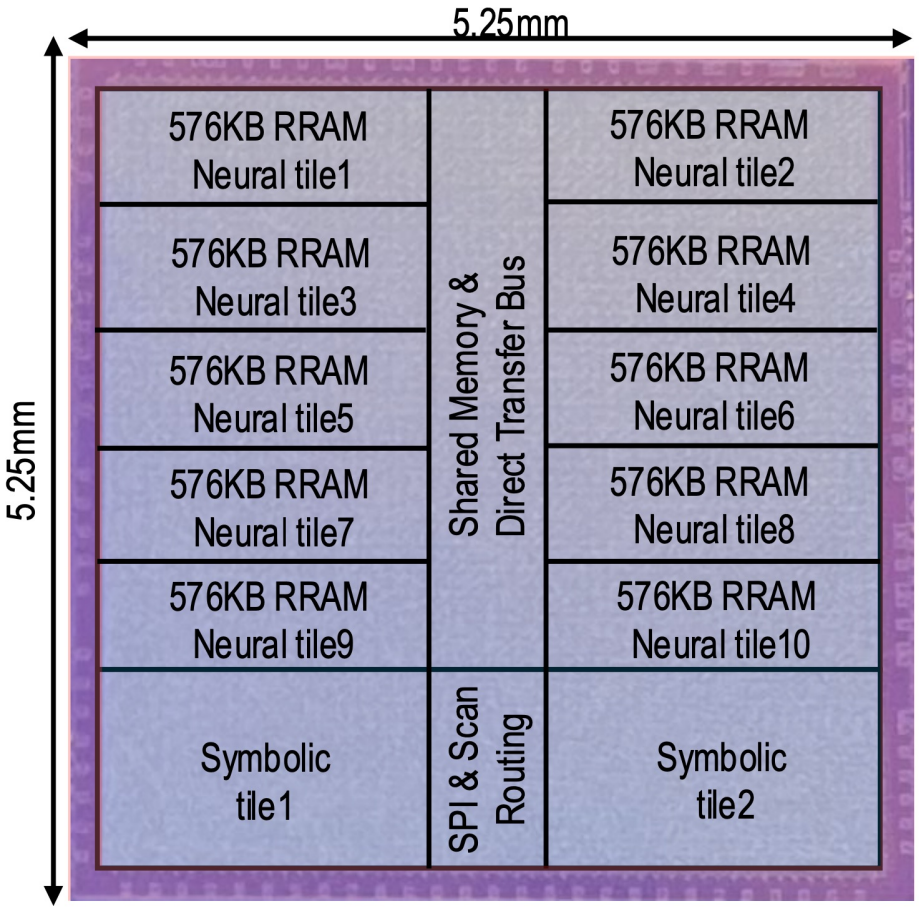
	First-Order Logic (FOL) /	Probabilistic Circuit	Hidden Markov Model	Neural Network
Core op				Mul, Conv, ttention
Struc				ered DAG
Contro				Static
Parall				ssive data rallelism
Memory				contiguous

Design Principles for Symbolic Reasoning Architecture

- Semantics-Aware Primitives
- Control-Data Decoupling
- Graph-Aware but Not Graph-Bound Execution
- Fine-Grained Dynamic Parallelism
- Composable with Neural Units

✗ GPU: warp underutilization + memory stalls
✗ CPU: high control overhead + low throughput
✗ ML Accelerator: structural mismatch
✗ Graph Accelerator: semantic mismatch

REASON: SoC Architecture & Chip for NeSy Computing



Key Architecture & Chip Features:

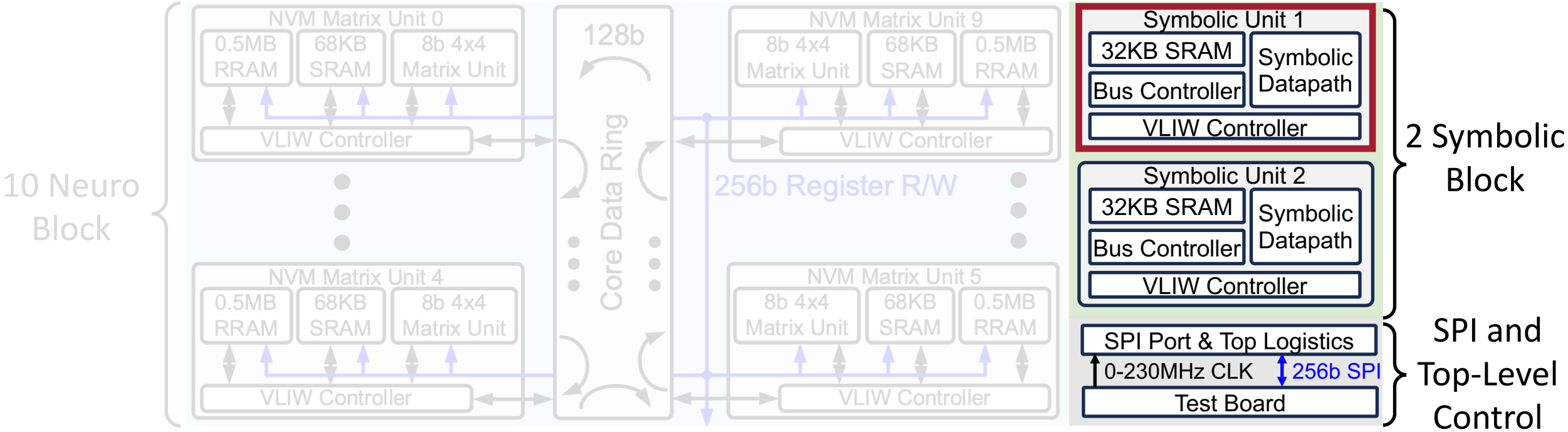
- **Unified Intermediate Representation**
- **Flexible Architecture and Dataflow**
- **Heterogenous Memory System**
- **Compiler-Hardware Co-Design**
- **Software-Defined Power Management**

Wan, Liu, Qian, Yang, Raychowdhury, Krishna, "*REASON: Accelerating Probabilistic Logical Reasoning for Neuro-Symbolic Intelligence*", **HPCA 2026**
(First Place, DAC PhD Forum 2025)

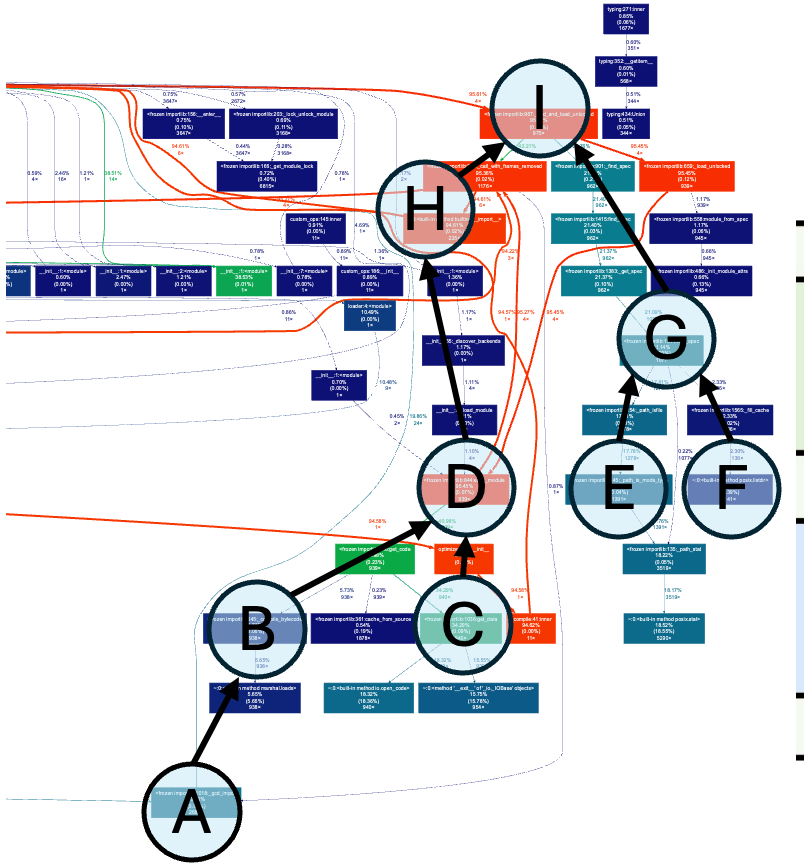
Liu*, **Wan***, Noh, Ibrahim, Spetalnick, Krishna, Khwa, Lele, Chih, Chang, Raychowdhury, "*A 40nm Programmable Heterogeneous SoC with RRAM/SRAM for Accelerating Neuro-Symbolic AI Models*", **JSSC 2026**

Top-Level SoC Architecture

- **Symbolic block:** 2 reconfigurable Reasoning Units
  - Supports neuro, vector, logical, probabilistic operations
- **Neuro block:** 10 Nonvolatile Matrix Units

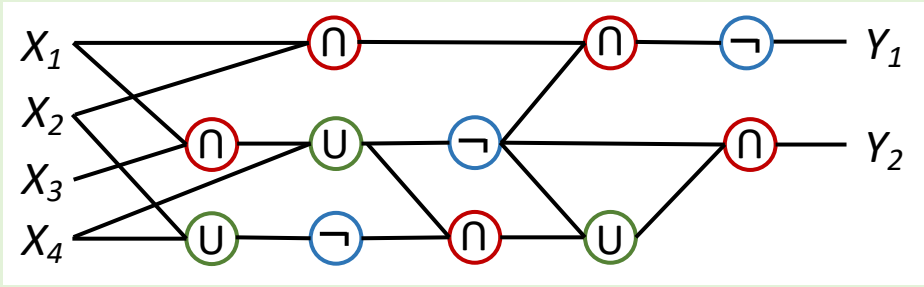


Unified Intermediate Representation (IR)

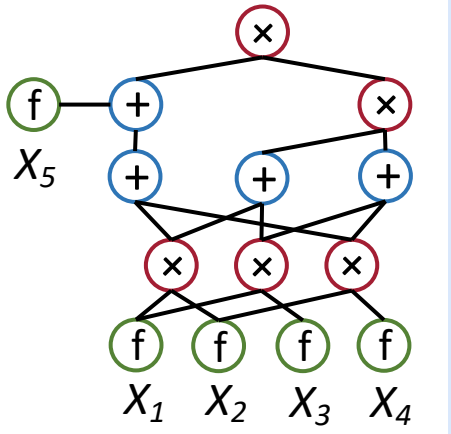


Directed Acyclic Graph (DAG)

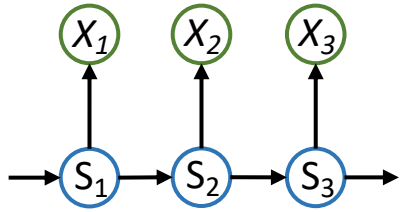
First-Order Logic (FOL)
Boolean Satisfiability (SAT)



Probabilistic Circuit (PC)

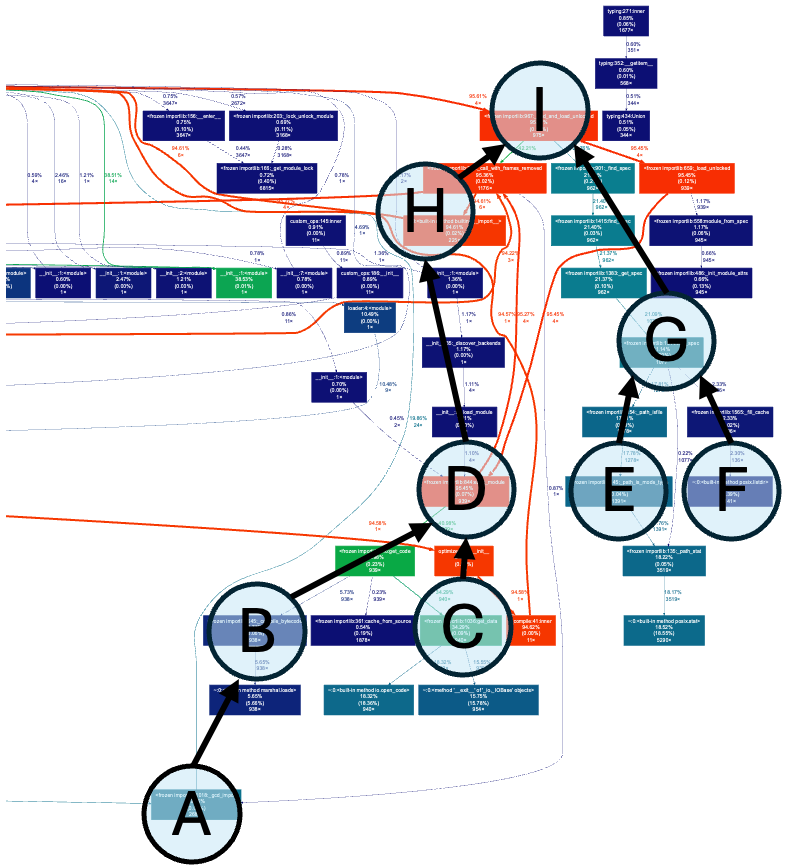


Hidden Markov Model (HMM)



Unified IR turns symbolic reasoning from an algorithmic problem into an architectural one

Unified Intermediate Representation (IR)



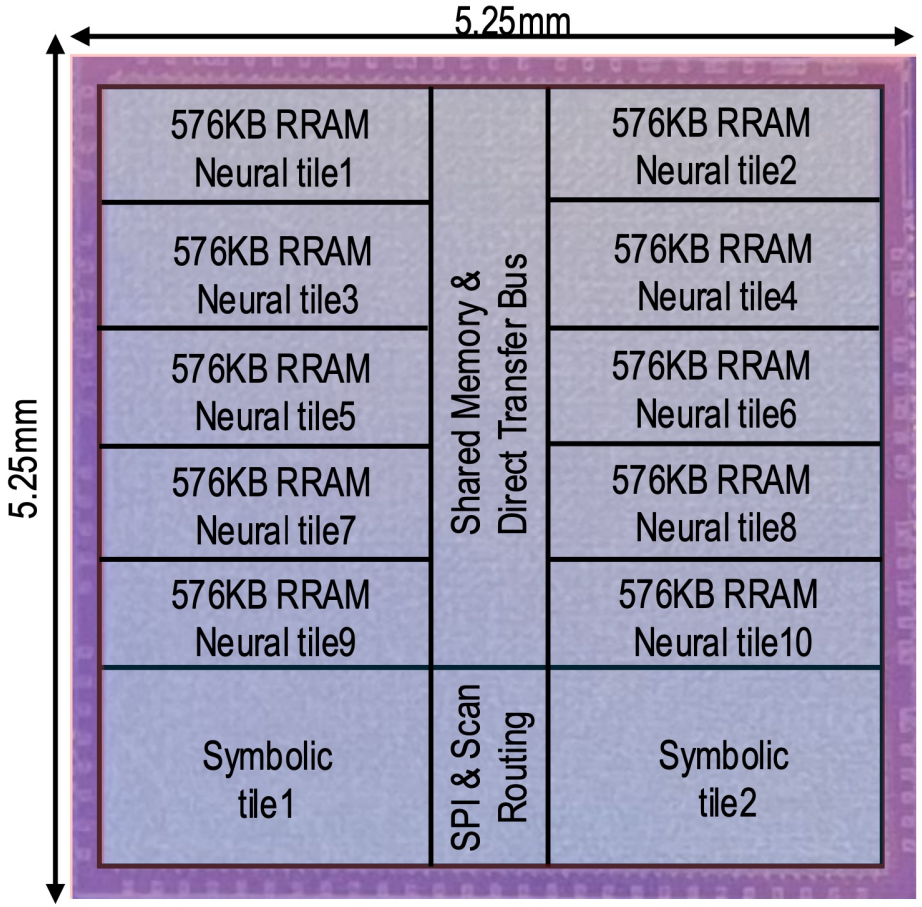
Directed Acyclic Graph (DAG)

Model	DAG Nodes	DAG Edges	DAG Execution
SAT/FOL	Literals and logical ops	Logic dependency between literals, clauses, formulas	Control-driven search and traversal
PC	Primitive distributions, sum, product	Weighted dependency for factorization	Dataflow-based sum-product propagation
HMM	Hidden state variables at each step	State transition and emission dependencies	Sequential message passing

Lowered into DAG-based IR

A unified DAG-based IR for symbolic (logical + probabilistic) reasoning

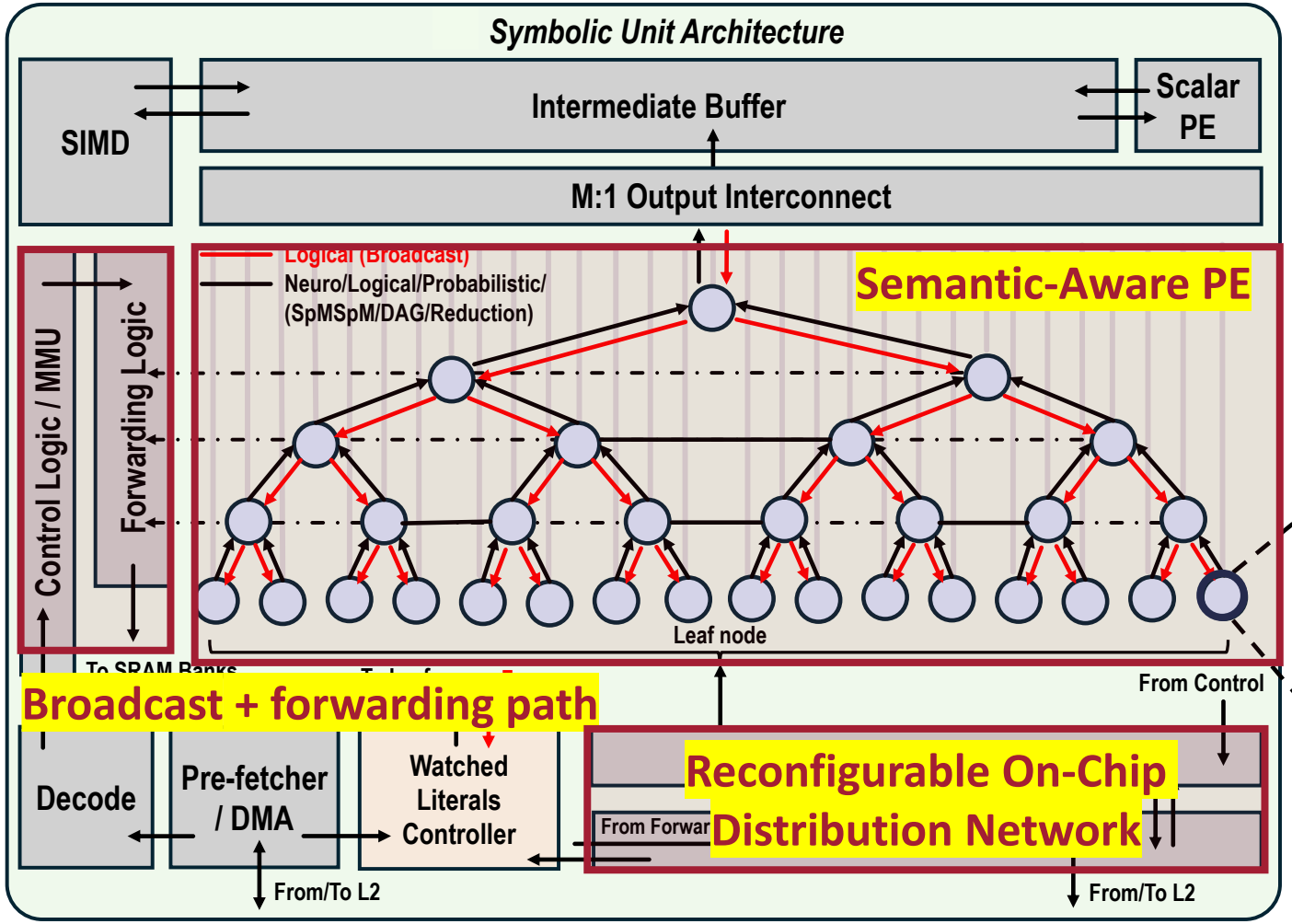
REASON: SoC Architecture & Chip for NeSy Computing



Key Architecture & Chip Features:

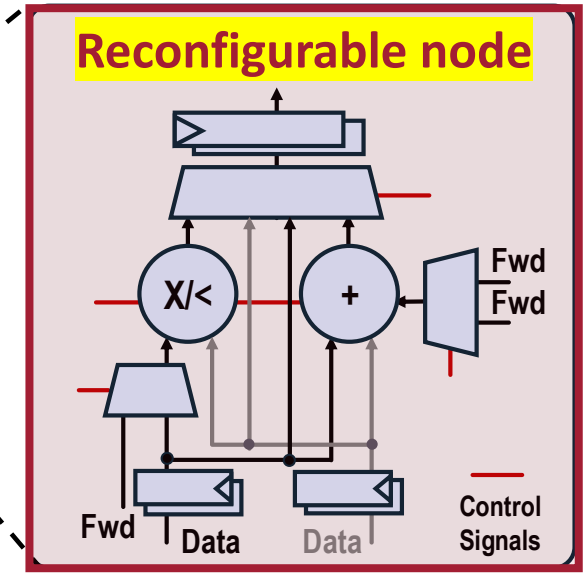
- **Unified Intermediate Representation**
- **Flexible Architecture and Dataflow**
- **Heterogenous Memory System**
- **Compiler-Hardware Co-Design**
- **Software-Defined Power Management**

Flexible Processing Element (PE) Architecture



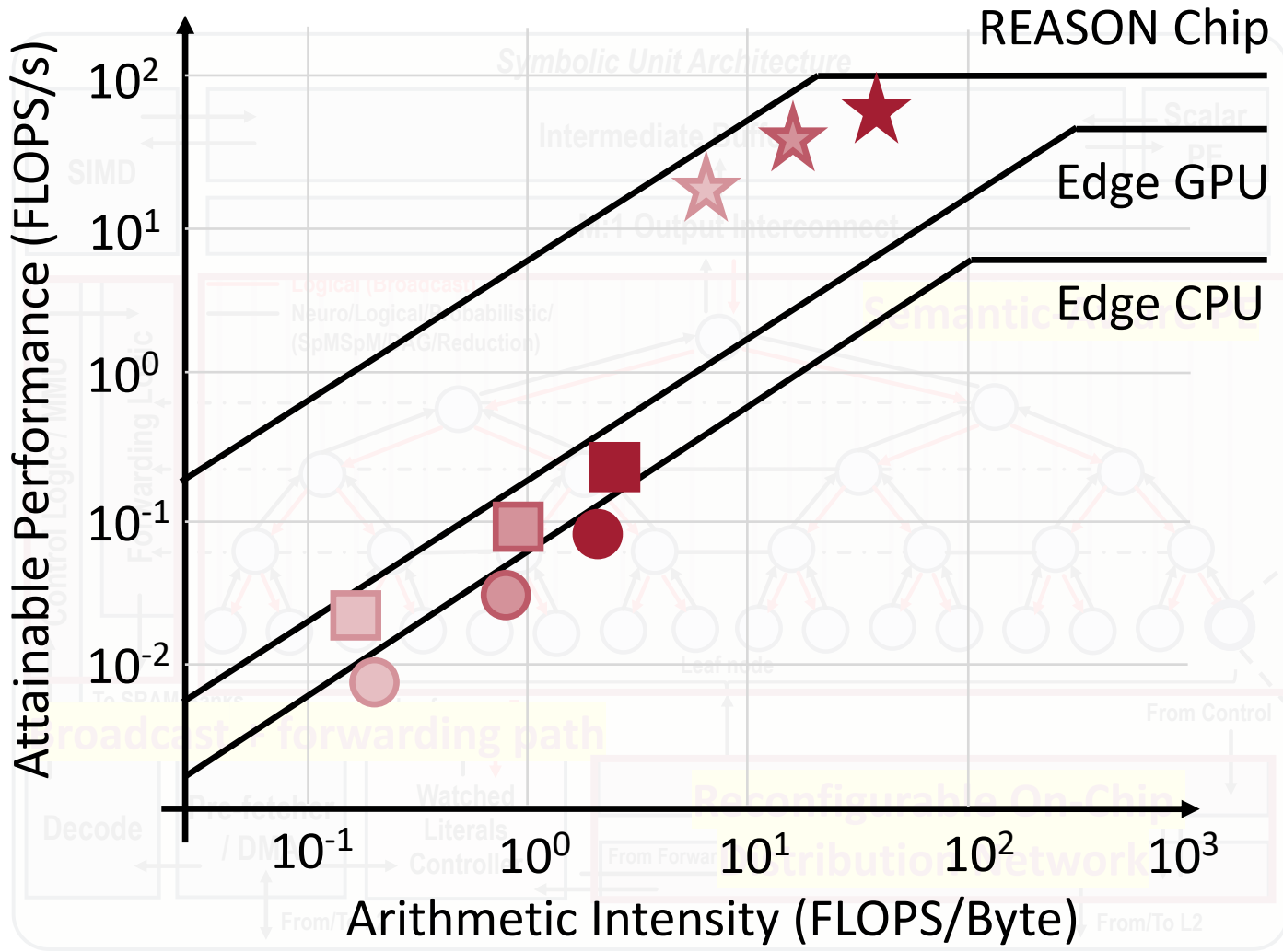
A symbolic unit is flexibly to support:

- Logical reasoning
- Probabilistic inference
- Sparse-Sparse MatMul



A flexible PE architecture enables fine-grained, semantics-aware symbolic reasoning

Flexible Processing Element (PE) Architecture



SAT @ Edge CPU
PC @ Edge CPU
HMM @ Edge CPU

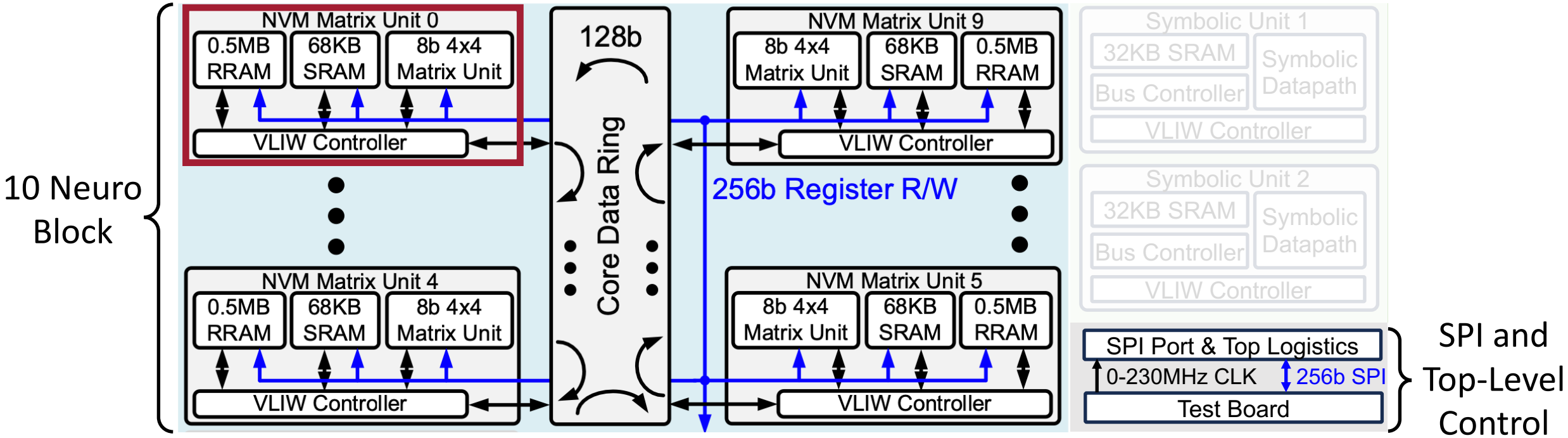
SAT @ Edge GPU
PC @ Edge GPU
HMM @ Edge GPU

SAT @ REASON
PC @ REASON
HMM @ REASON

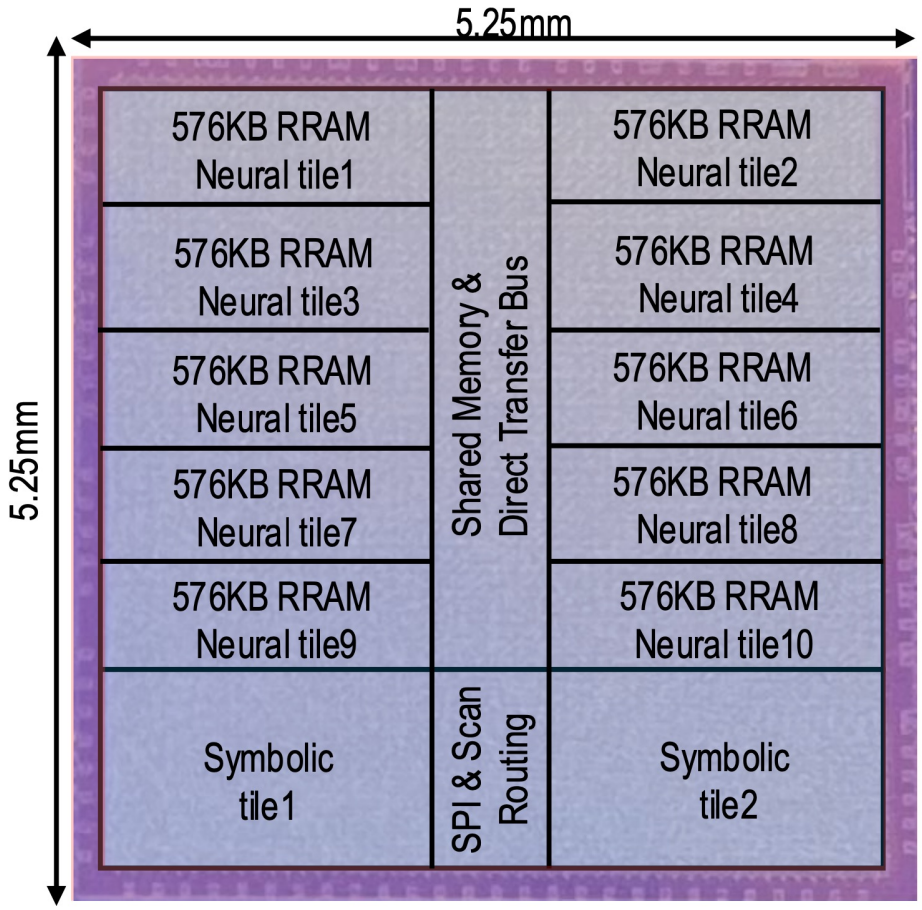
A flexible PE architecture enables fine-grained, semantics-aware symbolic reasoning

Top-Level SoC Architecture

- **Symbolic block:** 2 reconfigurable Reasoning Units
  - Supports neuro, vector, logical, probabilistic operations
- **Neuro block:** 10 Nonvolatile Matrix Units



REASON: SoC Architecture & Chip for NeSy Computing

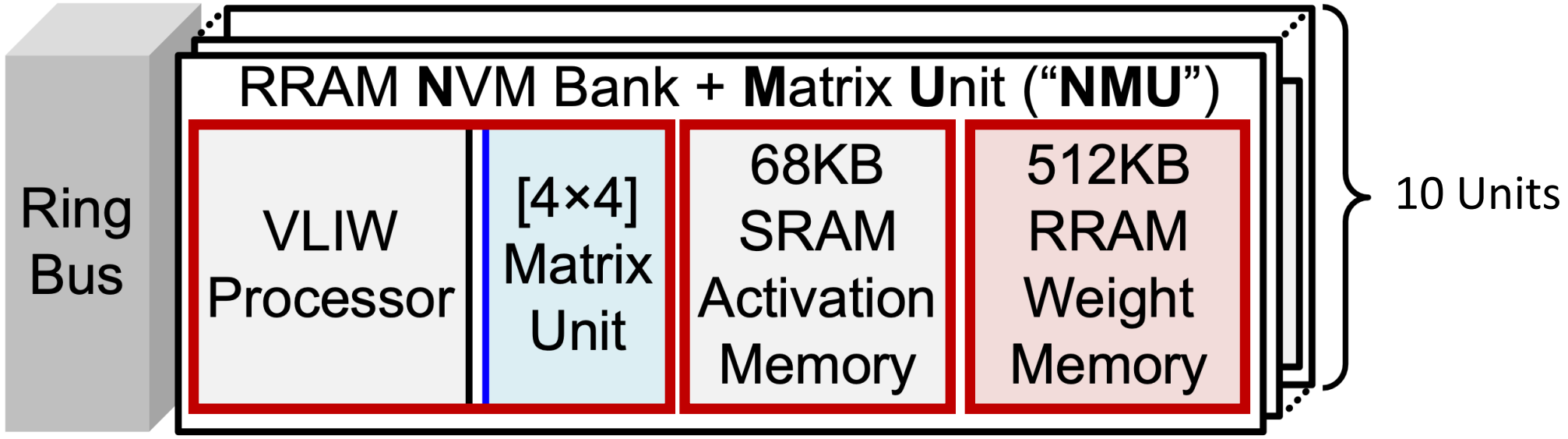


Key Architecture & Chip Features:

- **Unified Intermediate Representation**
- **Flexible Architecture and Dataflow**
- **Heterogenous Memory System**
- **Compiler-Hardware Co-Design**
- **Software-Defined Power Management**

Heterogenous RRAM/SRAM Memory Systems

- **RRAM** for on-chip weight storage provides high density and low power
- **SRAM** for frequent read/write operations
- **Digital MACs and RISC-V VLIW** for efficient computation and control

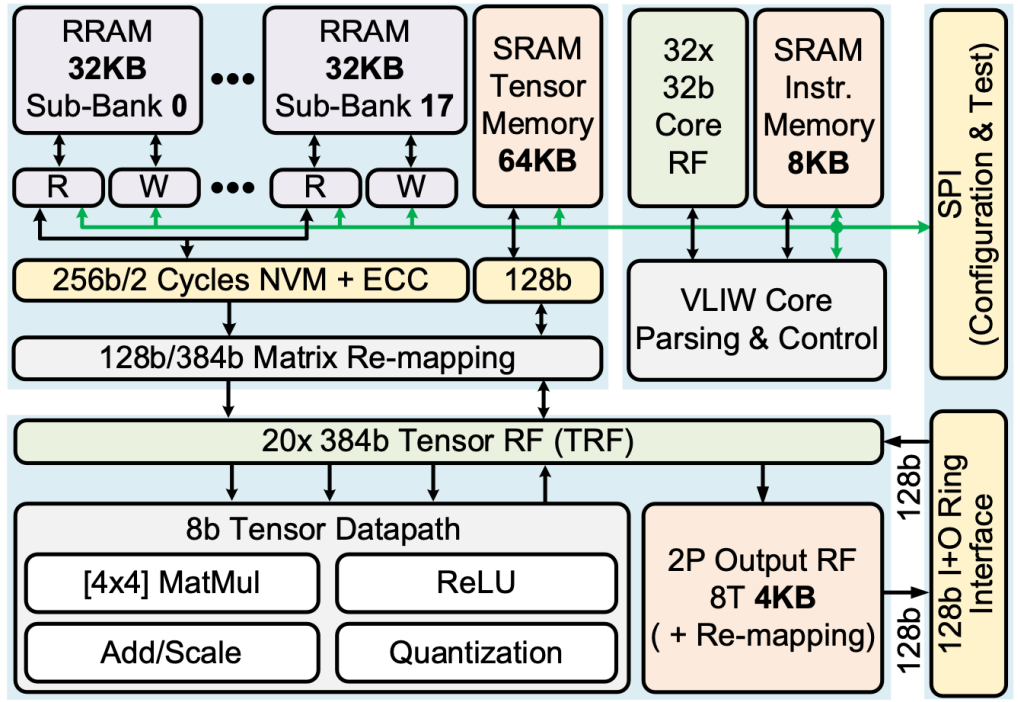


Neuro Unit Summary

Principled heterogeneous memory design for efficient and scalable computation

Neuro Architecture: Nonvolatile Matrix Unit

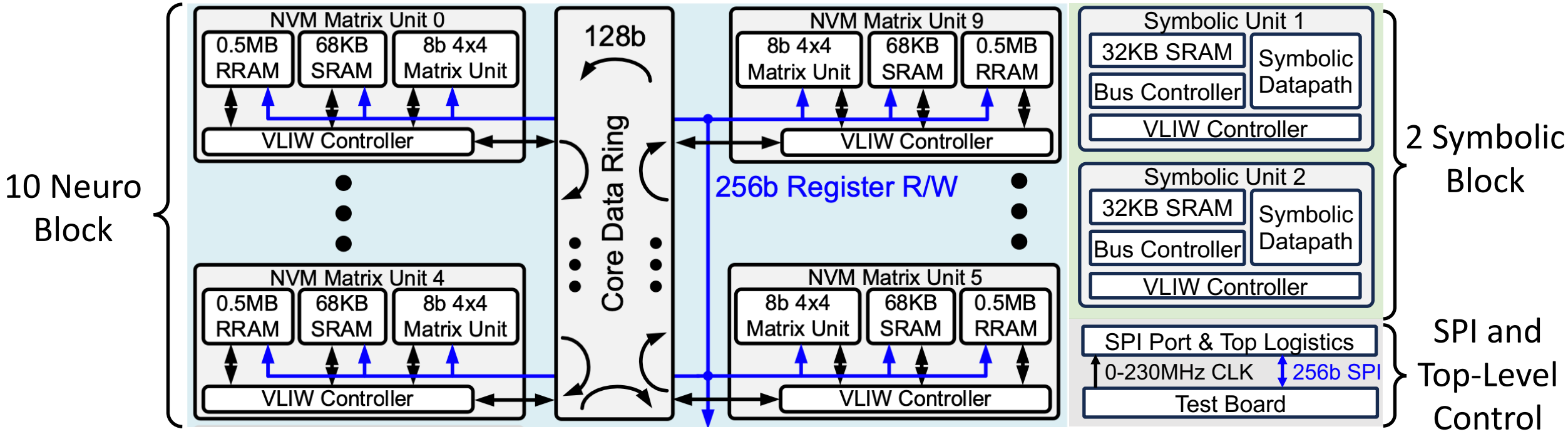
- [4x4] INT8 matrix compute datapath
- RISC-V-like VLIW control core with 32-bit register file (RF)
  - 20-entry tensor RF for [4x4] tiles with 24-bit accumulation



A fully digital, VLIW-controlled matrix unit with dense nonvolatile weight storage

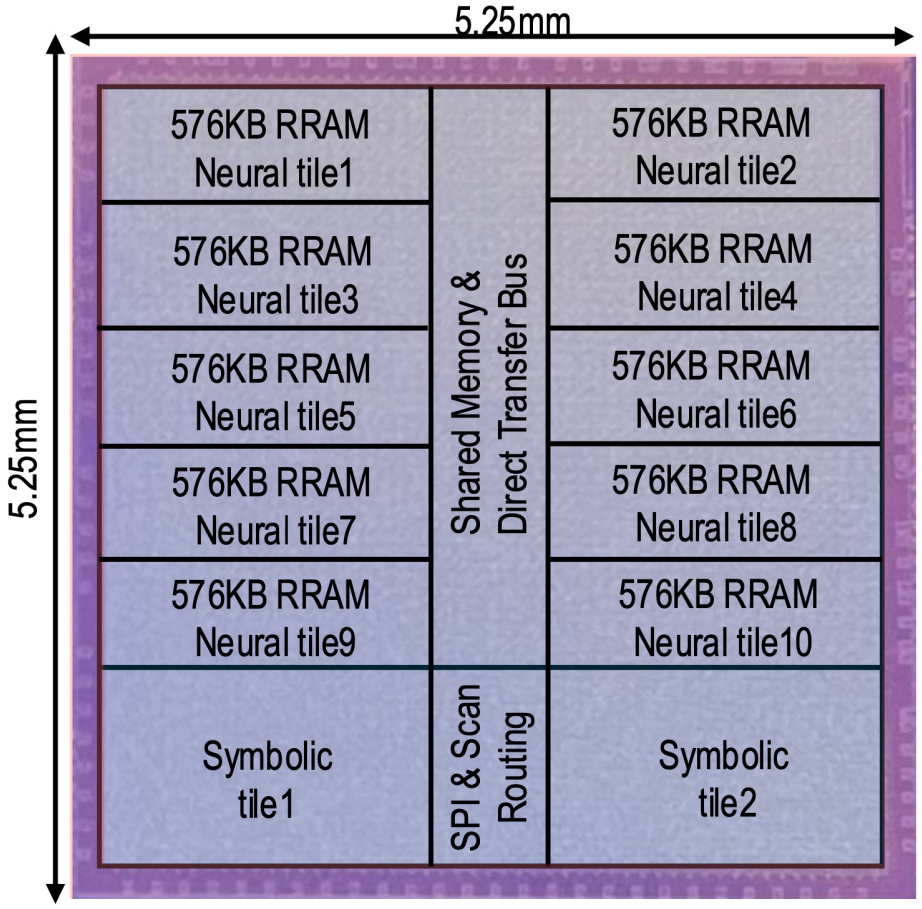
Top-Level SoC Architecture

- **Symbolic block:** 2 reconfigurable Reasoning Units
  - Supports neuro, vector, logical, probabilistic operations
- **Neuro block:** 10 Nonvolatile Matrix Units



How can we architect SoC to enable programmability and adaptive power efficiency?

REASON: SoC Architecture & Chip for NeSy Computing



Key Architecture & Chip Features:

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- **Compiler-Hardware Co-Design**
- **Software-Defined Power Management**

Instructions and Programmability

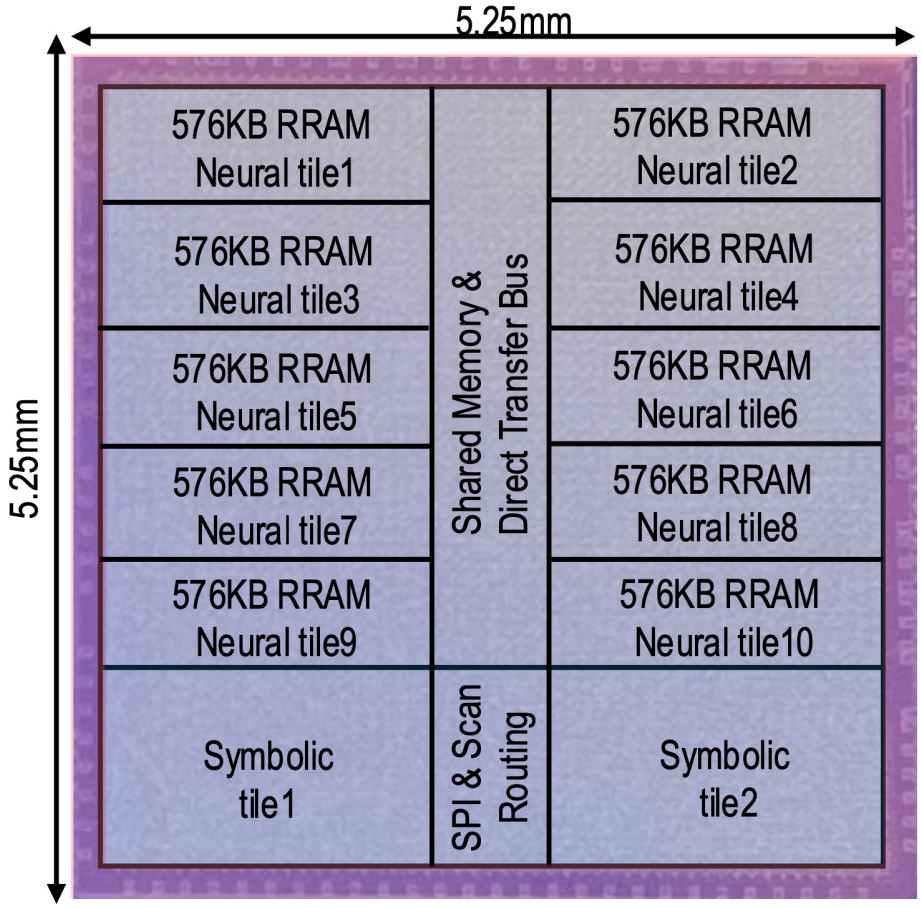
- 128-bit VLIW instruction coordinates memory, compute, data movement

	SRAM LD/ST	RRAM LD	Control	Compute Op	Ring R+W
Opcode	3b	1b	3b	4b	1b+1b
&RF0	5b	5b	5b	/	5b+5b
&RF1	/	/	5b	/	/
&TRF0	5b	5b	/	5b	5b+5b
&TRF1	/	5b	/	5b	/
&TRF2	/	/	/	5b	/
&TRF3	/	/	/	5b	/
Imm.	8b	8b	8b	/	8b+8b
Total	**21b**	**24b**	**21b**	**24b**	**19b+19b**

(RF: register file | TRF: tensor register file | VLIW: Very Long Instruction Word)

Fine-grained VLIW control enables tightly coupled memory, compute, and communication

REASON: SoC Architecture & Chip for NeSy Computing



Key Architecture & Chip Features:

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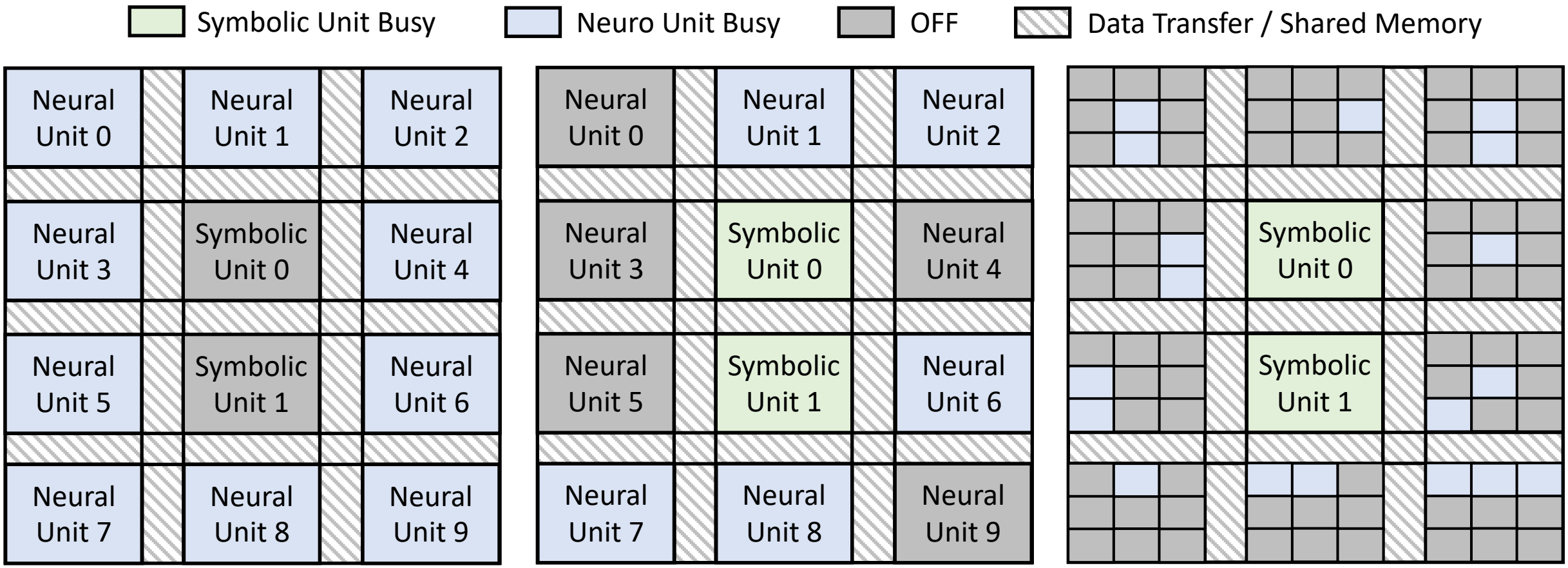
Software-Defined Power Management

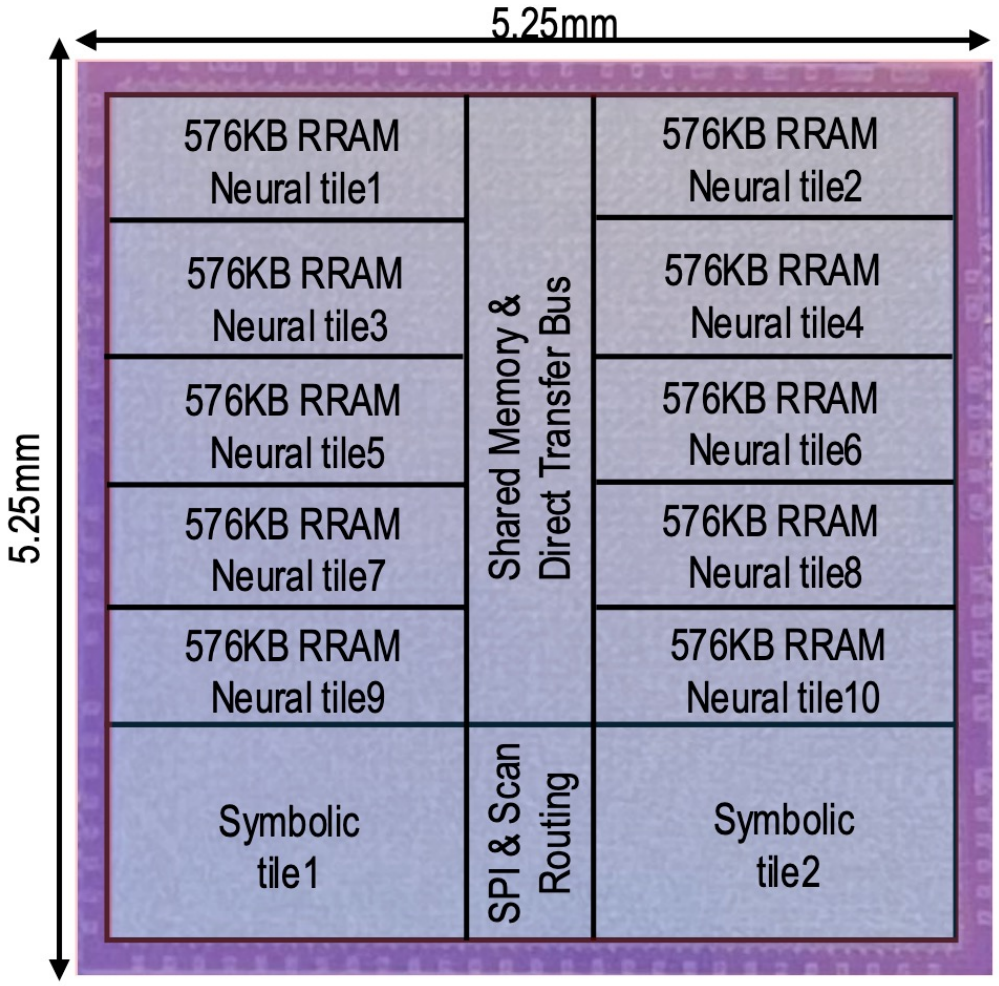


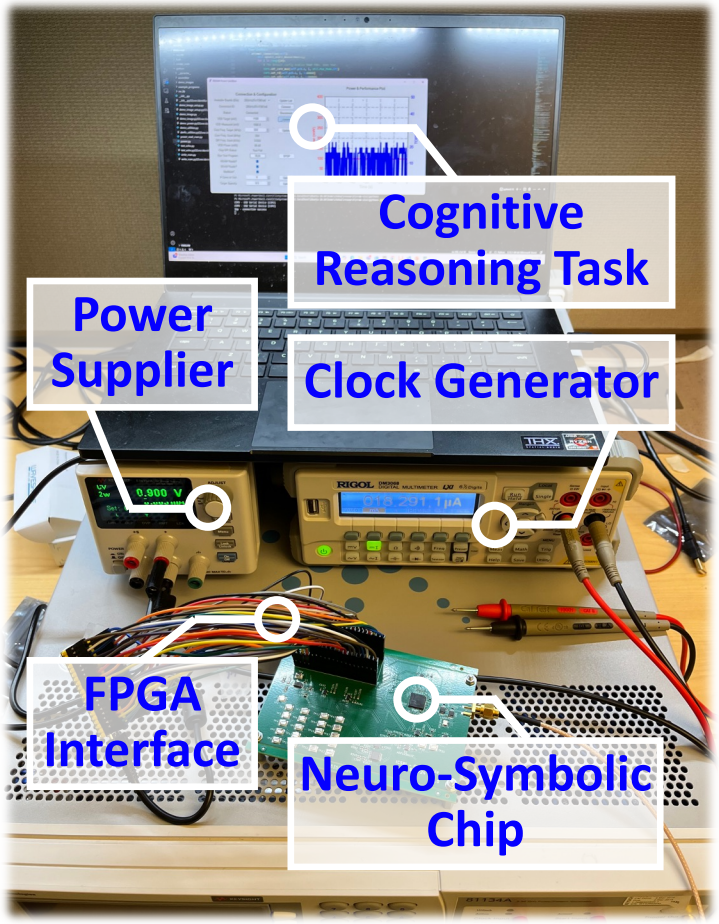
Symbolic core power down | Neuro core power down | Neuro core power down

Three software-defined power management configurations controlled by VLIW controller

Software-defined power management adapts energy use to compositional execution

Neuro-Symbolic SoC: Silicon Prototype

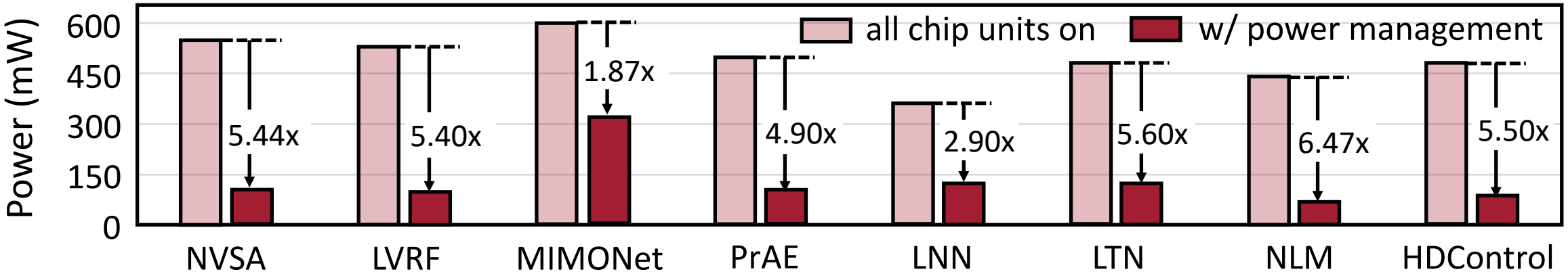




Technology	40 nm CMOS
Die Area	27.6 mm^2
Voltage	0.73 - 1.1 V VDD 3.3 V IO 1.5 – 4.0 V Write
Frequency	80 – 210 MHz
Memory	5.63 MB RRAM 0.85 MB SRAM
Interface	256b SPI
Sleep Mode	110µW@ 500mV

The first neuro-symbolic AI system-on-chip (SoC): from design to prototyping and testing

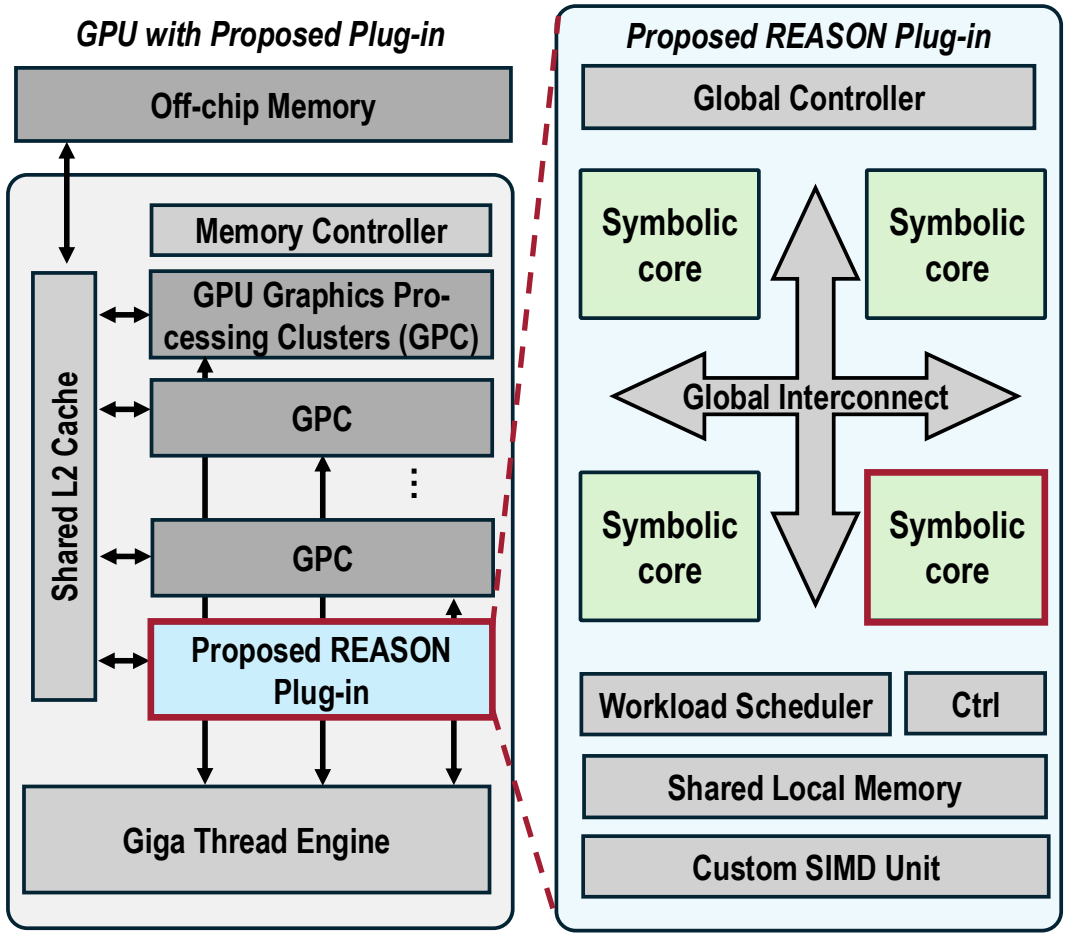
Measurement: Efficiency and Performance



Neuro-Symbolic Models	Applications	Main Operations	Memory Requirement
NVSA	Abstract Reasoning	NN, symbolic circular conv	5.4 MB
LVRF	Probabilistic Reasoning	NN, probabilistic circuits	4.8 MB
MIMONet	Simultaneous processing	NN, symbolic vector binding	5.6 MB
PrAE	Probabilistic Reasoning	NN, probabilistic markov model	1.9 MB
LNN	Full theorem prover	NN, symbolic first-order logic	2.5 MB
LTN	Querying and reasoning	NN, symbolic solver	3.0 MB
NLM	Relational reasoning	NN, symbolic first-order logic	4.6 MB

Consistent low-power execution across diverse neuro-symbolic models

From Edge to Agentic: GPU Co-Processor Integration



Neuro-Symbolic Models	Applications	Main Operations
AlphaGeometry	Math Reasoning	LLM, First-order logic, SAT
R^2-Guard	Unsafety detection	LLM, First-order logic, Probabilistic Circuit
GeLaTo	Constrained generation	LLM, First-order logic, SAT, Markov Model
Ctrl-G	Interactive editing	LLM, Markov Model, Probabilistic Circuit
LINC	Logical reasoning	LLM, First-order logic, SAT

GPU runs LLM inference, REASON co-processor runs symbolic and probabilistic components

Wan, Liu, Qian, Yang, Raychowdhury, Krishna, "*Accelerating Probabilistic Reasoning for Neuro-Symbolic Intelligence*", **HPCA 2026**

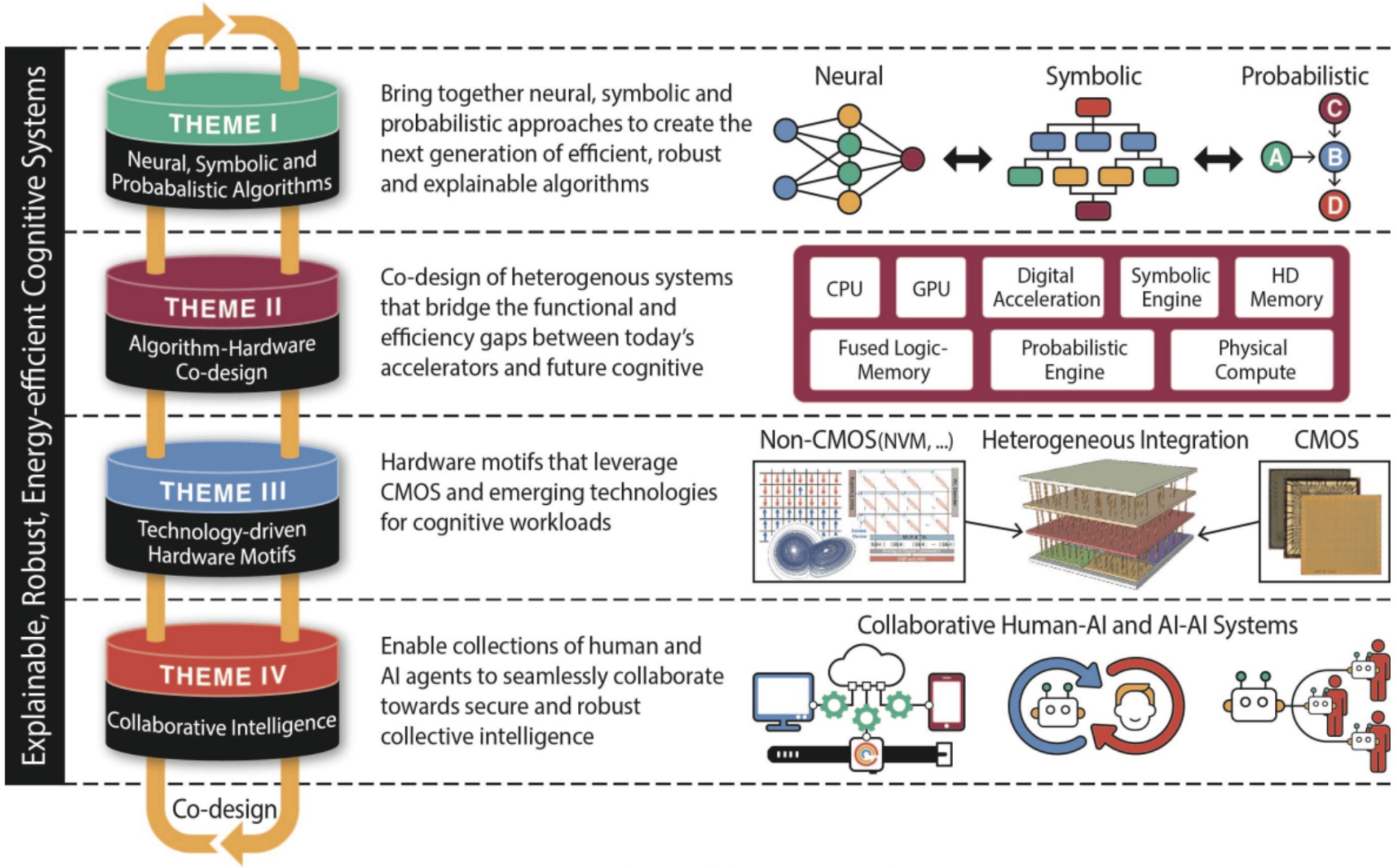
GPU co-processor integration enables efficient and versatile LLM-symbolic agentic processing

Thrust 1 Takeaways

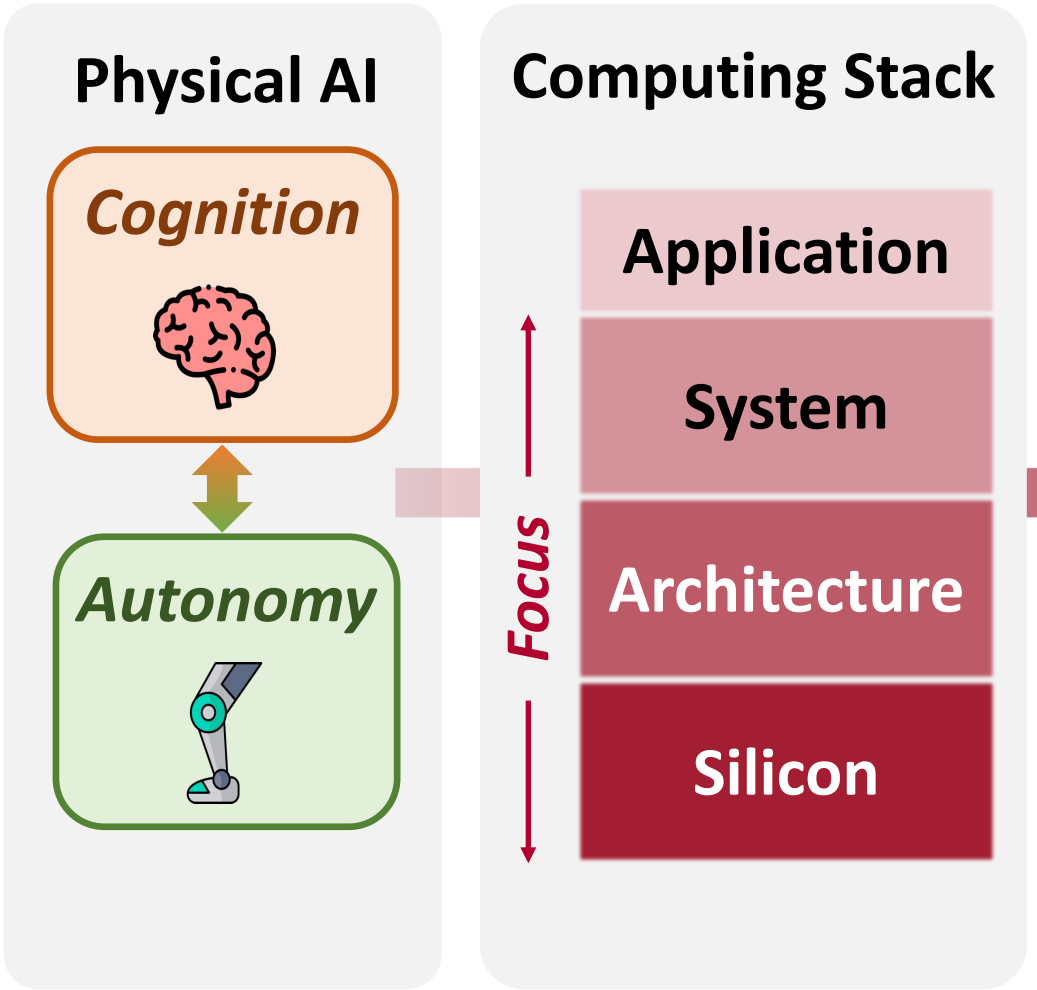
Building **system architecture** & **silicon** to make compositional AI deployable in physical world via principled cross-stack co-design:

- New **system-level abstractions** and **scheduling** mechanisms
- Novel **architectures** and **dataflows** that unify neuro and symbolic AI
- The first **programmable SoC tapeout** with memory-centric energy-adaptive design enabling neuro-symbolic cognition on silicon

CoCoSys: Center for Co-Design of Cognitive Systems

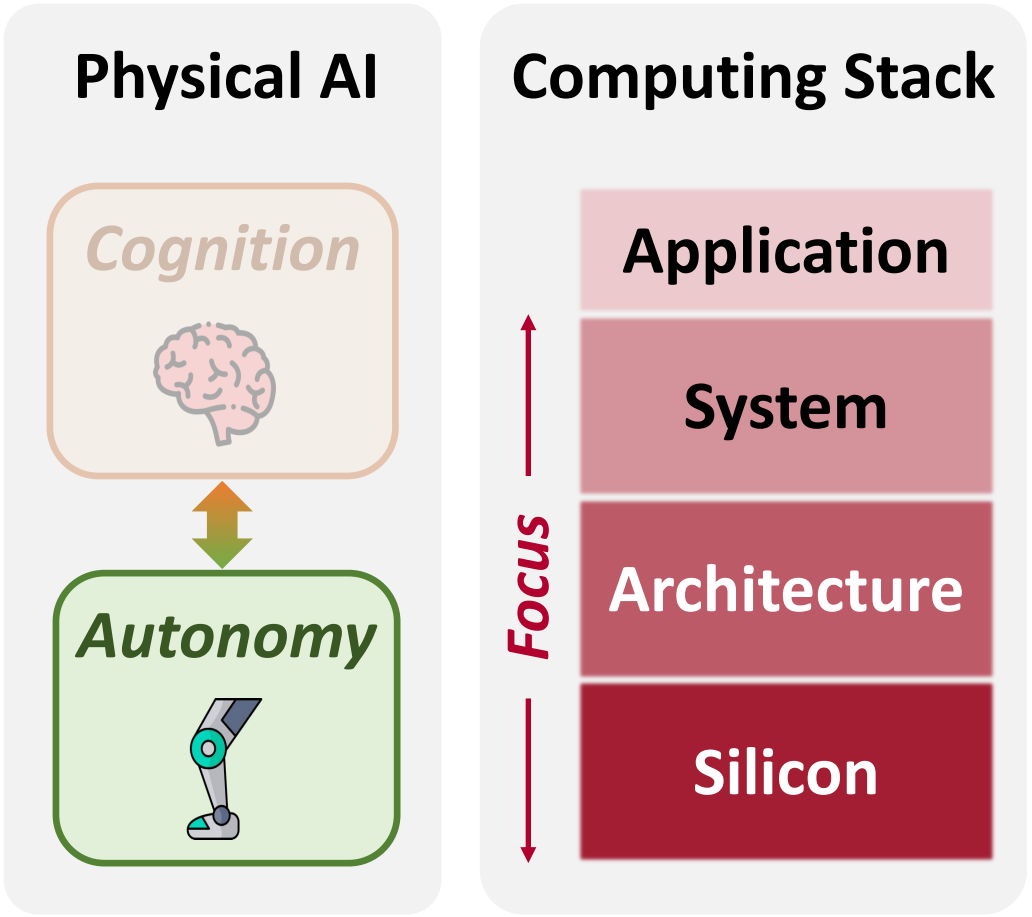


Outline



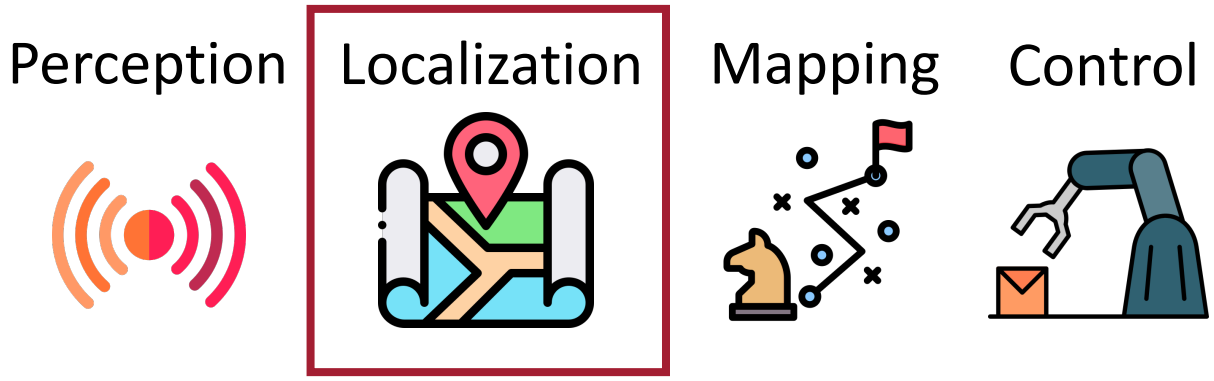
- Background & Motivation
- Thrust 1: Hardware for *Cognition*
- **Thrust 2: Hardware for *Autonomy***
  - ReCA: SoC for autonomy + cognition
  - MulBerry: agile SoC design flow for physical intelligence
- Future Research

What Does Physical Intelligence Compute?



How do we architect computing systems for energy-efficient and real-time autonomy?

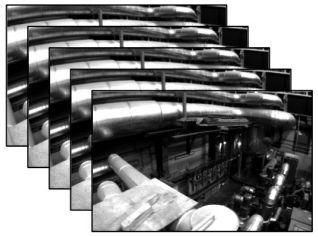
How do we architect computing systems to bridge cognition and autonomy efficiently?



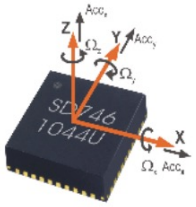
Wan, Du, Ibrahim, Qian, Jabbour, Zhao, Krishna, Raychowdhury, Reddi, "*ReCA: Integrated Acceleration for Embodied Agent Planning*", **ASPLOS 2025** **(Selected as Industry-Academia Partnership (IAP) Highlight)**

Localization and Mapping Using SLAM

Camera



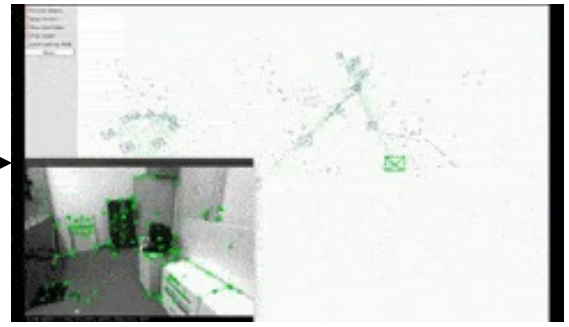
IMU
(Inertial Measurement Unit)



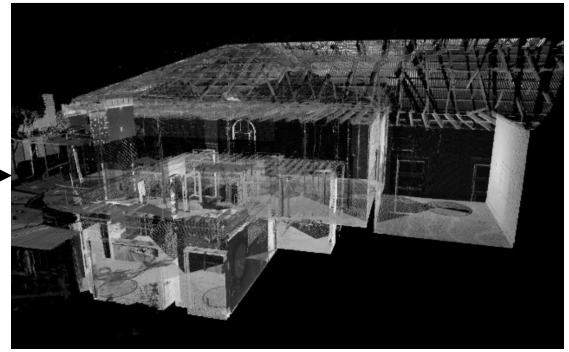
Feature Tracks

Estimated States

Simultaneous Localization and Mapping (SLAM)

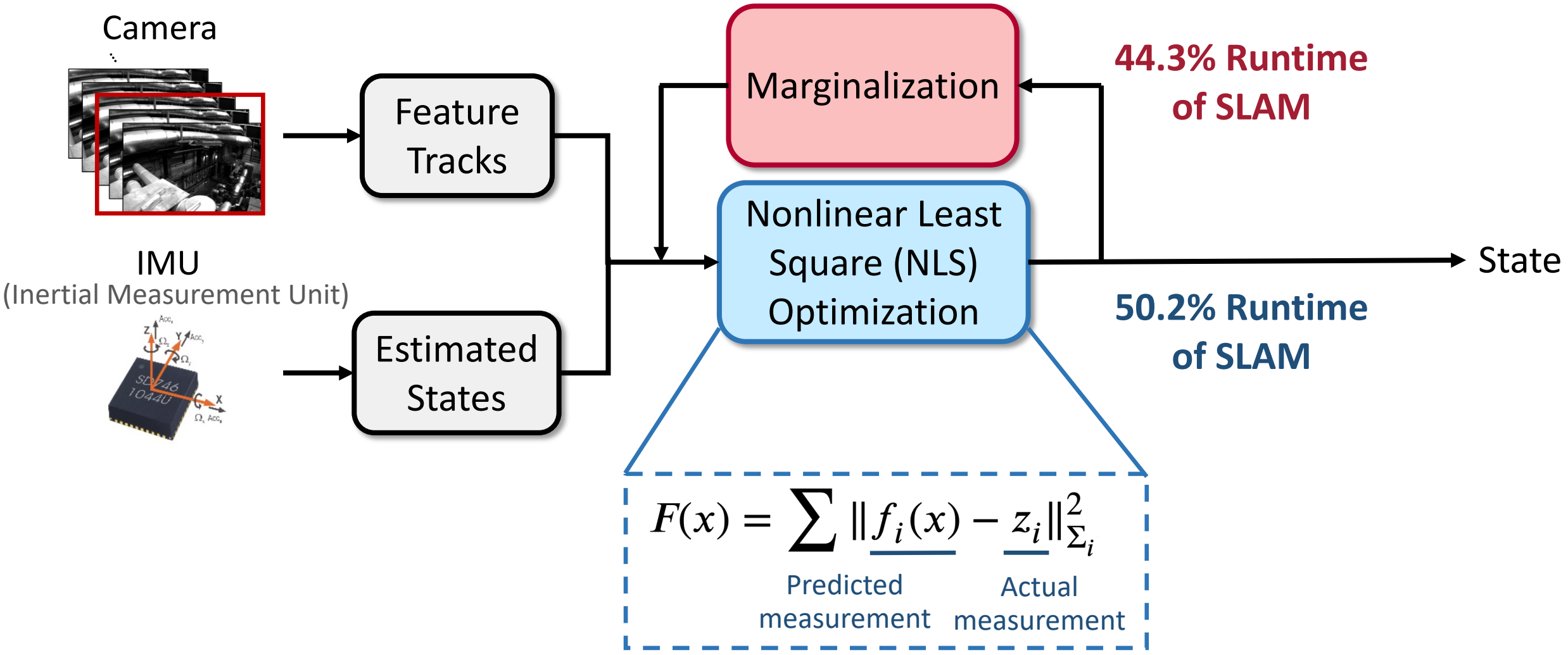


Localization
(6 DoF poses)



Mapping
(3D coordinates)

How Does SLAM Work?

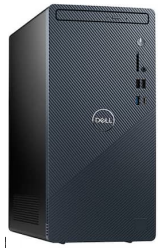


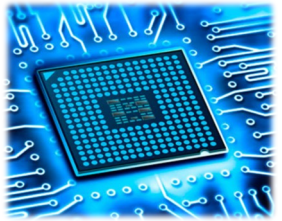
SLAM is dominated by iterative complex optimization

Challenge: Power and Speed

Goal

- **Power**: < 2 W
- **Latency**: < 0.02 s



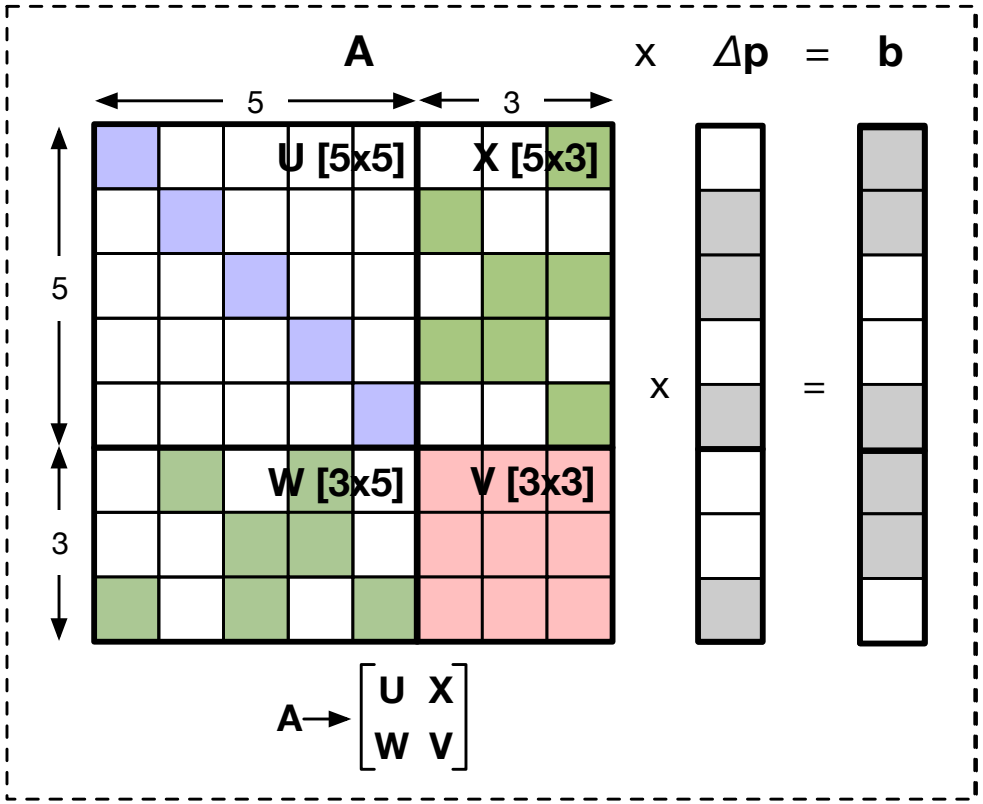


	Goal	Desktop CPU	Embedded CPU	FPGA / ASIC
Latency	< 0.02 s	**0.08 s**	**0.5 s**	Low power if only use on-chip memory (e.g., 3 MB on FPGA)
Power	< 2 W	**50 W**	**3 W**	Need algorithm-hardware co-design approach
		Too high power	**Too slow**	

General purpose computing not good enough

Compute Block Optimization

Iterative Optimization:



Make U as diagonal matrix:

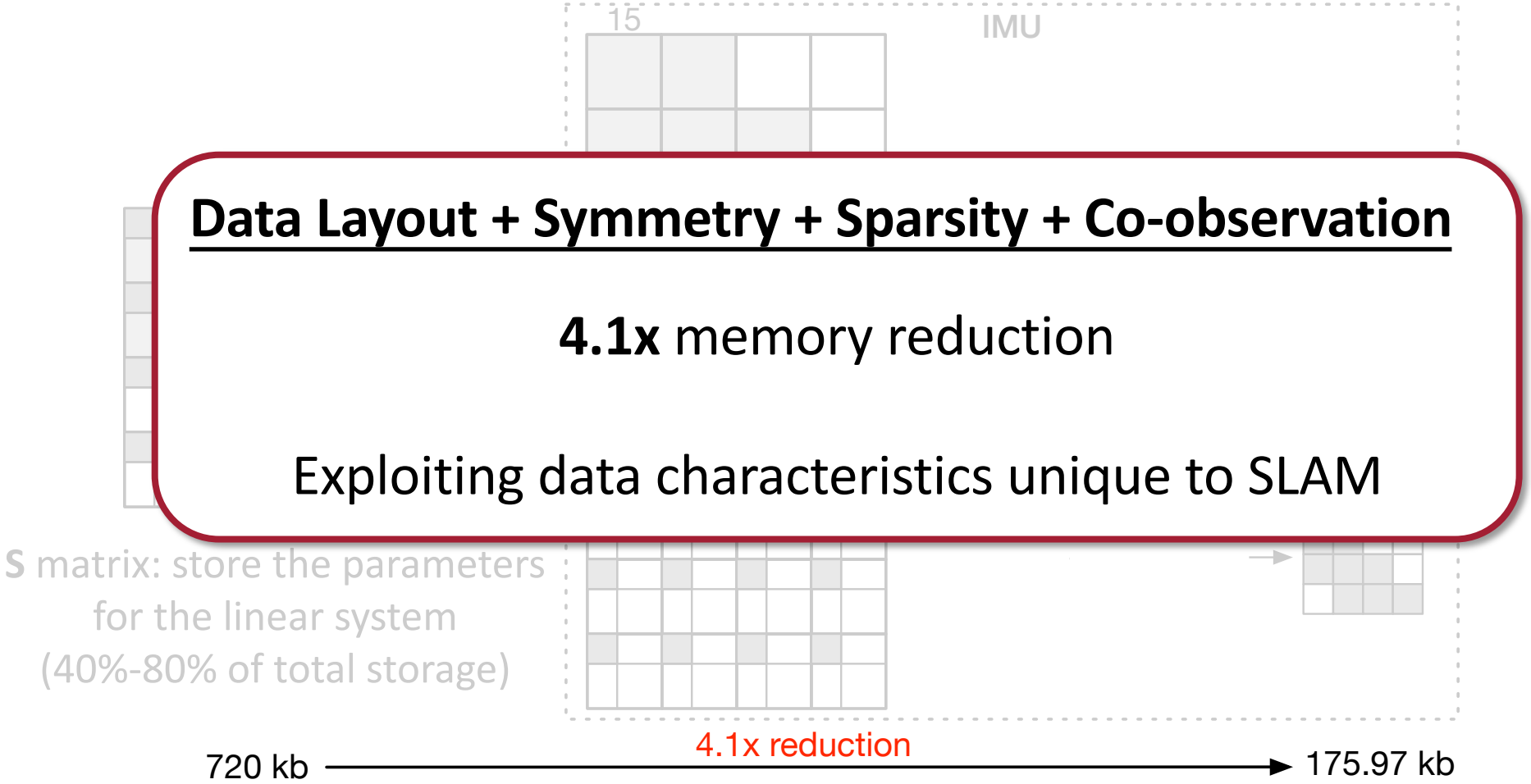
***O*(n³)->*O*(n)** computational complexity

X becomes the transpose of W:

1.34x on-chip memory reduction

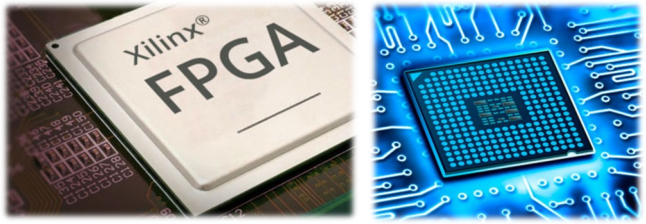
Leveraging diagonality, symmetry, and data reuse to reduce complexity & resource consumption

Memory Layout Optimization



Leveraging memory data layout, symmetry, sparsity to reduce memory consumption

Results: Low Latency and Power

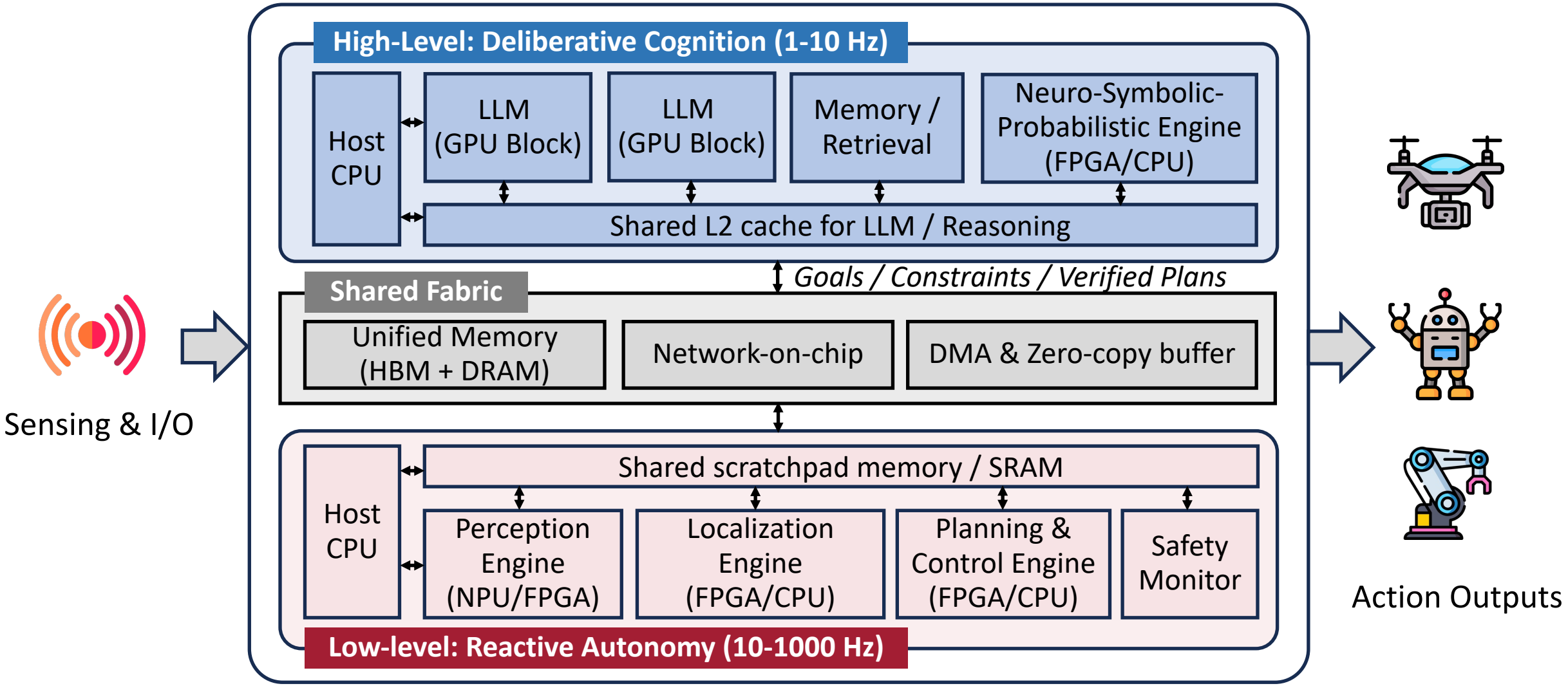


	Goal	Desktop CPU	Embedded CPU	FPGA (ours)
Error	< 0.1 m	**0.06 m**	**0.06 m**	**0.06 m**
Latency	< 0.02 s	**0.08 s**	**0.5 s**	**0.01 s**
Power	< 2 W	**50 W**	**3 W**	**1.5 W**
		Too high power	**Too slow**	**Best of Both Words**

The co-designed FPGA implementation only requires 2.7 MB memory

On-board FPGA on real-UAV test demonstrate improved computation latency and energy

SoC Bridges Cognition (Thrust 1) + Autonomy (Thrust 2)



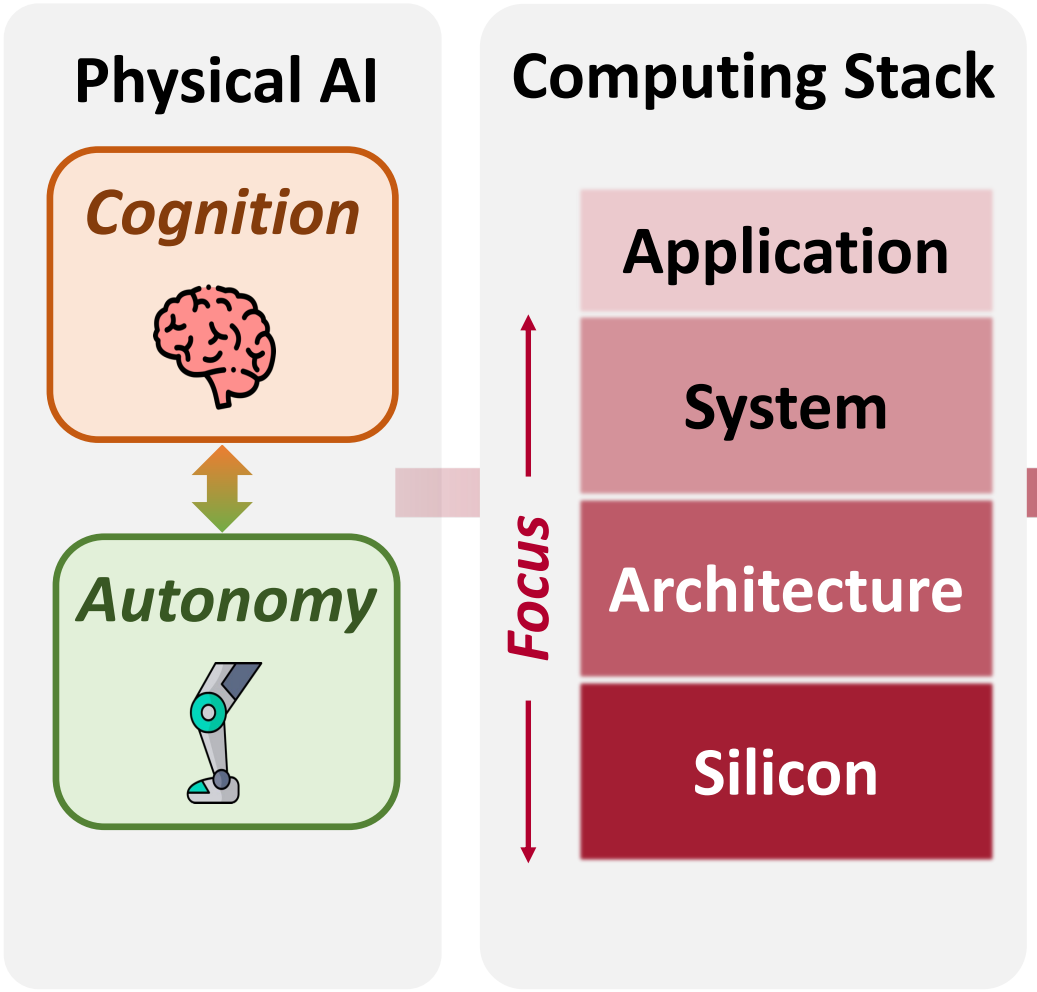
Heterogenous SoC bridges reactive autonomy and deliberative cognition for physical agents

Thrust 2A Takeaways

Hardware can influence algorithm design – exploit hardware capabilities (e.g., parallelism, reuse) and avoid hardware bottlenecks (e.g., memory)

Algorithm design can open up opportunities for hardware designers and computer architects by shifting the bottleneck

Outline



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  - **MulBerry: agile SoC design flow for physical intelligence**
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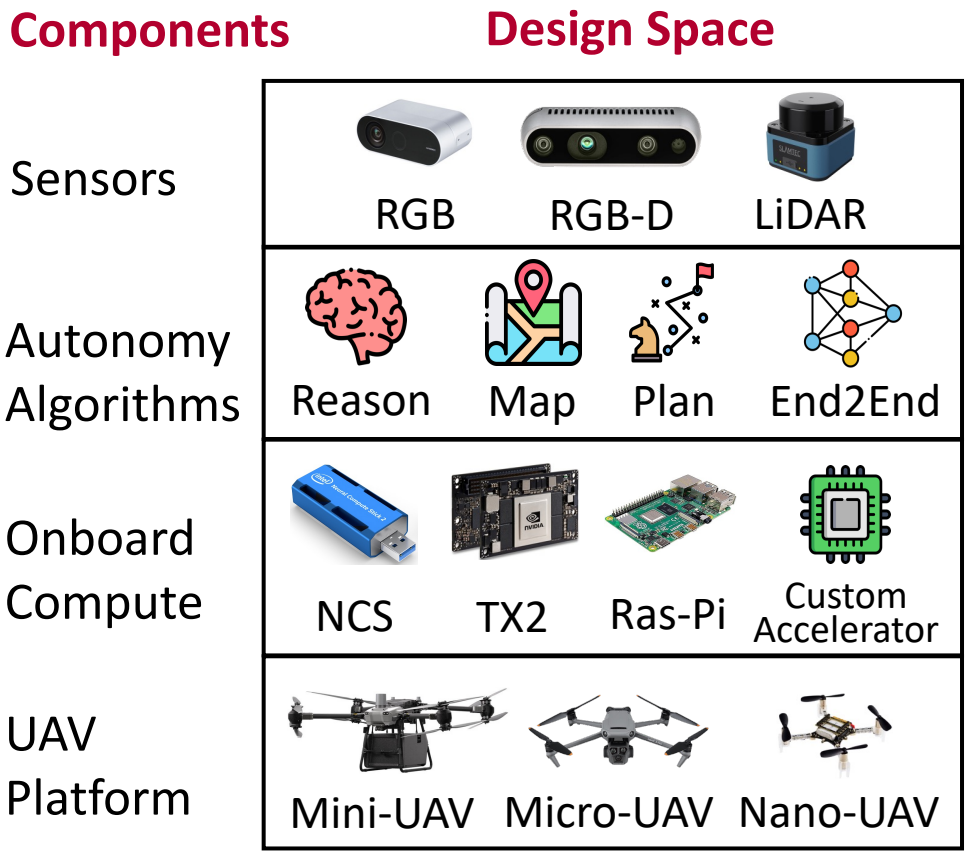
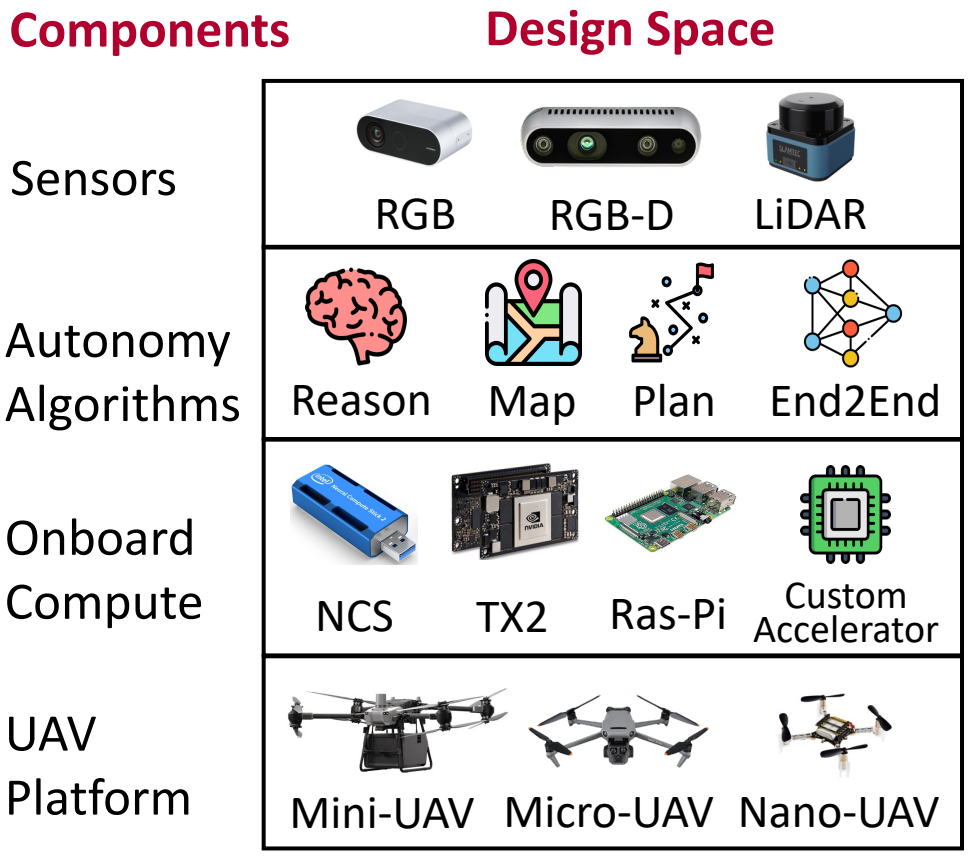
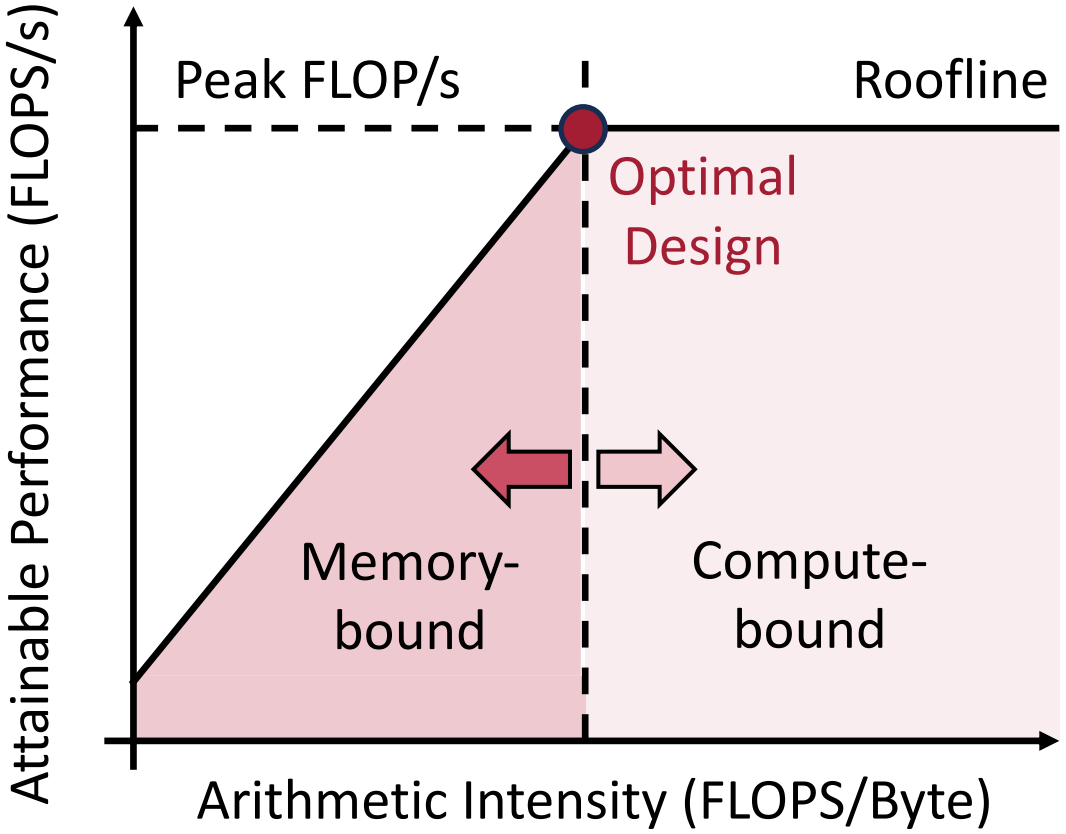
Compute is Only One Component



What's the role of computing in physical intelligence systems?

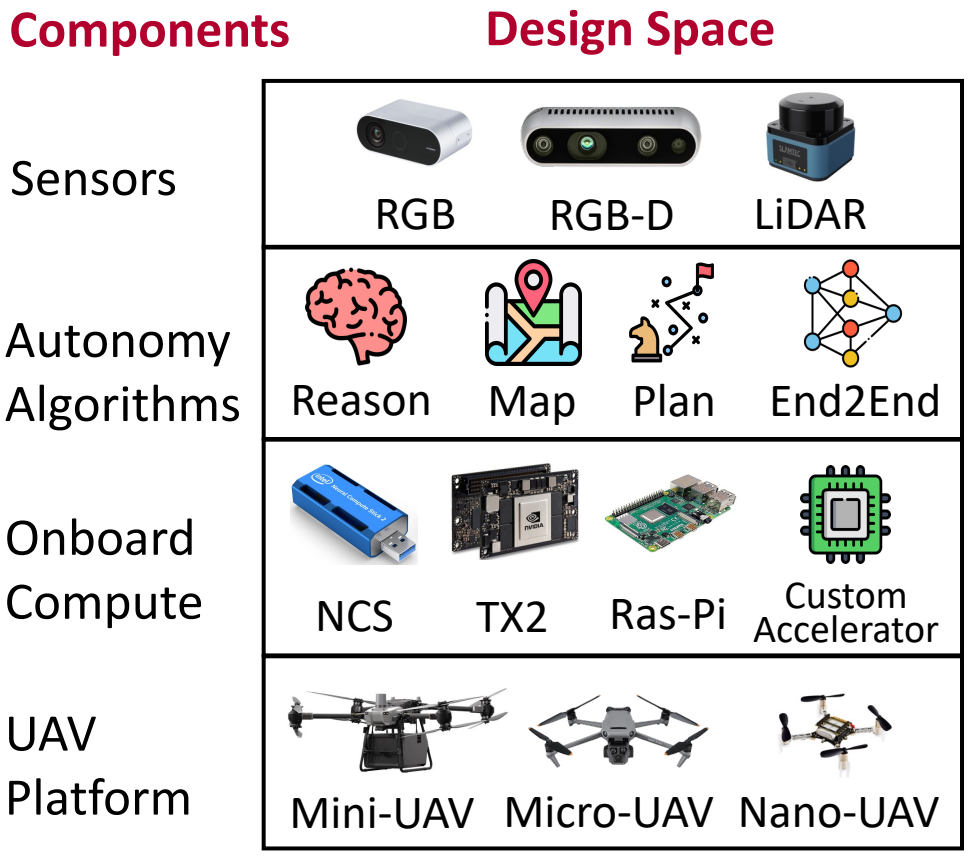
Roofline Model in Computer Architecture

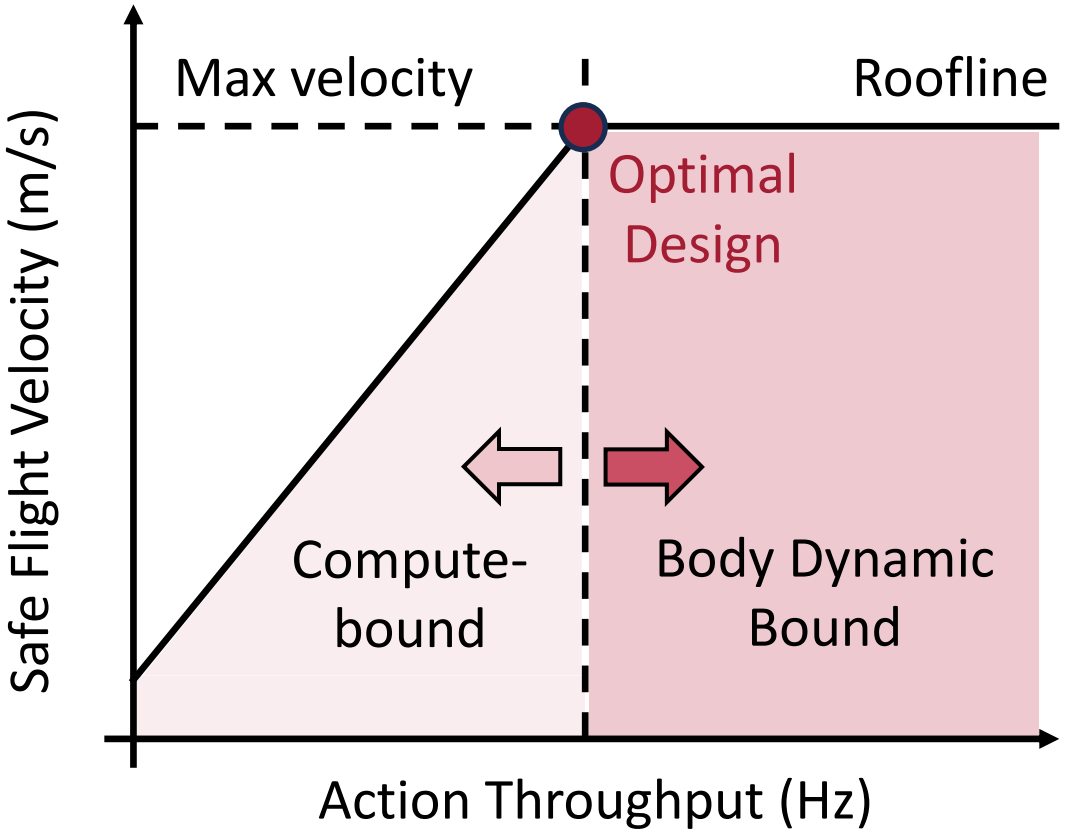




Performance model provides system insights

Roofline Model in Physical Intelligence System

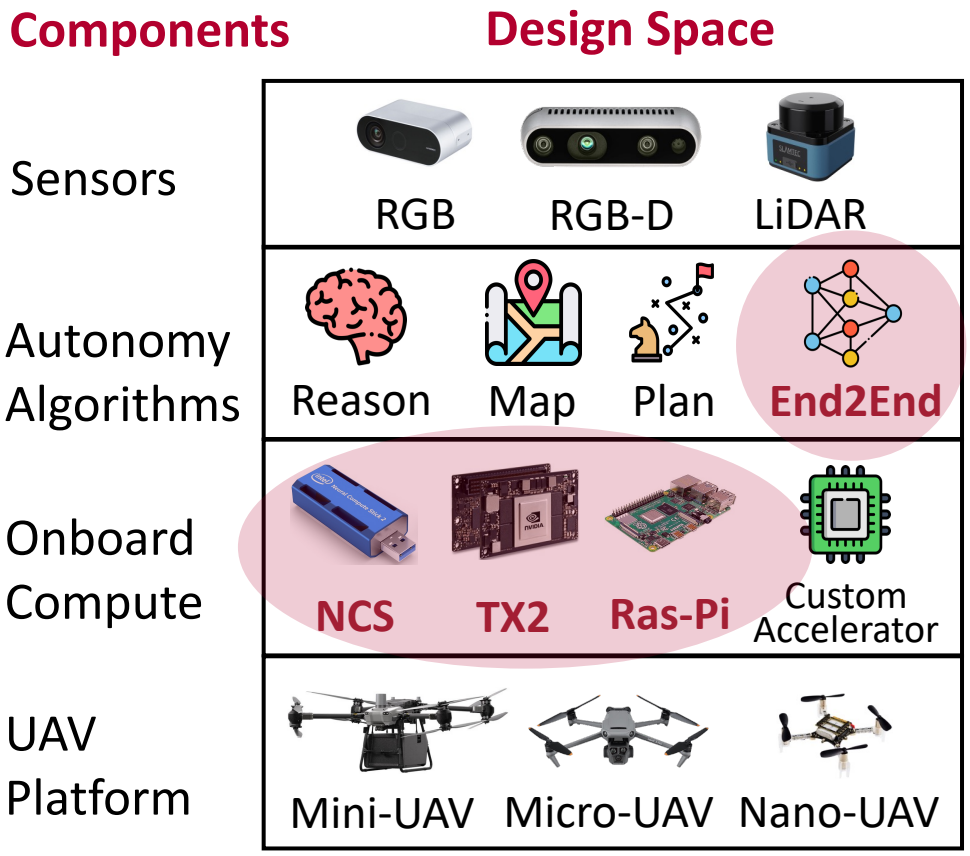


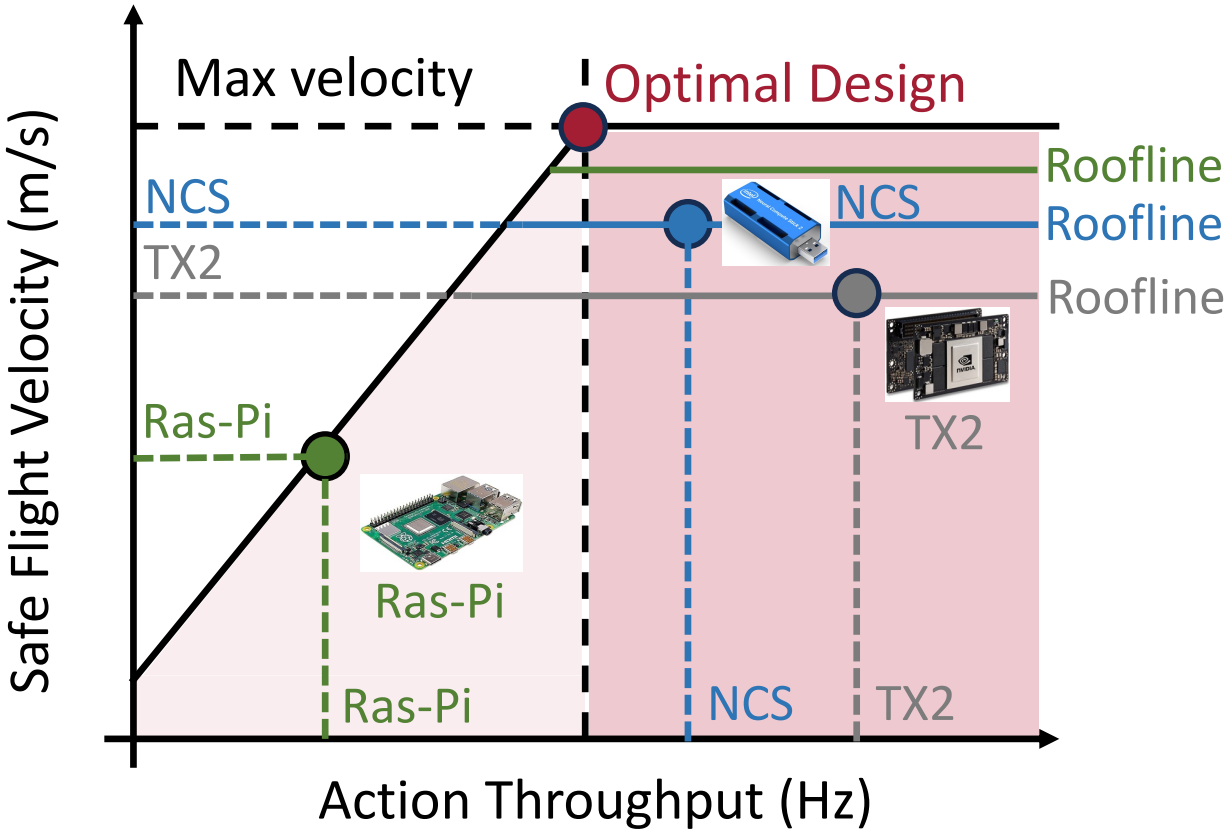


Validated via real-world UAV experiments

Performance model for UAVs: Cyber-physical co-design insights

Best Compute Design ≠ Best System Design



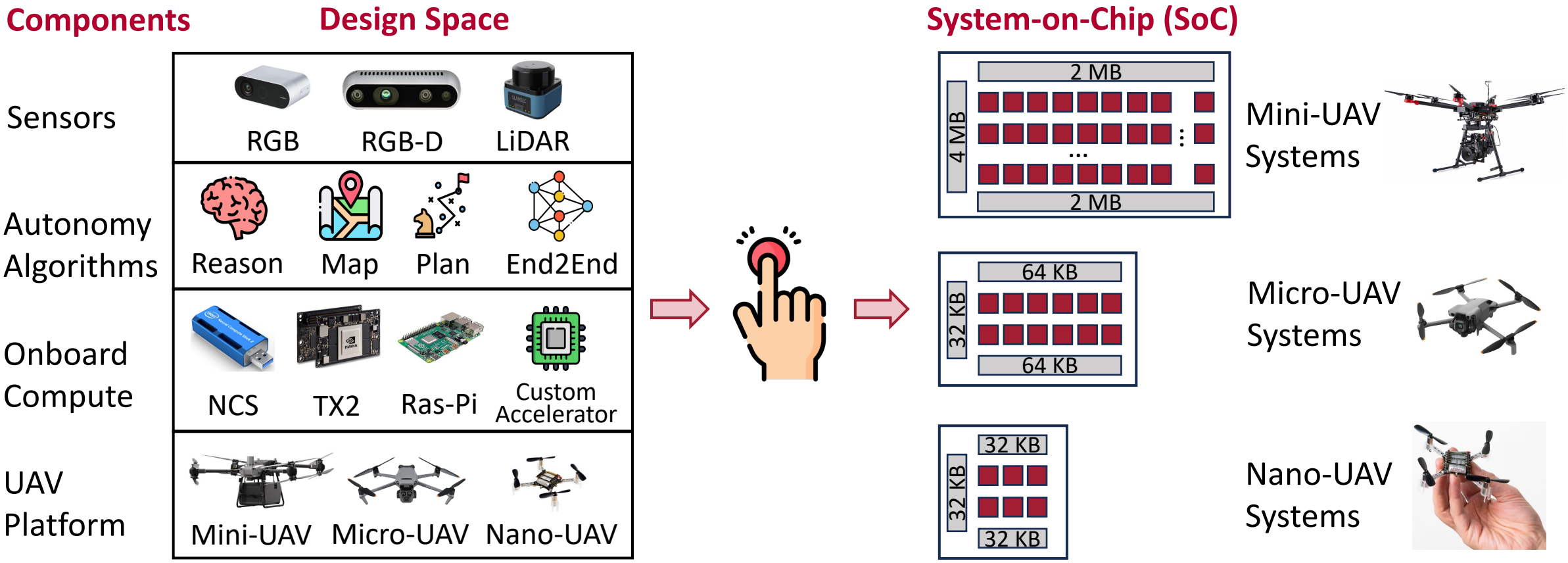


Action Throughput (Hz): Ras-Pi < NCS < TX2

Safe Flight Velocity (m/s): Ras-Pi < TX2 < NCS

How can we leverage this performance model to handle the increasing levels of complexity?

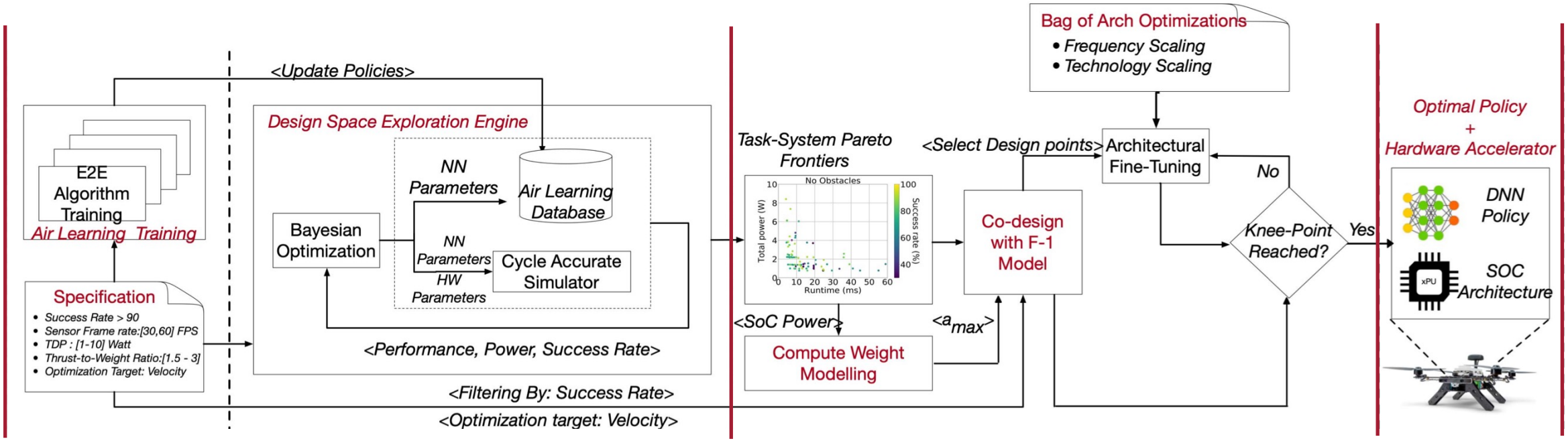
Agile SoC Design Framework for Autonomous UAV



Wan, Chandramoorthy, Swaminathan, Reddi, Raychowdhury, "*Robust Low-Voltage SoC for Auto Systems*", **ASPLOS 2024**
(Best Poster Award, IBM AI Compute Symposium 2023; First Place, ACM Student Research Competition 2023)

Krishnan, **Wan**, Bhardwaj, Whatmough, Faust, Neuman, Wei, Brooks, Reddi, "*Automatic Domain-Specific SoC Design for UAVs*", **MICRO 2023**
(IEEE Micro Top Picks, Honorable Mention)

Agile SoC Design Framework for Autonomous UAV



Optimization Phase

Domain Agnostic

Mapping Phase

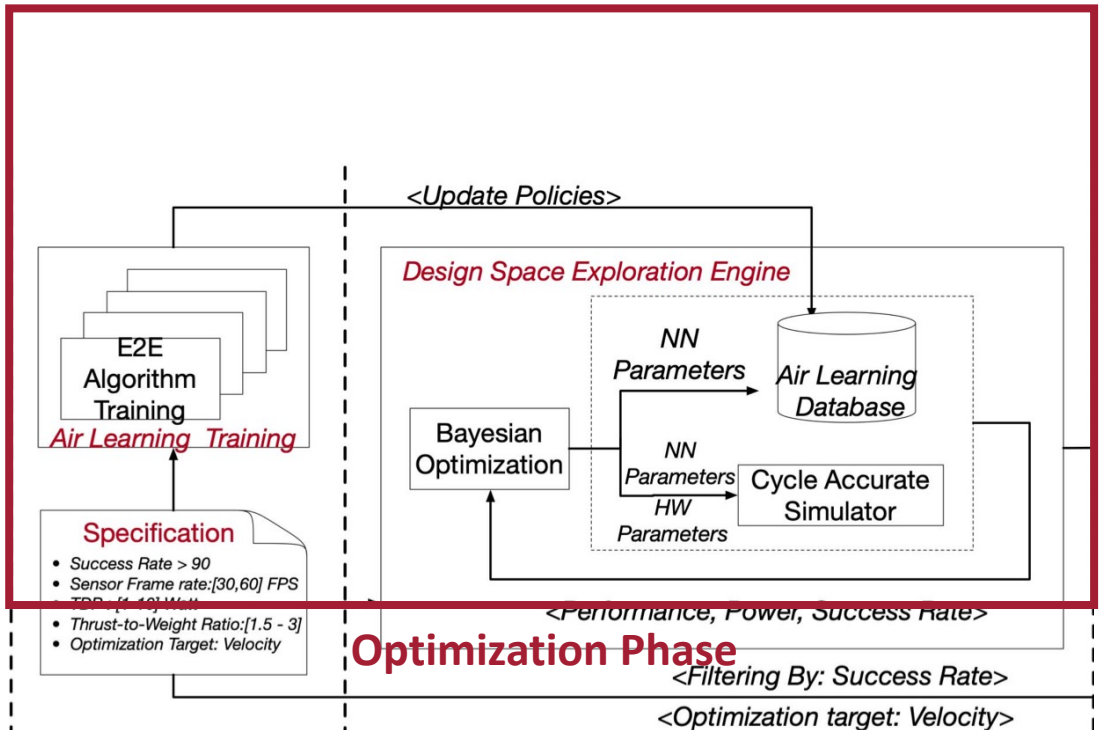
Domain Specific

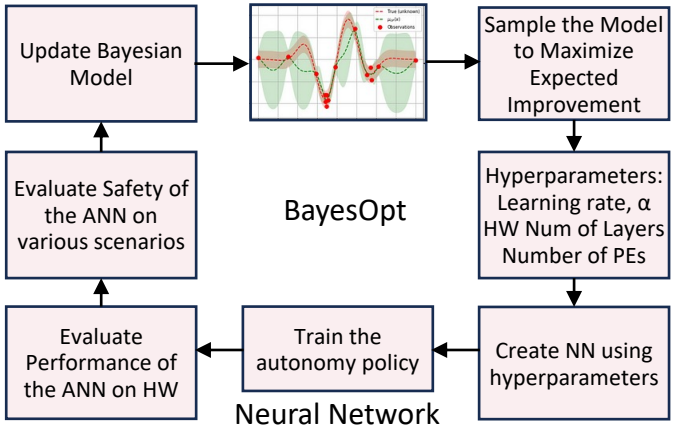
Deployment Phase

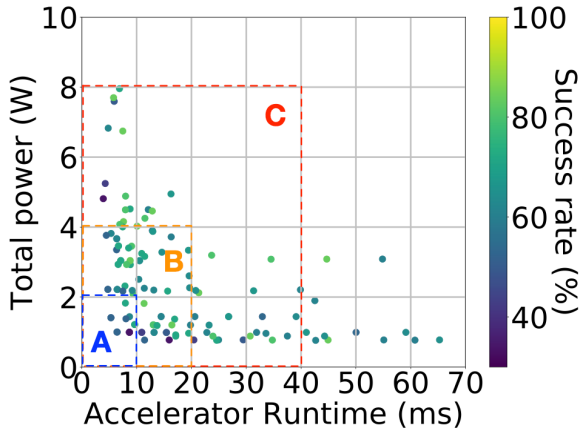
Target System

Agile SoC development flow breaks the domain-specific SoC design into three phases

(Domain-agnostic) Algorithm-Hardware Co-Design Optimization

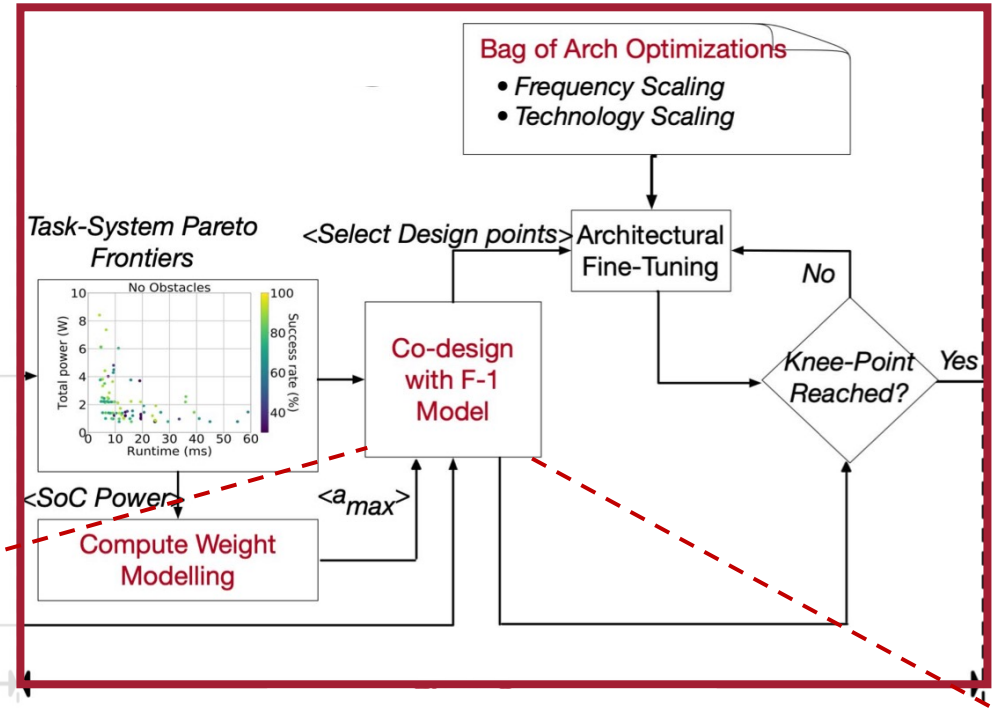




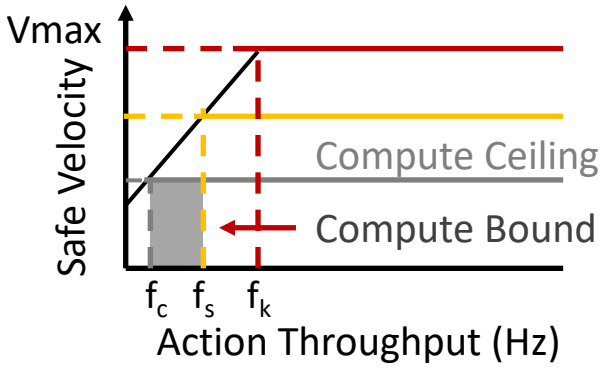


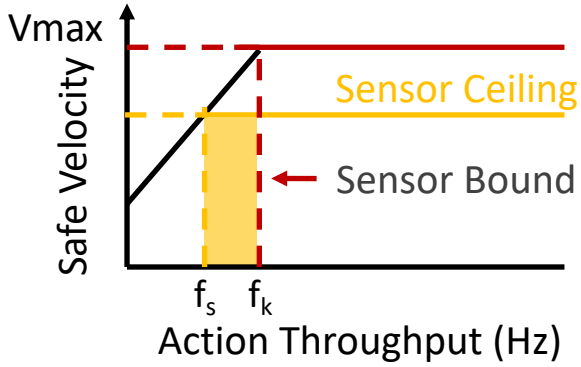
Phase 1: Enable automated algorithm-hardware co-design optimization

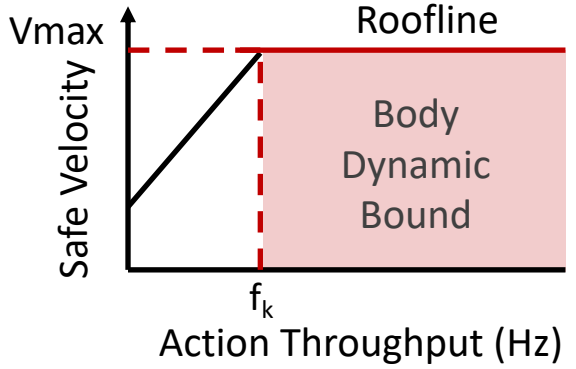
(Domain-specific) Cyber-Physical Mapping

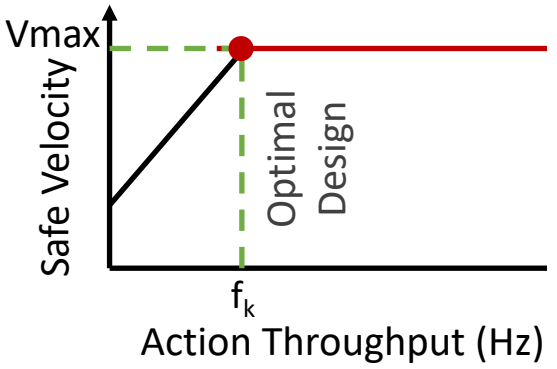


Mapping Phase

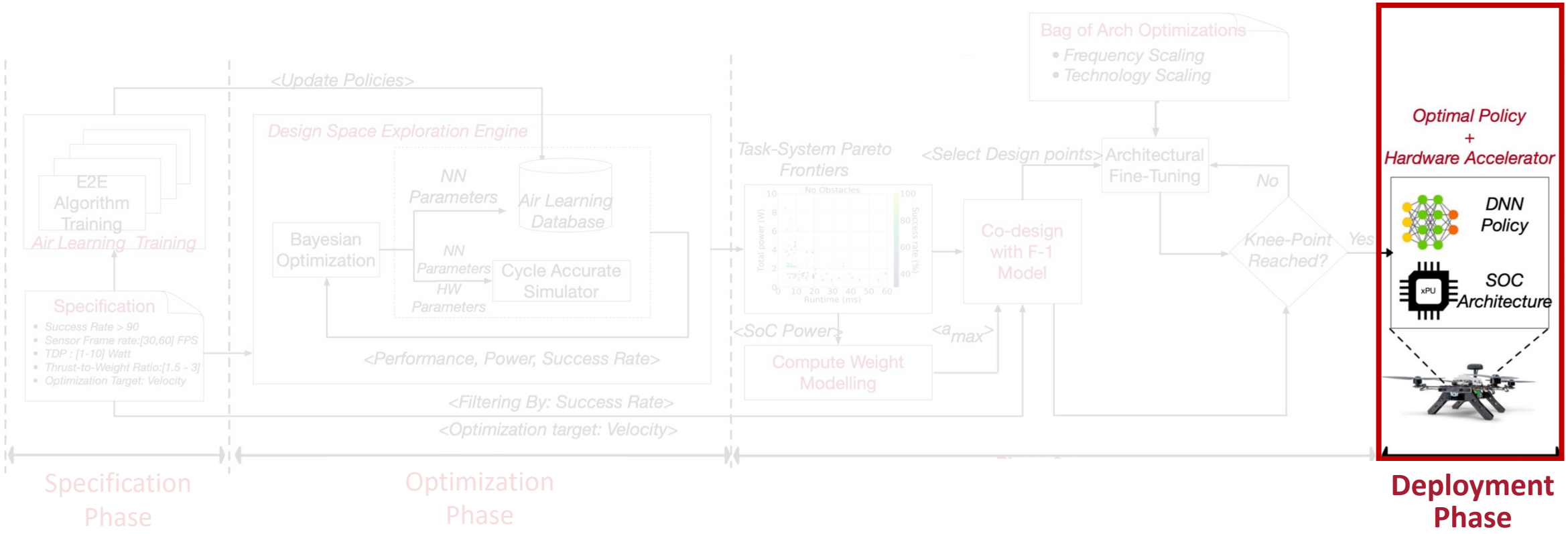






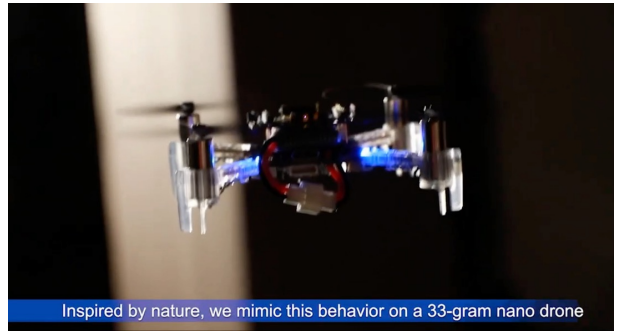


Phase 2: Enable precisely capture of cyber-physical interactions to get best system performance



Mini UAV

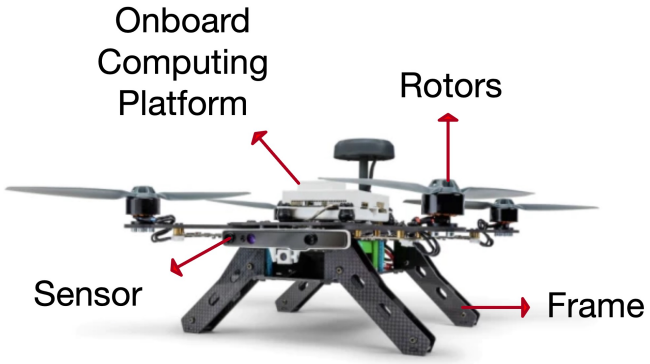
Micro UAV

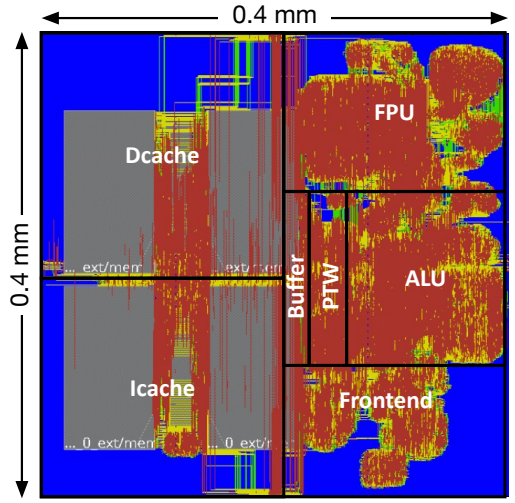


Nano UAV

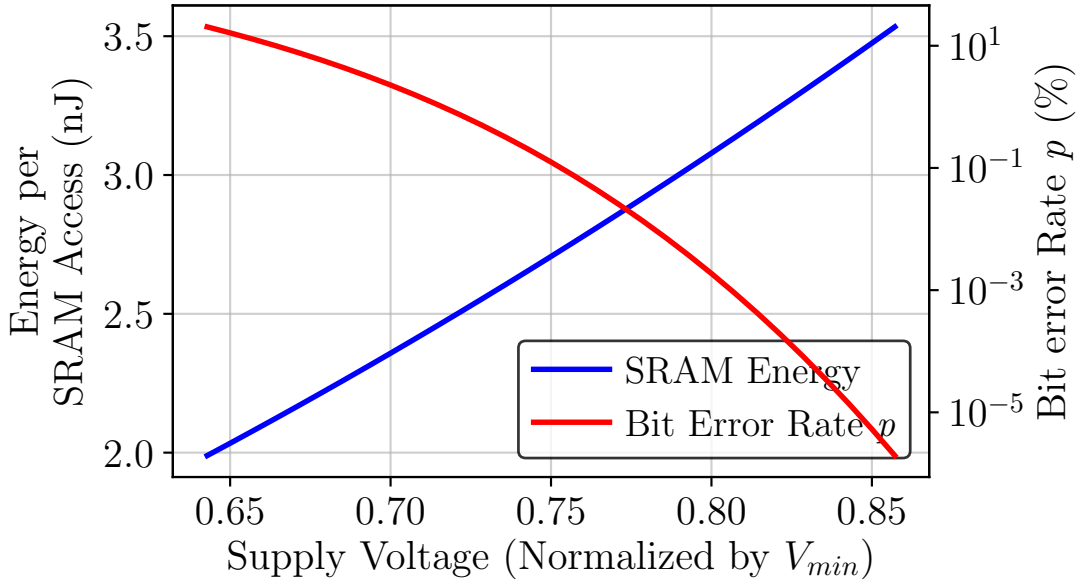
Phase 3: Enable real-word deployment across diverse scenarios

SoC Voltage Scaling Further Reduces System Energy





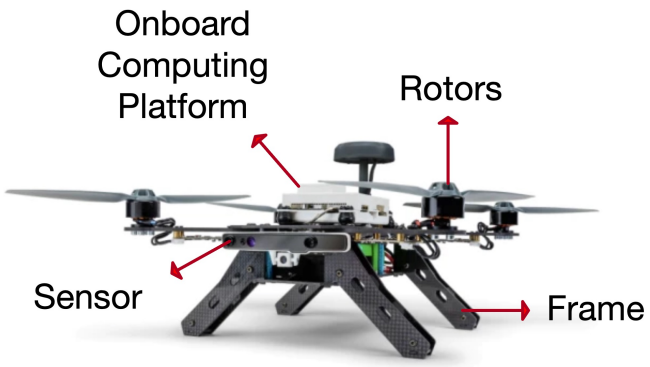
SRAM Access Energy / Bit Error Rate vs. Operating Voltage

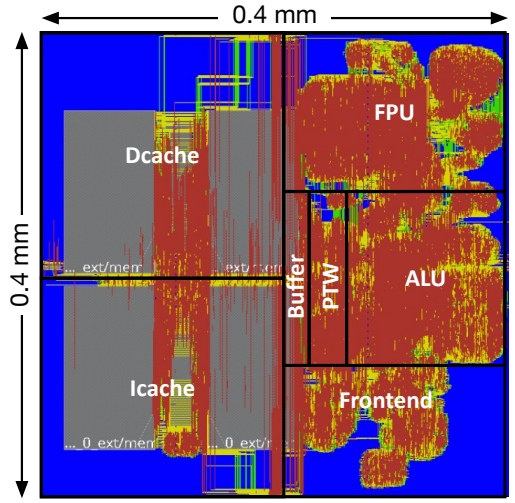


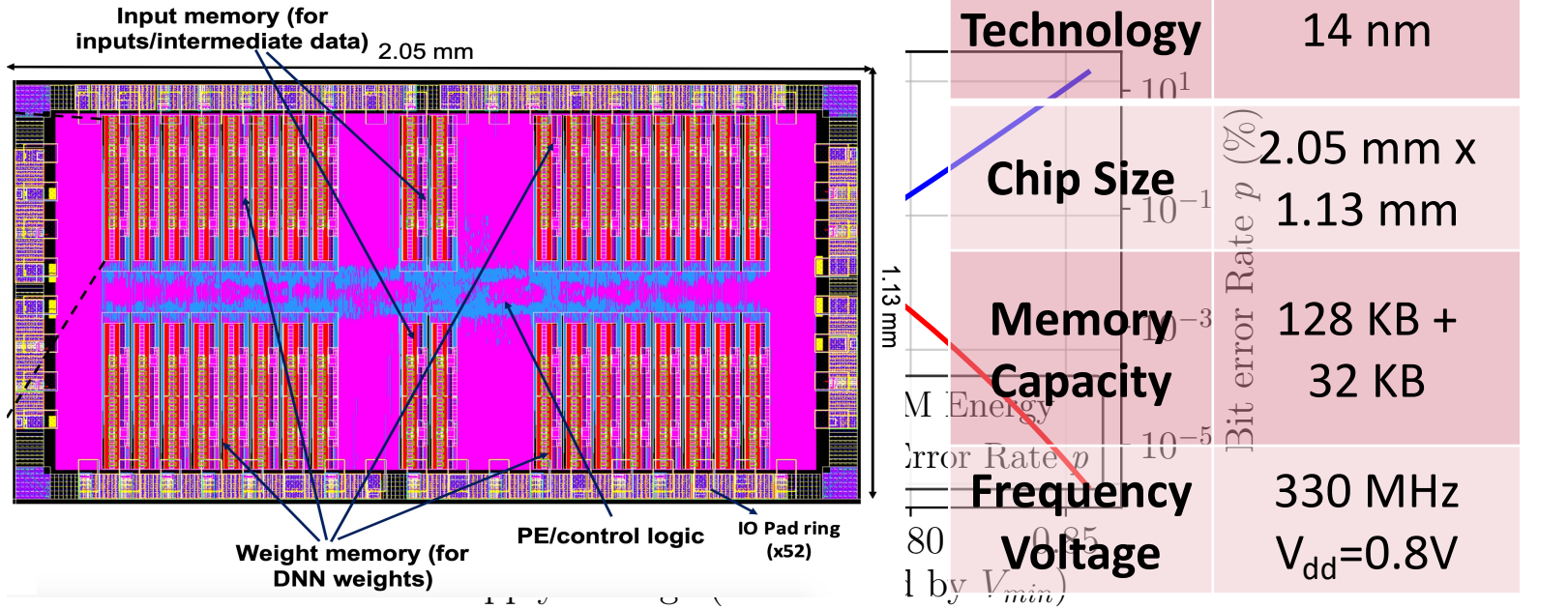
Data measured from 14nm FinFET SRAM chips

SoC compute-level voltage scaling enables system-level energy reduction

SoC Voltage Scaling Further Reduces System Energy







Data measured from 14nm FinFET SRAM chips

	Normal Vol	Lower Vol
Average flight energy per task	366 J	**299 J**
Number of completed missions	40	**49**

SoC compute-level voltage scaling enables system-level energy reduction

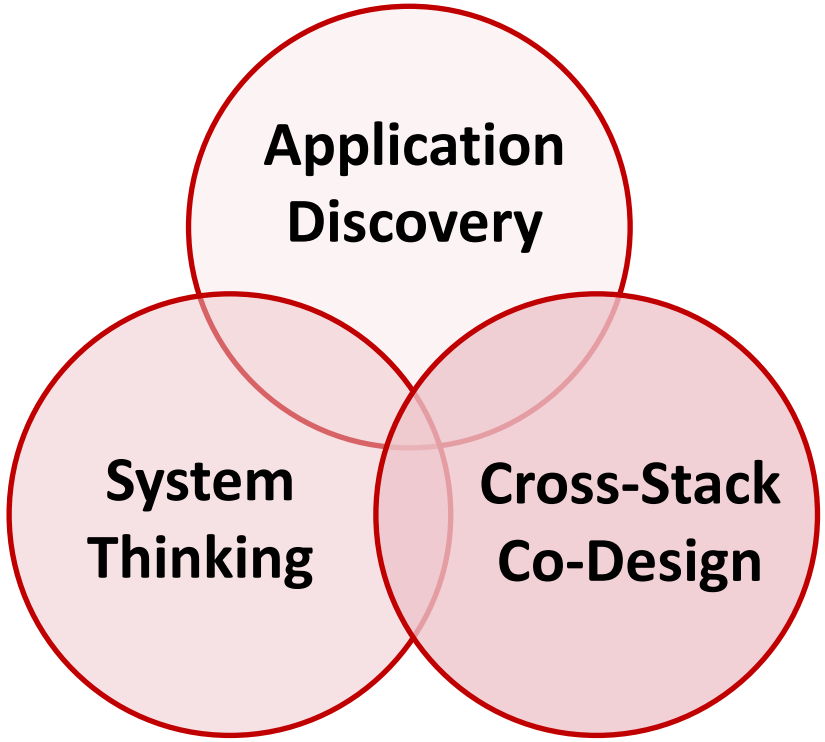
Thrust 2B Takeaways

Expand design space to include choice of **energy allocation** and **optimization efforts** across compute, sensing, and actuation

Use efficient computing to reduce energy consumption of other components in system **-> magnify savings beyond compute**

Concluding Remark

Achieving **scalable and efficient hardware architectures** demands a holistic approach that integrates **system thinking**, **automation**, **co-design**, and **silicon realization and validation**



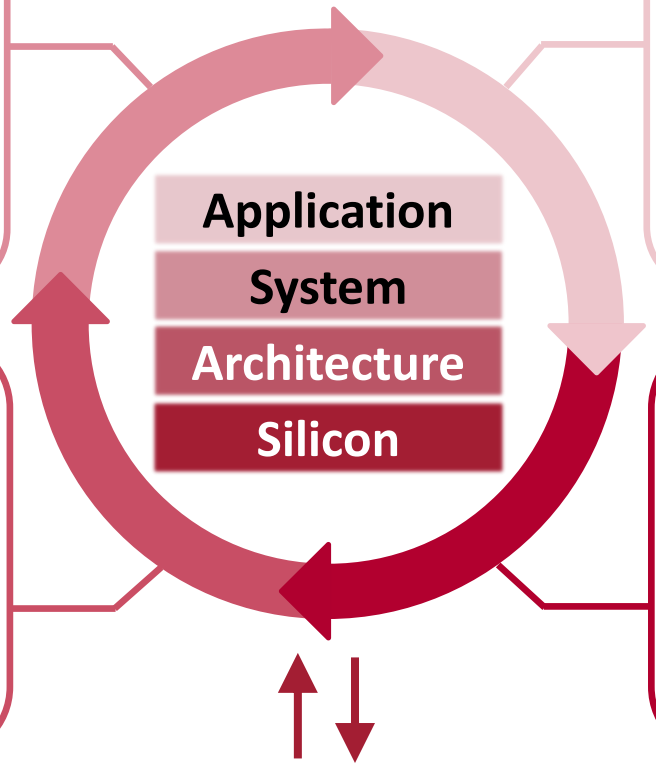
Outline

- Background & Motivation
- Thrust 1: Architecting Computing Systems for *Cognition*
- Thrust 2: Architecting Computing Systems for *Autonomy*
- **Ongoing and Future Research**

Vision: Advancing the Future of Physical AI Computing

System
- Benchmark and toolchain
- Runtime and orchestration
- Agentic-centric memory

Applications
- Robotics, VLA, World model
- Spatial intelligence, XR
- Healthcare, Bioinformatics



Architecture
- Domain-specific architecture
- Composable architecture
- Chiplet system architecture
- Agile SoC design flow

Hardware in Silicon
- In/Near-memory computing
- Emerging memory tech
- Heterogenous integration
- Smart power management

Physical and Agentic AI

Cross-stack system-architecture-silicon co-design for Physical AI computing

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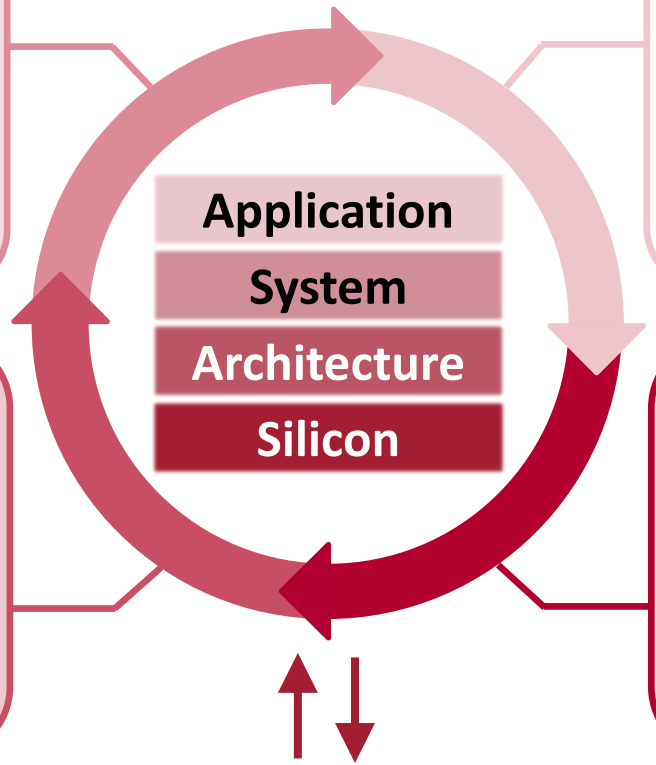

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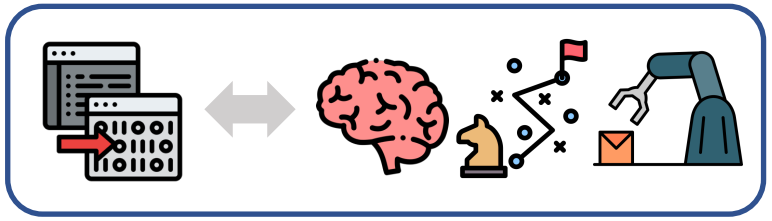
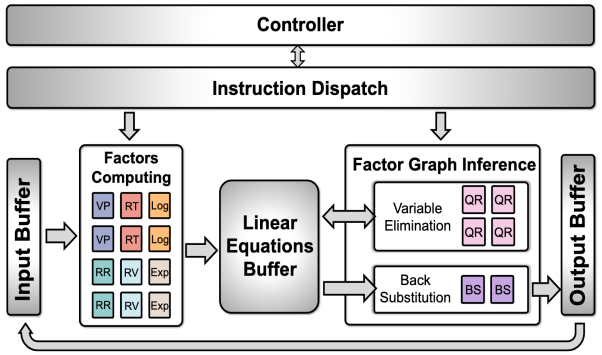
Cross-stack system-architecture-silicon co-design for Physical AI computing

Direction 1: System & Architecture for Physical AI

What kind of computing systems and architectures are required to support agentic, compositional intelligence operating in the physical world?

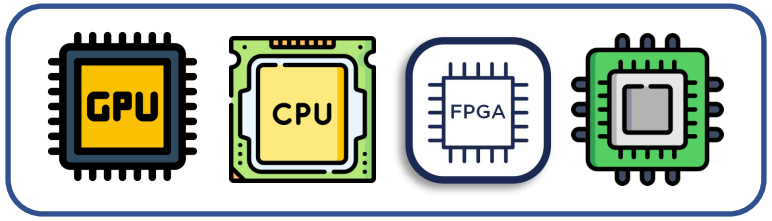
Runtime & Compiler

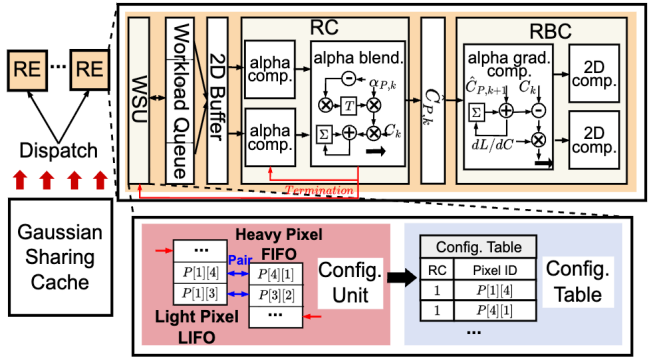




"ORIANNA: Runtimes and Compilation for Optimization-based Robotics", **ASPLOS'24**

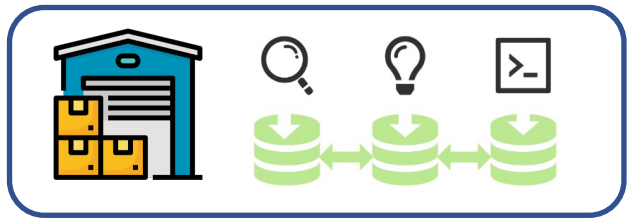
Heterogenous architecture

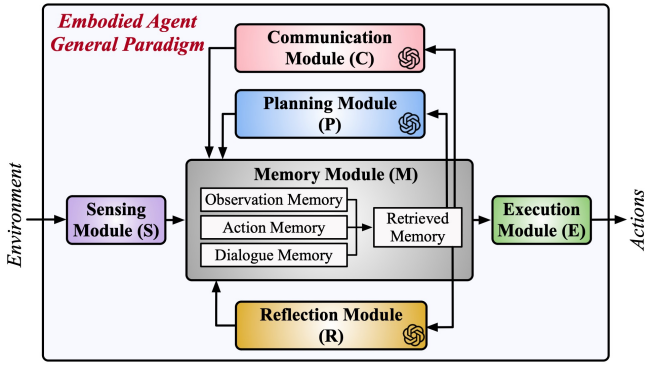




"RTGS: Architecture for 3D Gaussian Splatting SLAM", **MICRO'25**

Memory Systems





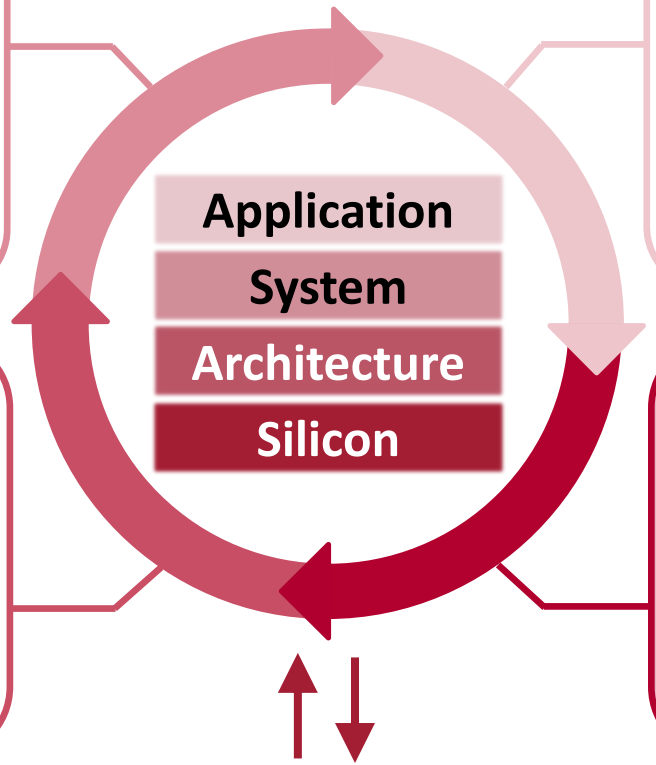
"EmbodiedPerf: Memory for Embodied Systems", **ISPASS'25**

Goal: Enable composable and scalable execution substrates for agentic intelligence

Vision: Advancing the Future of Physical AI Computing

System
- Benchmark and toolchain
- Runtime and orchestration
- Agentic-centric memory

Applications
- Robotics, VLA, World model
- Spatial intelligence, XR
- Healthcare, Bioinformatics



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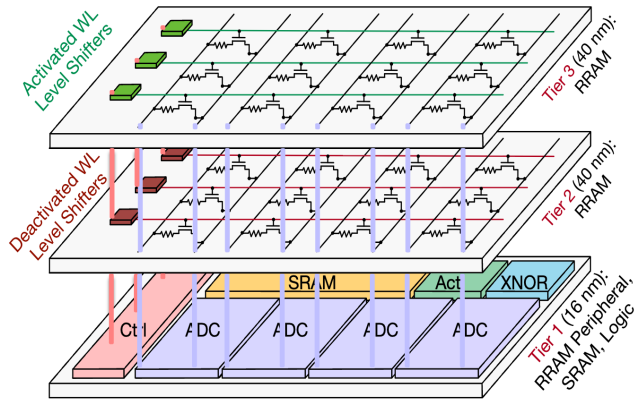
Physical and Agentic AI

Cross-stack system-architecture-silicon co-design for Physical AI computing

Direction 2: System-Tech Co-Opt for Physical AI

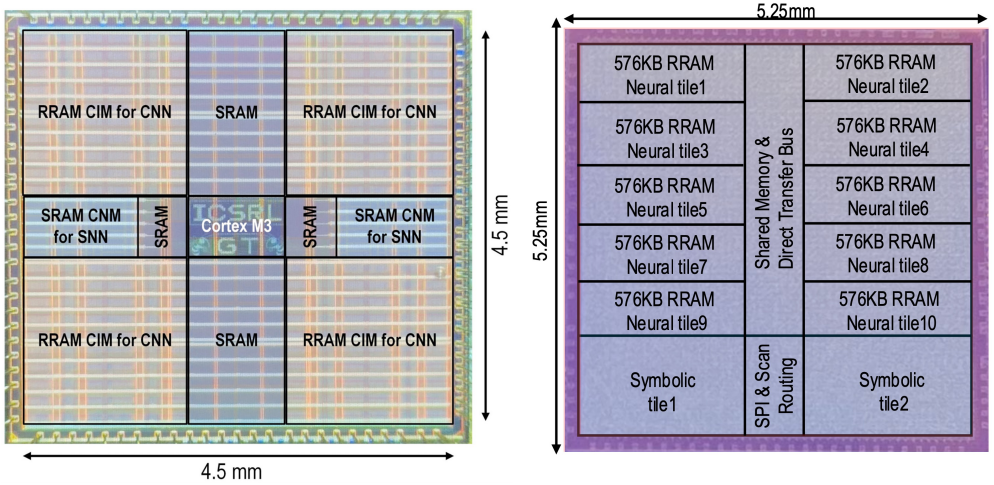
How should we design computing systems for physical intelligence when performance, energy, and reliability are shaped by emerging technologies?

Chiplet heterogenous integration



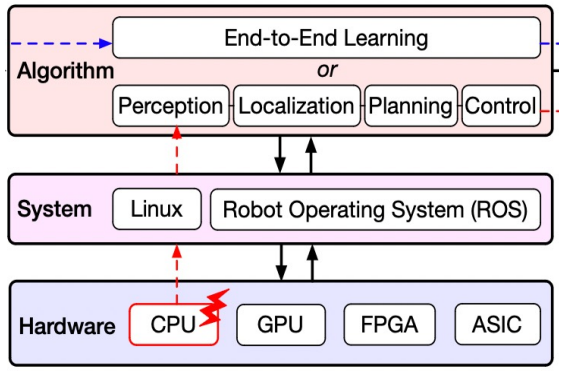
"Heterogeneous 3D Integrated CIM for Neuro-symbolic AI", **DATE'24**

Memory-centric computing (in/near-mem, emerging mem)



"SoC for Neuro-Symbolic AI", **JSSC'26**
"Heterogeneous RRAM/SRAM for Tracking", **ISSCC'23**, **JSSC'23**

Reliability-efficiency optimization



"Reliability Optimization for Autonomous Systems", **DAC'23**

Goal: Enable energy-efficient and reliable physical intelligence at the edge of silicon limits

Vision: Advancing the Future of Physical AI Computing

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Applications
- Robotics, VLA, World model
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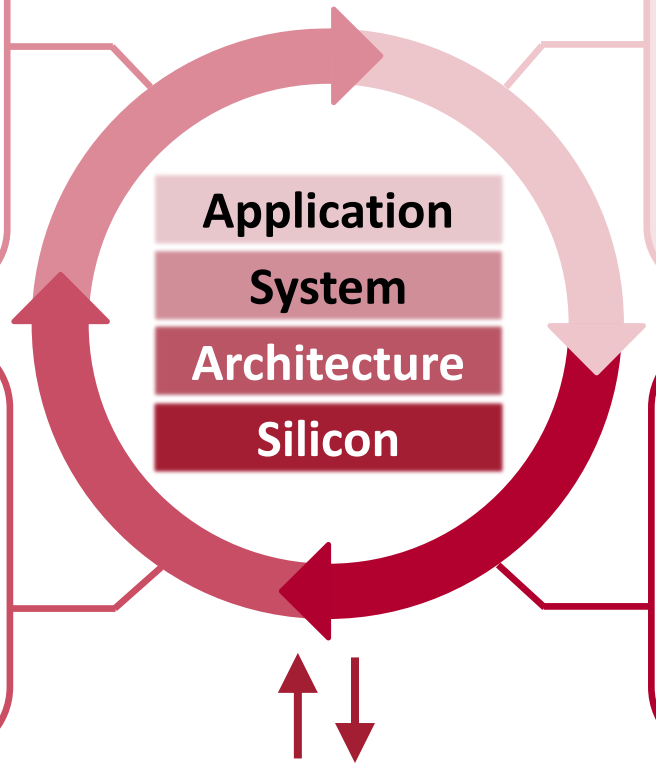

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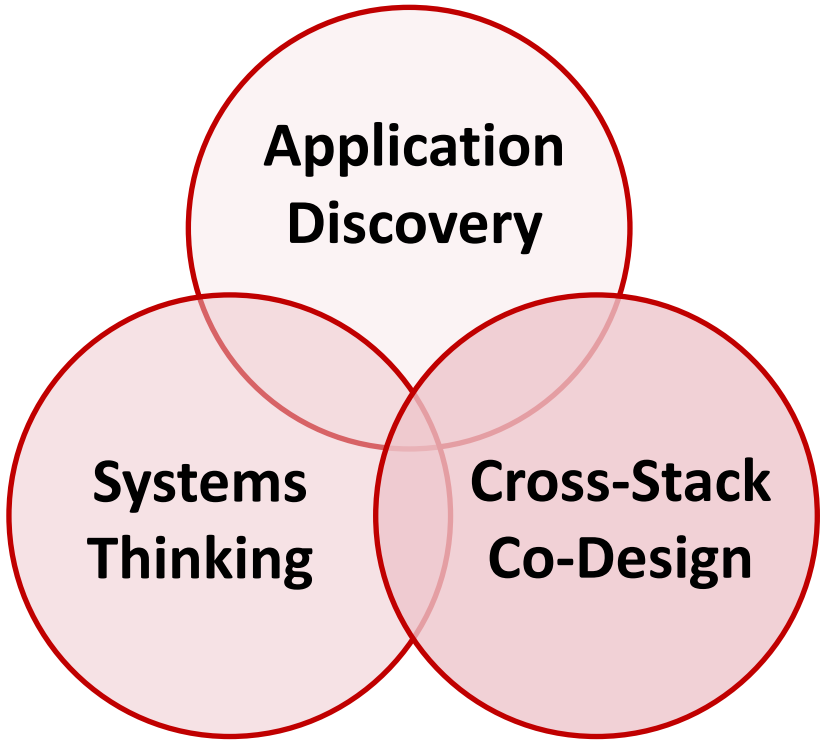
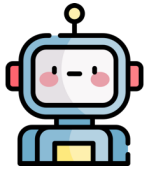
Physical and Agentic AI

Cross-stack system-architecture-silicon co-design for Physical AI computing

Direction 3: Generalized Design Methodology

How can we systematically design efficient, reliable computing systems across diverse emerging application domains?





Autonomous & Cyber-physical Systems
- **VLA**: map vision + language to actions
- **World Model**: predict physical futures
- **World-Action Model**: couple prediction & action
- **Agentic Physical AI**: coordinate reasoning & skills

Edge & Embedded AI Systems
- AR/VR, 3D vision, IoT, mobile sensing

Emerging Intelligent Infrastructure
- sensor-compute fabrics, medical systems

Goal: Enable generalizable and reusable cross-stack design methodology across domains

Vision: Advancing the Future of Physical AI Computing

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- Agentic-centric memory

Applications
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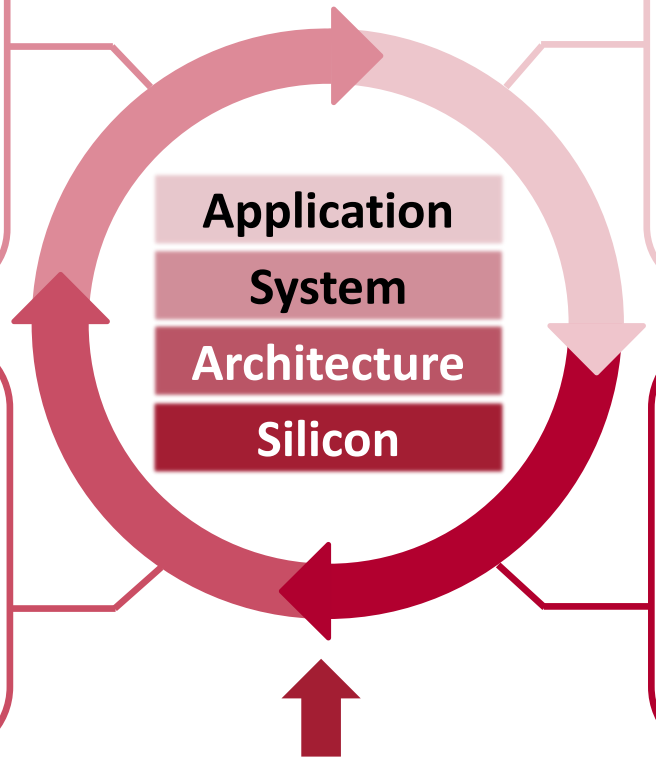

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Physical and Agentic AI

Cross-stack system-architecture-silicon co-design for Physical AI computing

Direction 4: Agentic AI for Computing System Design

What if intelligence not only runs on computing systems, but also designs them?

Where AI designs: the computing stack

co-design across layers

AI for System

AI agents reason ***what*** to compute

HawkEye [NeurIPS'26]

GenAI for Systems [arXiv'26]

AI for Architecture

AI agents design ***how*** to compute

QuArch [ICML'26] · **ArchEval** [ISCA-WS'26] · **AgentDSE** [NeurIPS-WS'26]

AI for Chip

AI tools implement it ***physically***

CARVE [arXiv'26] · **HDLxGraph** [ASP-DAC'27] · **LACE** [ICLAD'26]

What it takes to build an AI designer

1. **Datasets & Knowledge**
   make tacit expertise explicit
2. **Benchmarks**
   measure reasoning, not recall
3. **Evaluation Platforms**
   simulators and tests as verifiers
4. **Agentic Workflows**
   reasoning inside the tool loop
5. **Verification & Evidence**
   measured, auditable evidence
6. **Cross-Layer Co-Design**
   shared contracts, software → silicon

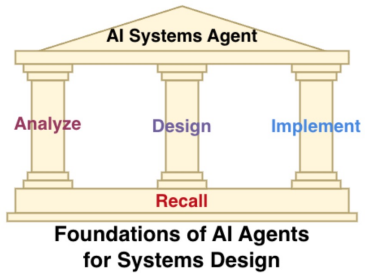
Goal: Enable hardware, system, and AI agents to co-design, co-reason, and co-evolve

AI for Architecture: From Reasoning to Discovery

① Benchmark → ② Evaluate → ③ Explore → ④ Discover

QuArch [ICML'26]

Can models reason about architecture?



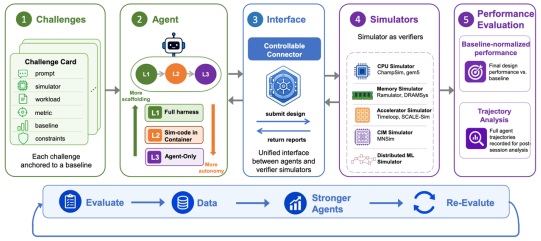
3K QAs

recall 84-91% vs. reasoning 44-66%

Knowing is not reasoning

ArchEval [ISCA'26 WS]

Can agents act as computer architects?



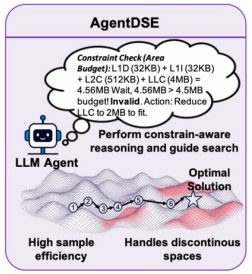
20 tasks · 8 simulators

memory, accelerators, CPU, CIM, Distributed ML

Good simulator operators, not arch judges

AgentDSE [ISCA'26 WS]

Can reasoning make search efficient?



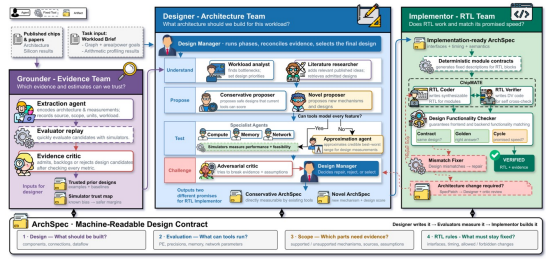
37-165× fewer

simulator calls vs. BO

DSE: from black-box sampling to hypothesis-test-refine process

ArchOrchestra

Can agents design and discover novel arch?



2.06x ADP improve

vs. published DNN Accs

Trustworthy design requires proposal, evidence, realization

Benchmark -> Evaluate -> Explore -> Discover: AI agents as computer architects

AI for System and Chip: Same Lessons Recur

AI for System AI agents reason what to compute

AI for Chip AI tools implement it physically

HawkEye NeurIPS'26

Hardware-aware GPU kernel optimization

1.22× expert Triton; new GPU = **10 tests**

Insight: ***testable knowledge beats more compute***

HDLxGraph ASP-DAC'27

Graph (AST + DFG) retrieval over HDL repositories

+12% search · **+12%** debugging · **+5%** completion

Insight: ***HDL is not software text, represent it as a graph***

GenAI for Systems arXiv'26

275+ papers · 11 areas · software -> silicon

5 recurring challenges <-> **5** design principles

Insight: ***the same difficulties recur, make progress compound***

LACE ICLAD'26

Multi-agent RISC-V ISA extension, formally checked

pass@1: near-zero -> **72.8%** across 4 RISC-V cores

Insight: ***decompose, localize, verify***

Same recipe across the stack: structured knowledge, closed feedback loops, and verification

Agentic AI for Computing System Design

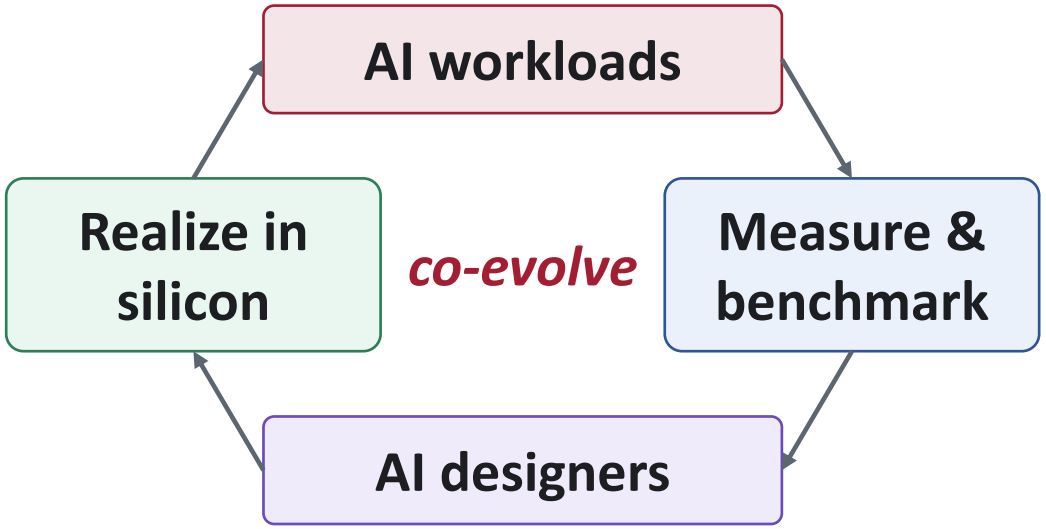
Three lessons that recur from system to silicon

1. **Evaluators: the bottleneck and the boundary**
   Make every evaluation count, bound what cannot be measured

2. **Knowledge ≠ reasoning ≠ judgment**
   Models know, agents operate tools, neither judges designs yet

3. **Trust comes from separating roles**
   Proposal, evidence, realization, no agent certifies its own work

Vision: self-evolving computing systems



AI as a research collaborator

Full-stack closure: shared contracts

A design flywheel: years -> days

From AI that writes hardware to AI that participates in computing system research

Agentic AI for Computing System Design

MODULE 2: AI FOR COMPUTING

8	Oct 30	● **AI for Software Systems: Compilers and GPU Kernels** Required reading: AlphaDev · AlphaEvolve · KernelBench Optional reading & more →
9	Nov 6	● **Midterm Project Presentations** No assigned readings.
10	Nov 13	● **AI for Computer Architecture I: Measuring and Exploring** Guest lecture: **Prof. Sagar Karandikar** (UC Berkeley) Required reading: ArchEval · CHIA Optional reading & more →
11	Nov 20	● **AI for Computer Architecture II: Architecture 2.0** Guest lecture: **Prof. Vijay Janapa Reddi** (Harvard) Required reading: ArchOrchestra · Architecture 2.0 (one chapter) Optional reading & more →
-	*Nov 27*	*No class - Thanksgiving*
12	Dec 4	● **AI for RTL and Chip Physical Design** Guest lecture: **Dr. Zhengqi Gao** (Google DeepMind / Anthropic) Required reading: AlphaChip · ChipNeMo · Comprehensive Verilog Design

Workshops @ ISCA 26, ASPLOS 26

We will discuss in the second half of this semester!

Architecting Physical Intelligence:

Cross-Stack Co-Design from Systems to Silicon

Zishen Wan
Assistant Professor
Department of CS
Columbia University

CS6998 Week 7: Accelerating Neuro-Symbolic and Compositional AI